



ELEC 5690 - Advanced Topics in Artificial Intelligence for Medical Image Analysis

Lecture 03: Basic Vision Models (Segmentation)

Sep 22, 2026



- Review what we have learned last time
- Basic vision models: segmentation
- Advanced models for higher-dimensional (3D) data
- Case studies of CNNs for medical image segmentation
- Advanced models for higher-dimensional (video) data



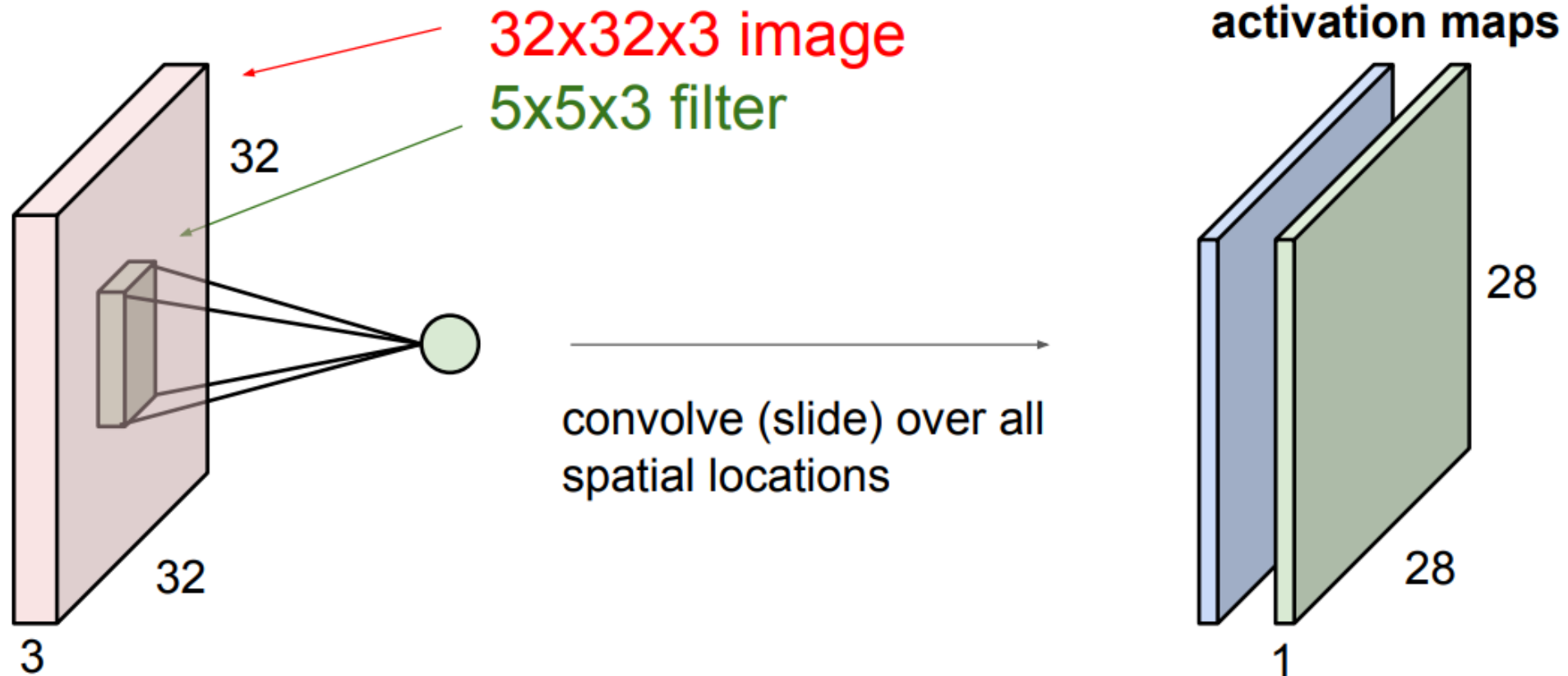
- Review what we have learned last time
- Basic vision models: segmentation
- Advanced models for higher-dimensional (3D) data
- Case studies of CNNs for medical image segmentation
- Advanced models for higher-dimensional (video) data

1. Last time



Convolutional layer

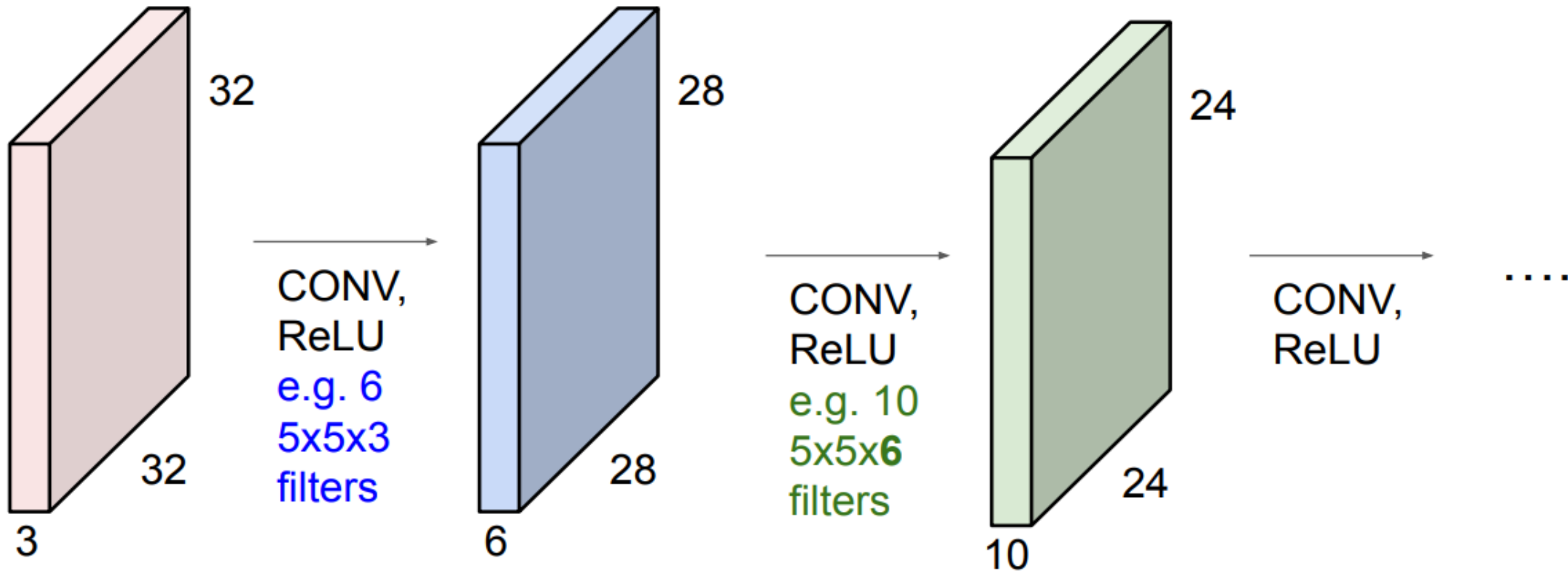
consider a second, **green** filter



1. Last time



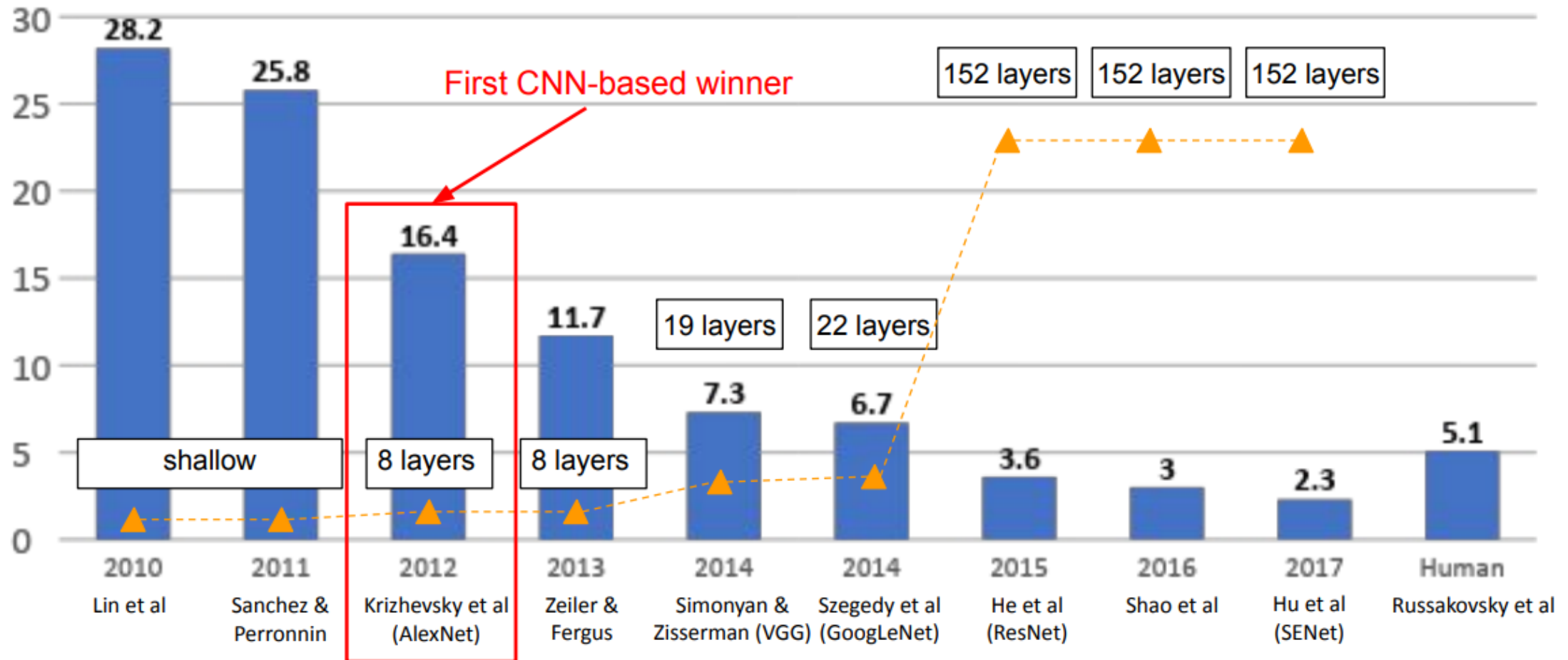
Preview: ConvNet (or CNN) is a sequence of Convolution Layers, interspersed with activation functions



1. Last time



ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



Slide credit: CS231n

1. Last time



VGGNet

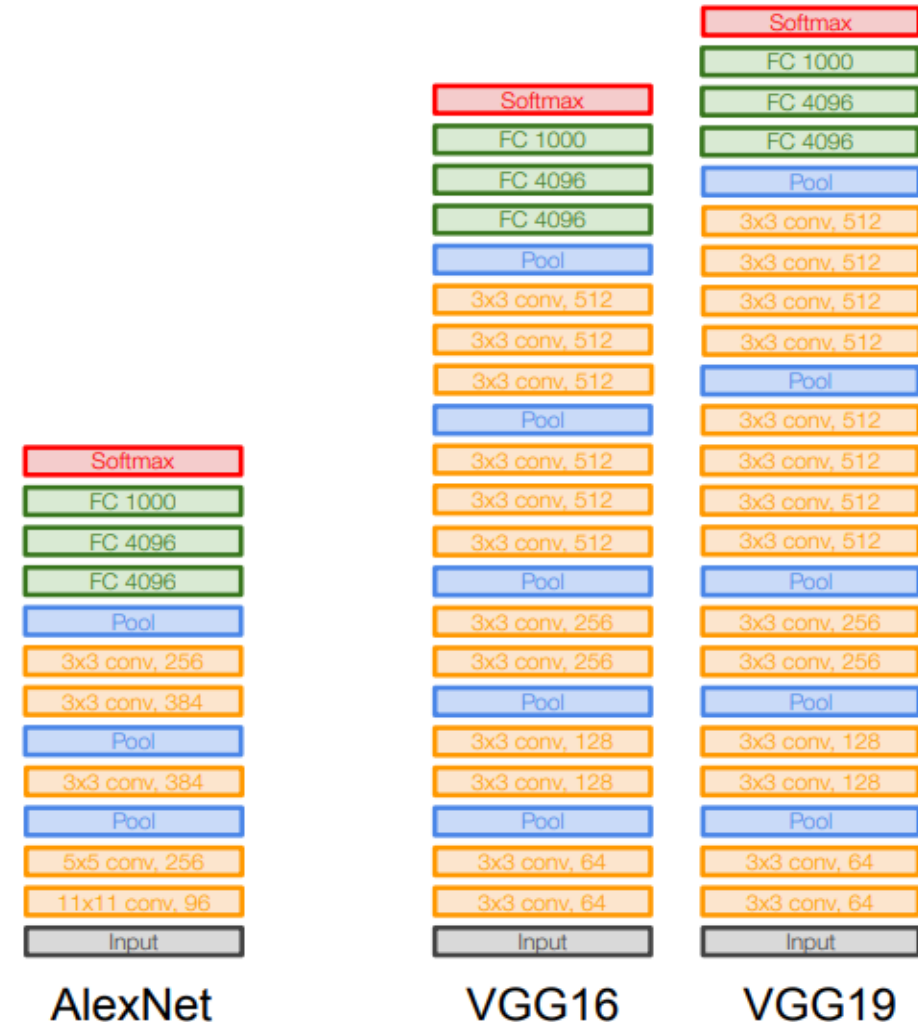
[Simonyan and Zisserman, 2014]

Small filters, Deeper networks

8 layers (AlexNet)

-> 16 - 19 layers (VGG16Net)

Only 3x3 CONV stride 1, pad 1
and 2x2 MAX POOL stride 2



1. Last time



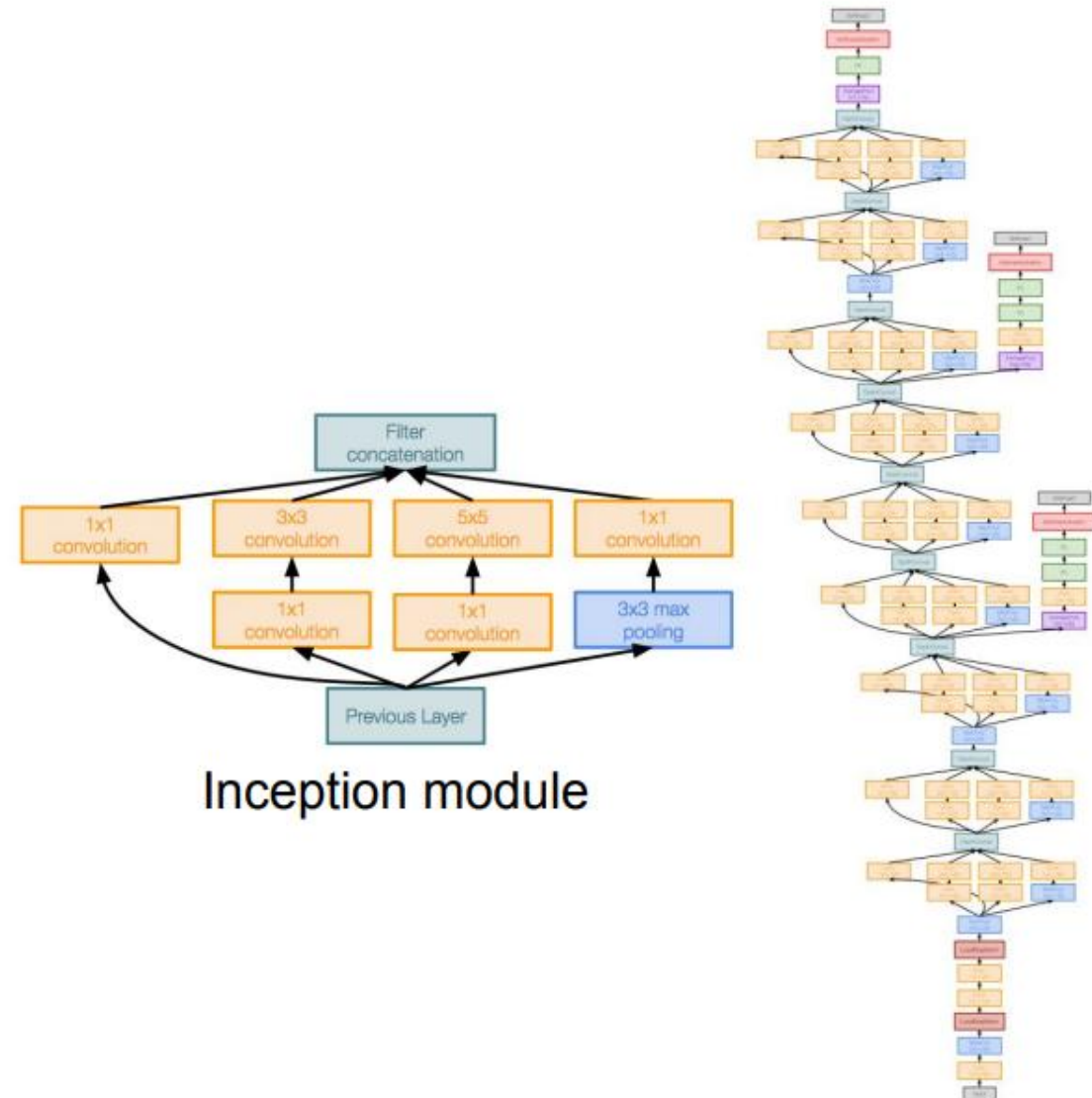
GoogLeNet

[Szegedy et al., 2014]

Deeper networks, with computational efficiency

- 22 layers
- Efficient “Inception” module
- Avoids expensive FC layers using a global averaging layer
- 12x less params than AlexNet

Also called “Inception Network”



1. Last time



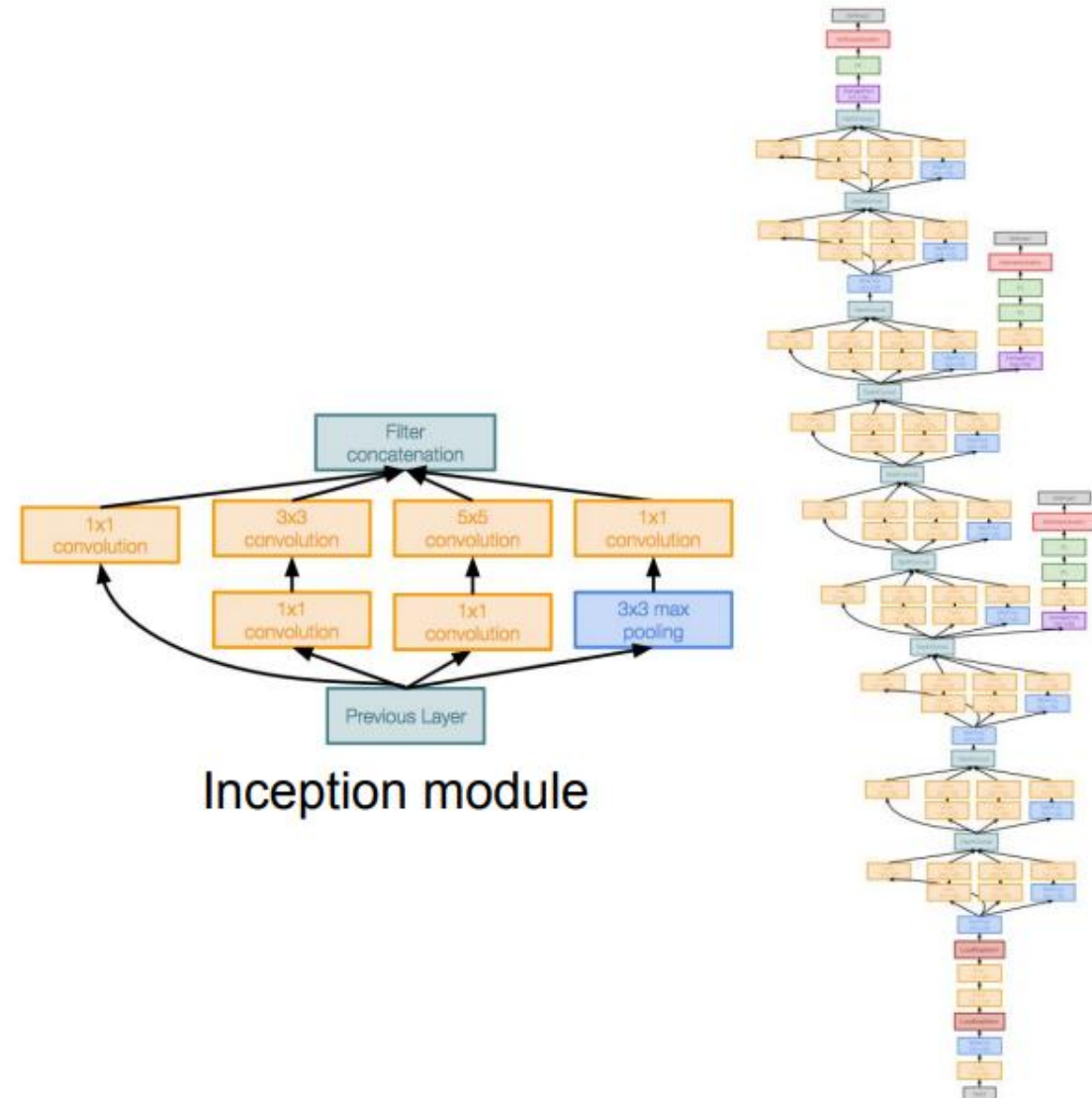
GoogLeNet

[Szegedy et al., 2014]

Deeper networks, with computational efficiency

- 22 layers
- Efficient “Inception” module
- Avoids expensive FC layers using a global averaging layer
- 12x less params than AlexNet

Also called “Inception Network”



1. Last time

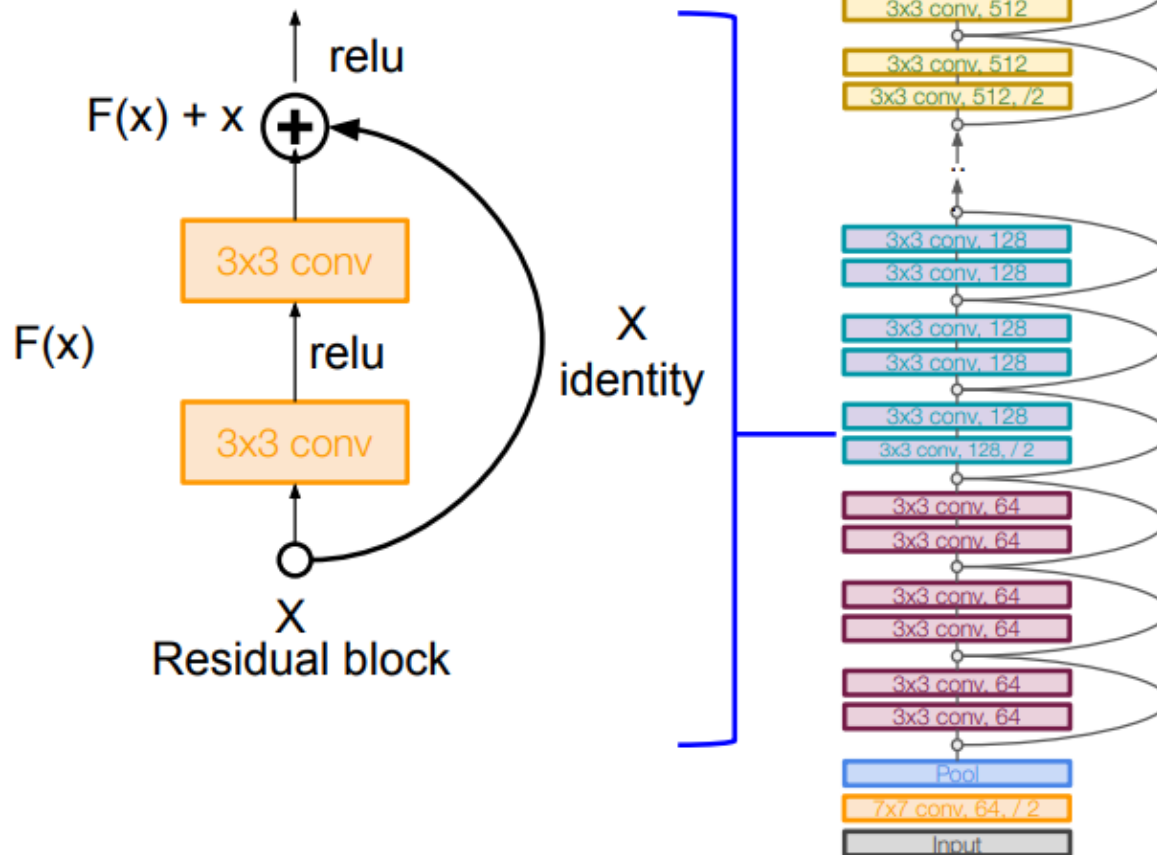


ResNet

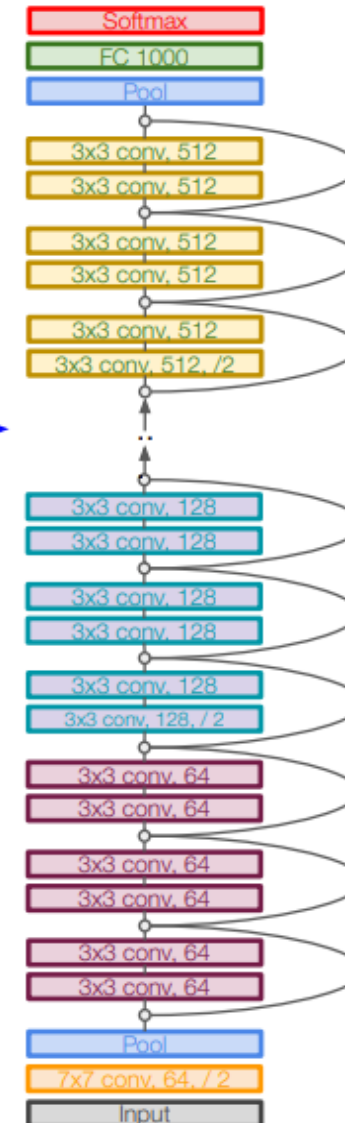
[He et al., 2015]

Full ResNet architecture:

- Stack residual blocks
- Every residual block has two 3x3 conv layers



[He et al., 2015]

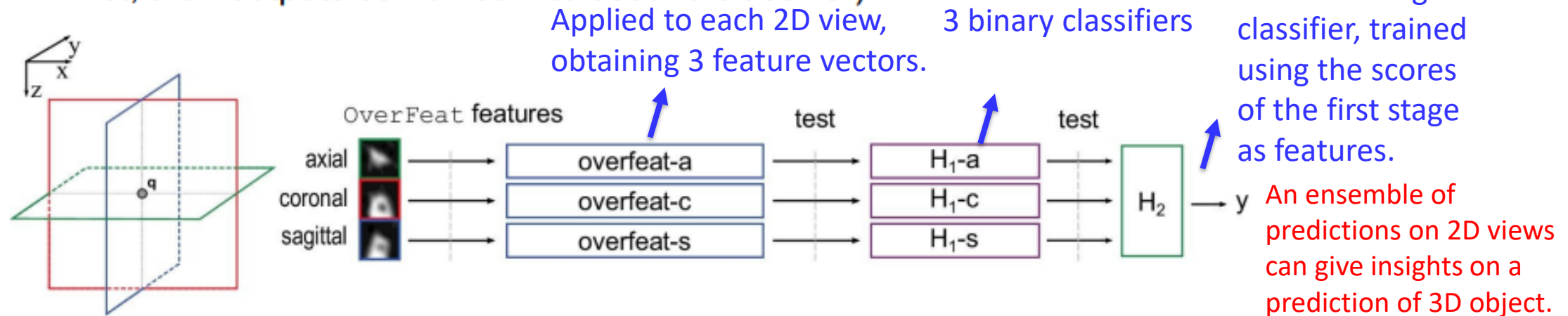


1. Last time



Ciompi et al. 2015

- Also extracted features from a deep learning model trained on ImageNet
 - Overfeat feature extractor (similar to AlexNet, but trained using additional losses for localization and detection)
 - To capture 3D information, extracted features from 3 different 2D views of each nodule, then input into 2-stage classifier (independent predictions on each view first, then outputs combined into second classifier).



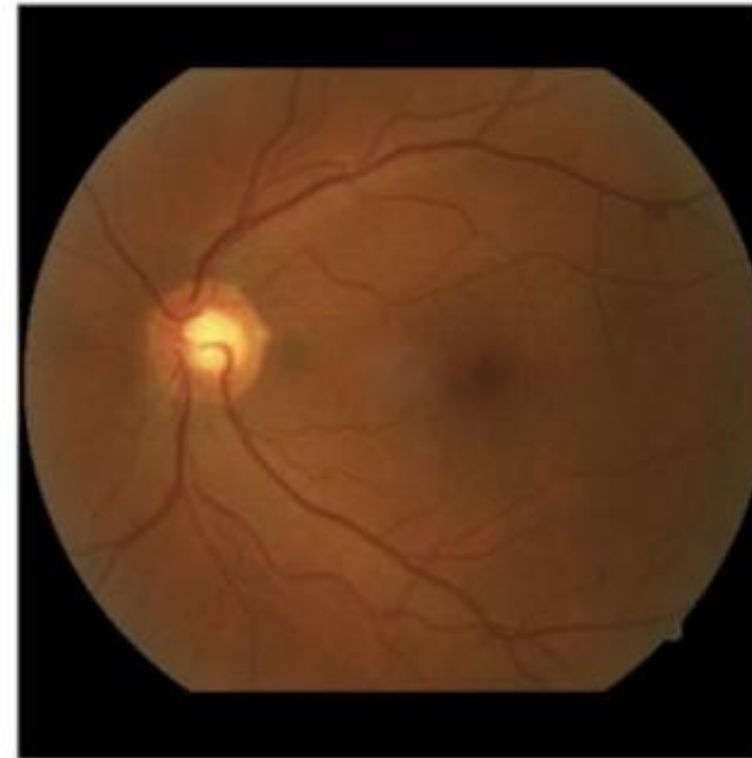
Ciompi et al. Automatic classification of pulmonary peri-fissural nodules in computed tomography using an ensemble of 2D views and a convolutional neural network out-of-the-box. Medical Image Analysis, 2015.

1. Last time



Gulshan et al. 2016

- **Task:** Binary classification of referable diabetic retinopathy from **retinal fundus photographs**
- **Input:** Retinal fundus photographs
- **Output:** Binary classification of referable diabetic retinopathy (y in $\{0,1\}$)
 - Defined as moderate and worse diabetic retinopathy, referable diabetic macular edema, or both



Gulshan, et al. Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs. JAMA, 2016.



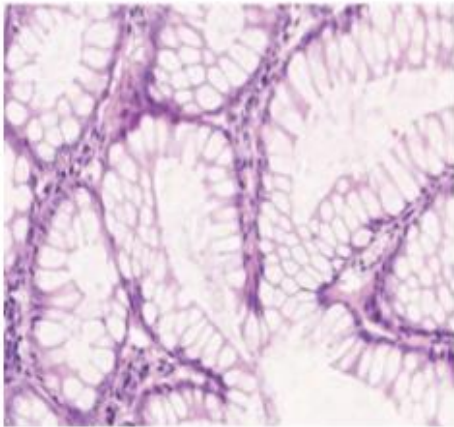
- Review what we have learned last time
- Basic vision models: segmentation
- Advanced models for higher-dimensional (3D) data
- Case studies of CNNs for medical image segmentation
- Advanced models for higher-dimensional (video) data

2. Basic vision models: segmentation



Richer visual recognition tasks: segmentation and detection

Classification



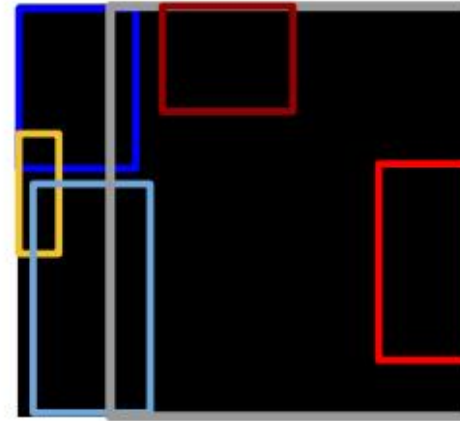
Output:
one category label for
image (e.g., colorectal
glands)

**Semantic
Segmentation**



Output:
category label for each pixel
in the image

Detection



Output:
Spatial bounding box for
each **instance** of a
category object in the
image

**Instance
Segmentation**



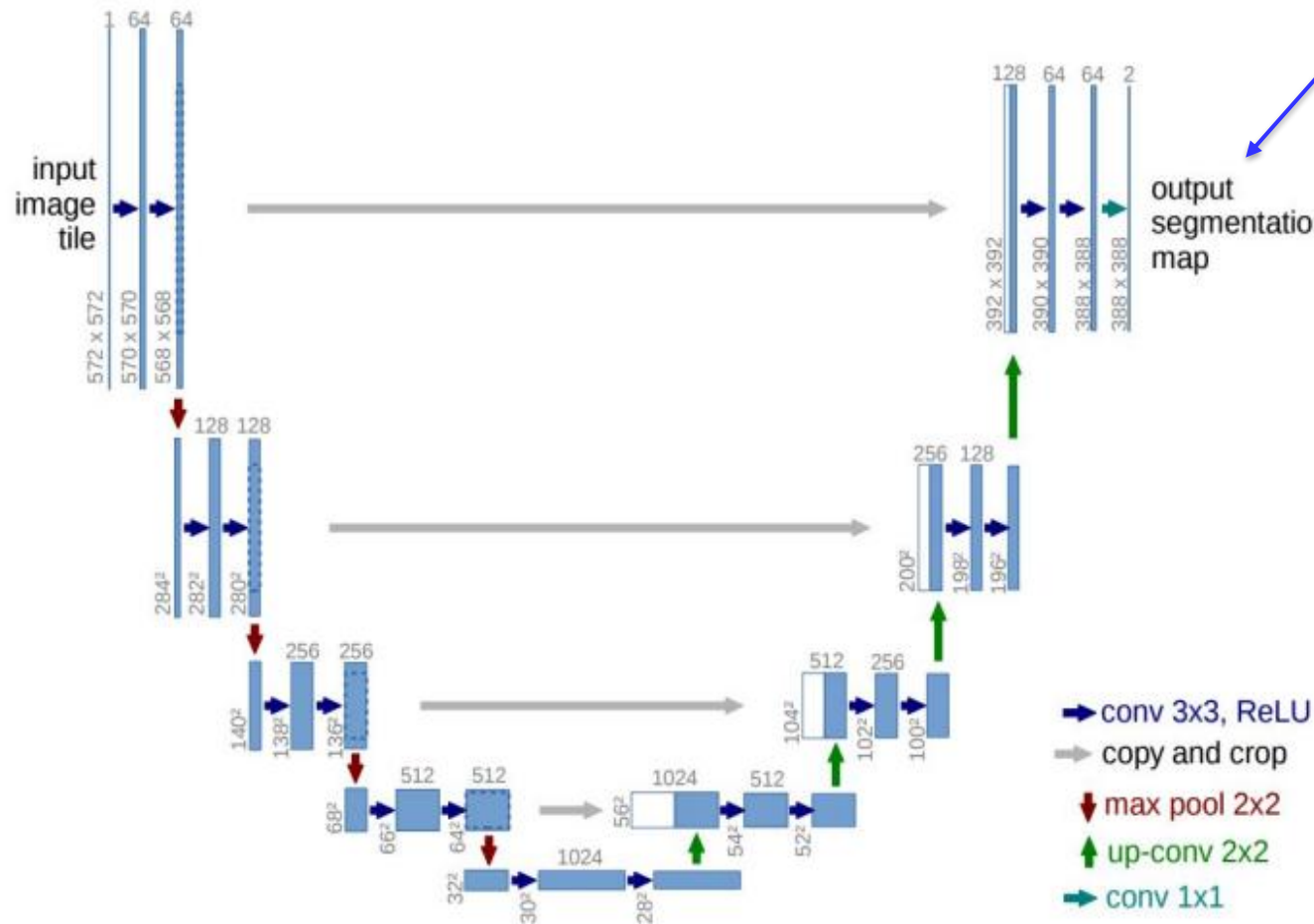
Output:
Category label and instance
label for each pixel in the
image

Distinguishes between different instances of an object

2. Basic vision models: segmentation



Semantic segmentation: U-Net



Output is an image mask:
width x height x # classes

Output image size a little smaller than
original, due to convolutional
operations w/o padding



Gives more “true” context for reasoning
over each image area. Can tile to make
predictions for arbitrarily large images

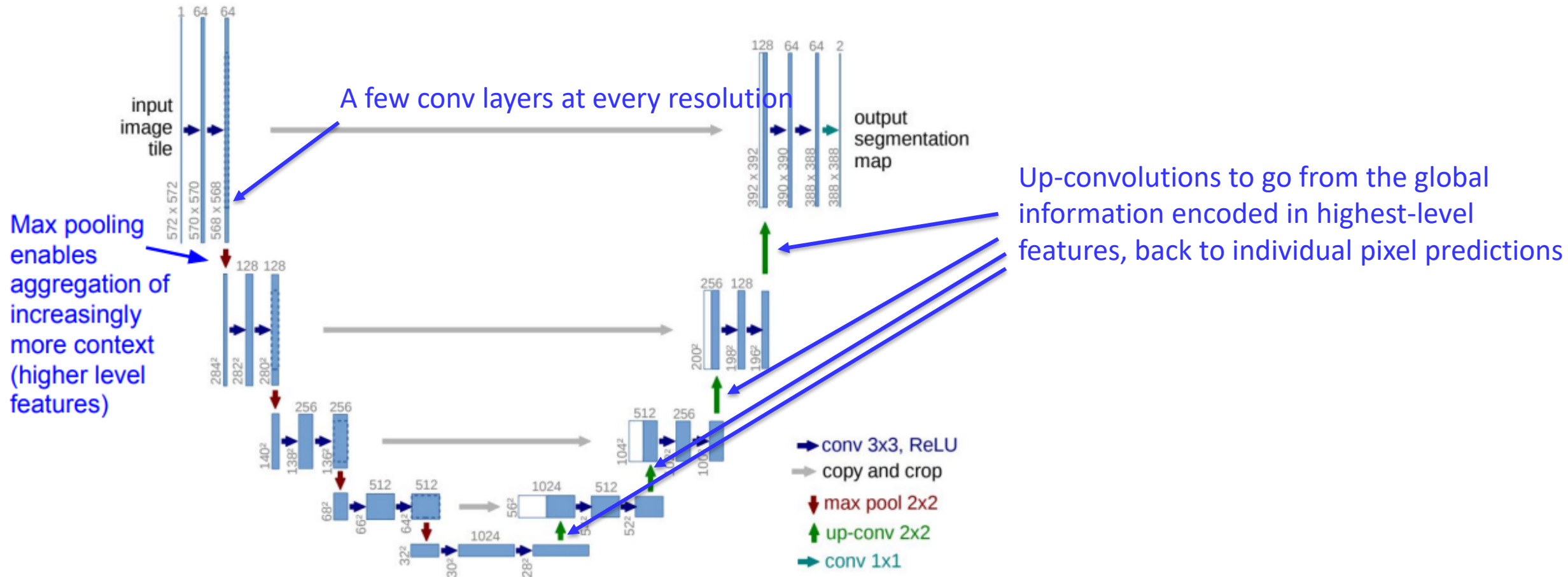
Semantic segmentation: U-Net



2. Basic vision models: segmentation



Semantic segmentation: U-Net

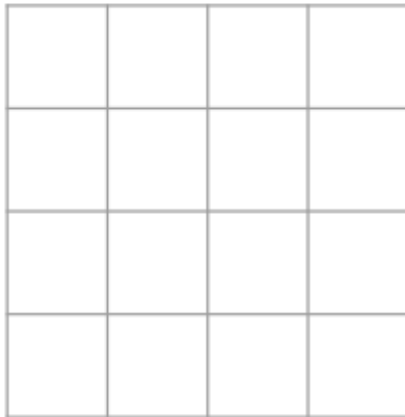


2. Basic vision models: segmentation

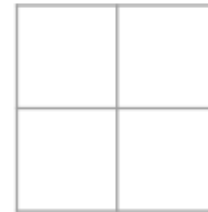


Up-convolutions

Recall: Normal 3 x 3 convolution, stride 2 pad 1



Input: 4 x 4



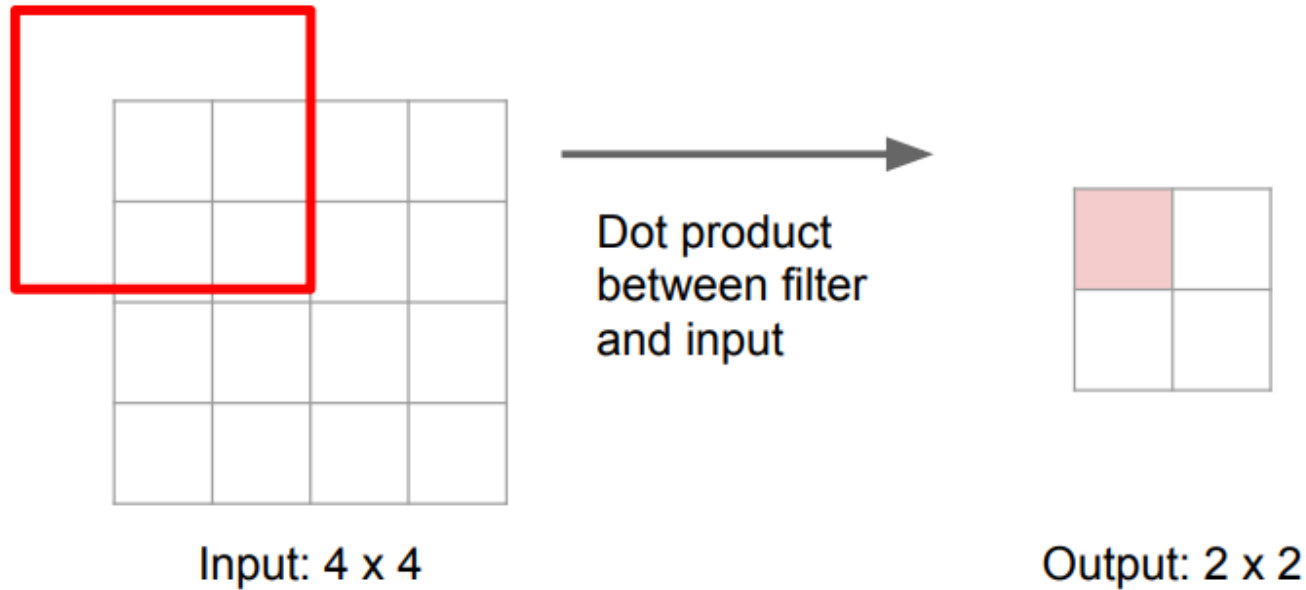
Output: 2 x 2

2. Basic vision models: segmentation



Up-convolutions

Recall: Normal 3 x 3 convolution, stride 2 pad 1

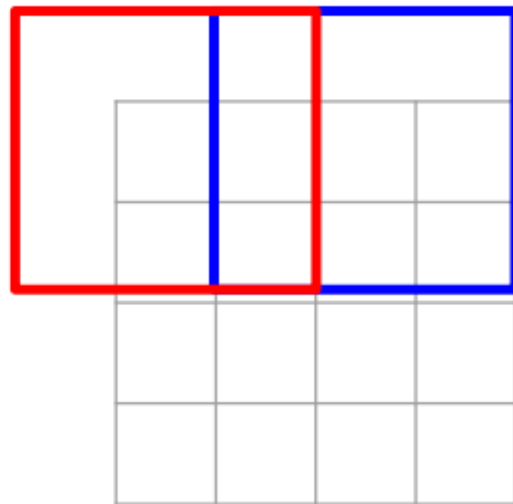


2. Basic vision models: segmentation



Up-convolutions

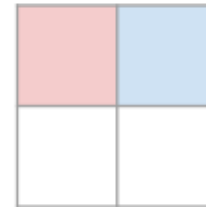
Recall: Normal 3 x 3 convolution, stride 2 pad 1



Input: 4 x 4



Dot product
between filter
and input



Output: 2 x 2

Filter moves 2 pixels in
the input for every one
pixel in the output

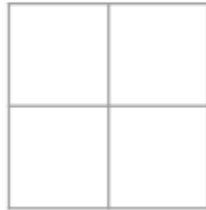
Stride gives ratio between
movement in input and
output

2. Basic vision models: segmentation

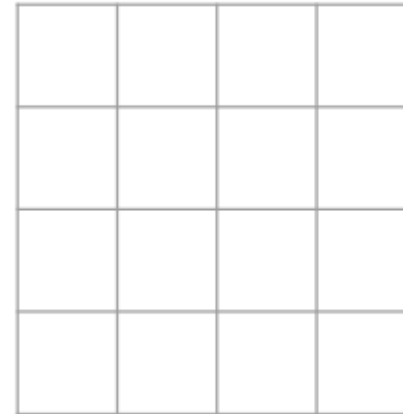


Up-convolutions

3 x 3 **transpose** convolution, stride 2 pad 1



Input: 2 x 2



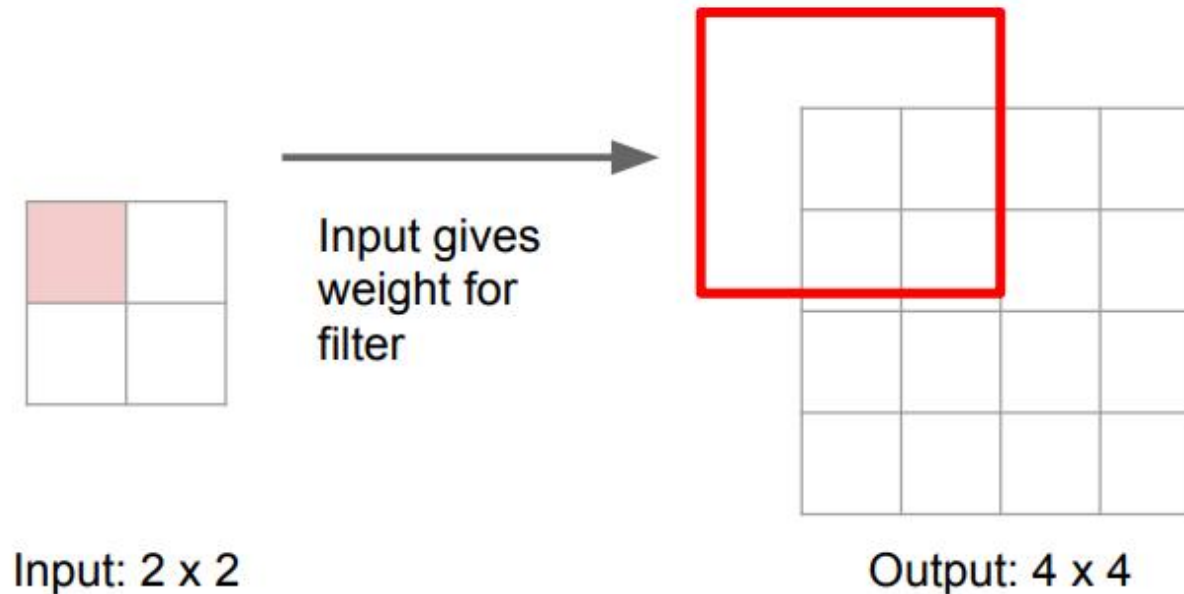
Output: 4 x 4

2. Basic vision models: segmentation



Up-convolutions

3 x 3 **up-convolution**, stride 2 pad 1

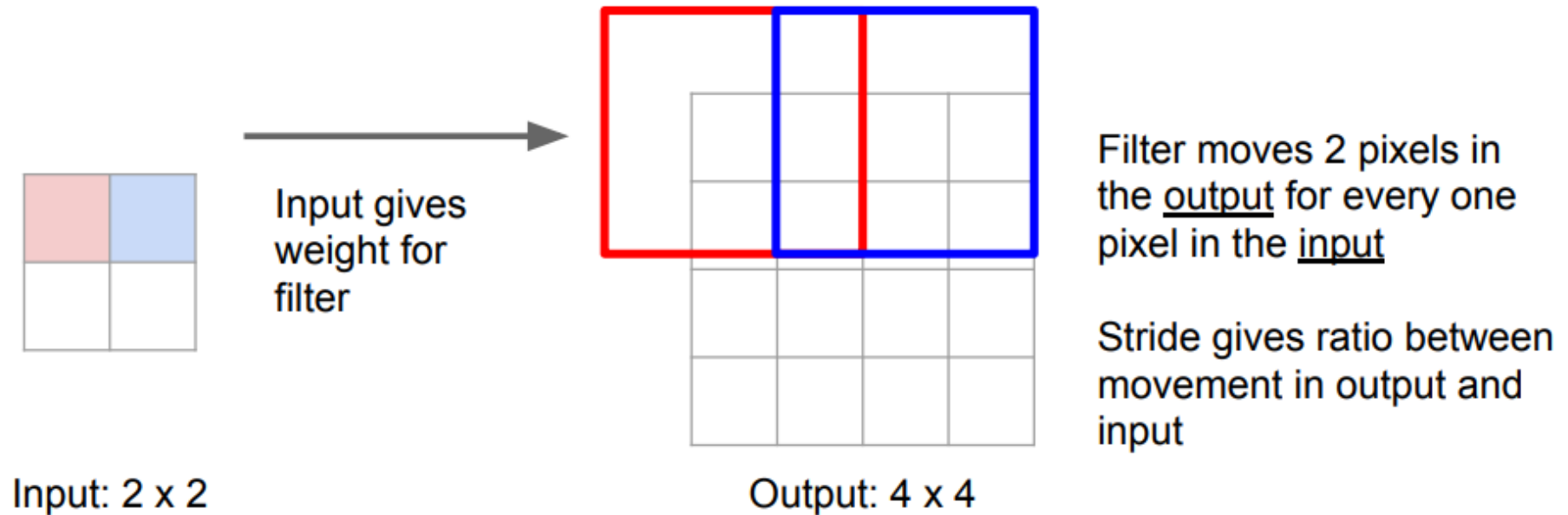


2. Basic vision models: segmentation



Up-convolutions

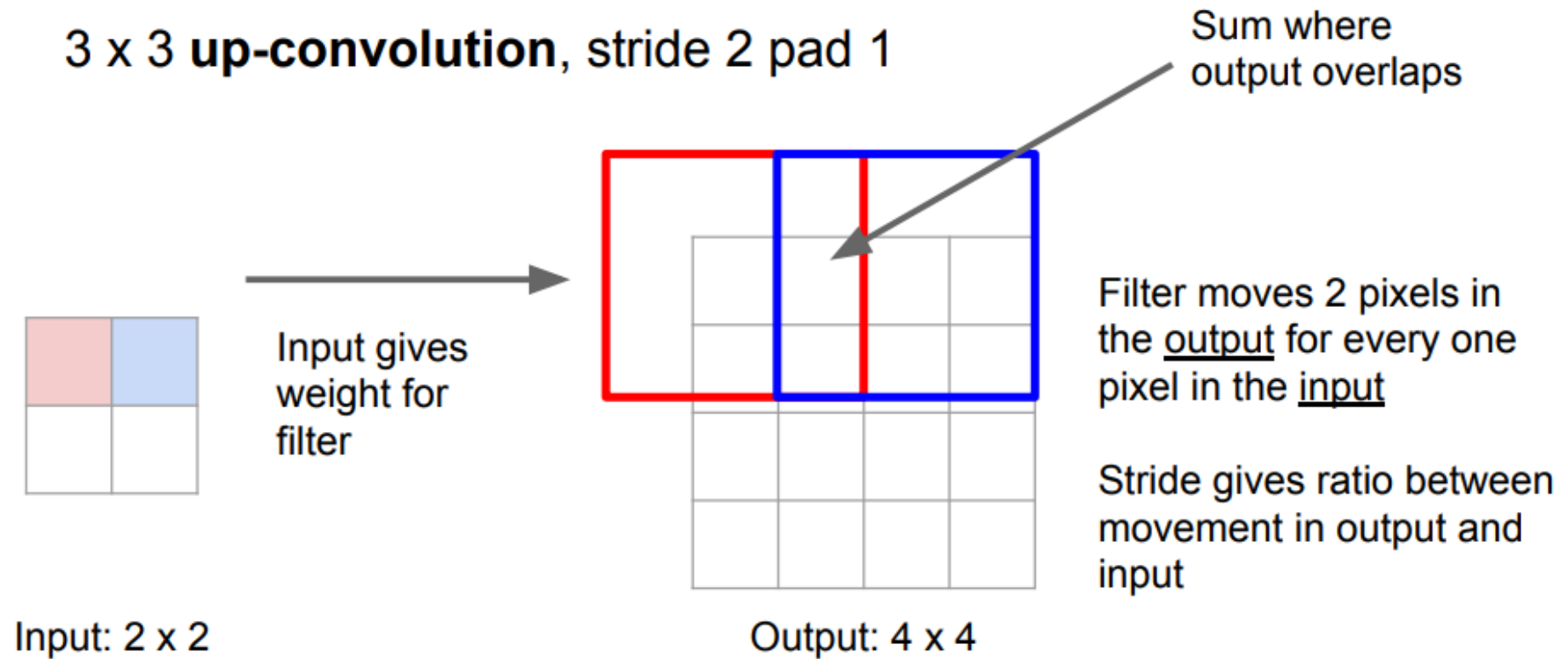
3 x 3 **up-convolution**, stride 2 pad 1



2. Basic vision models: segmentation



Up-convolutions



2. Basic vision models: segmentation

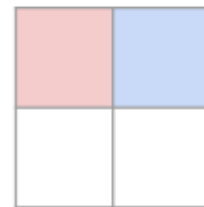


Up-convolutions

Other names:

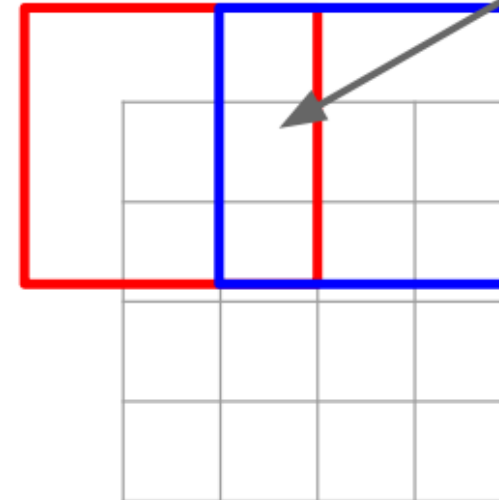
- Transpose convolution
- Fractionally strided convolution
- Backward strided convolution

3 x 3 up-convolution, stride 2 pad 1



Input: 2 x 2

Input gives weight for filter



Output: 4 x 4

Sum where output overlaps

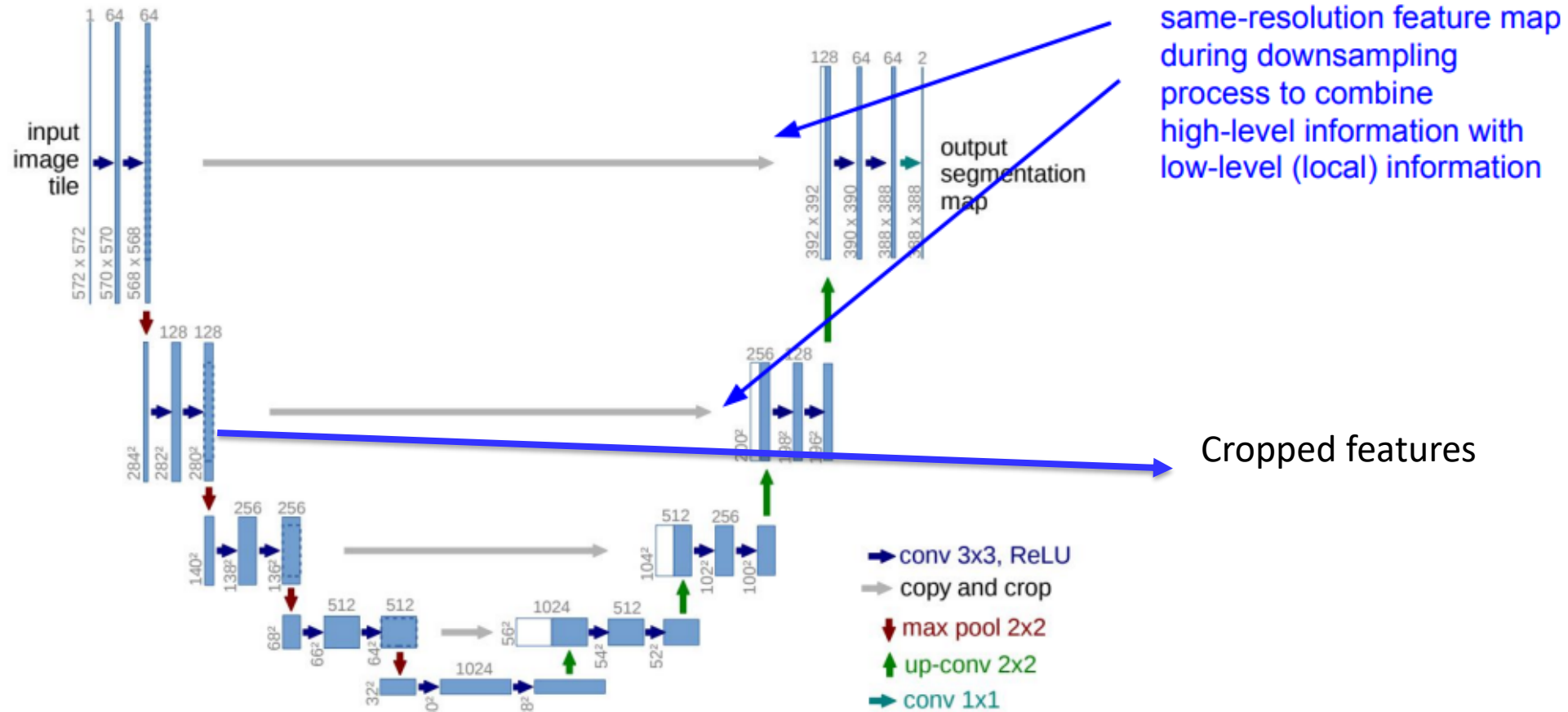
Filter moves 2 pixels in the output for every one pixel in the input

Stride gives ratio between movement in output and input

2. Basic vision models: segmentation



Semantic segmentation: U-Net

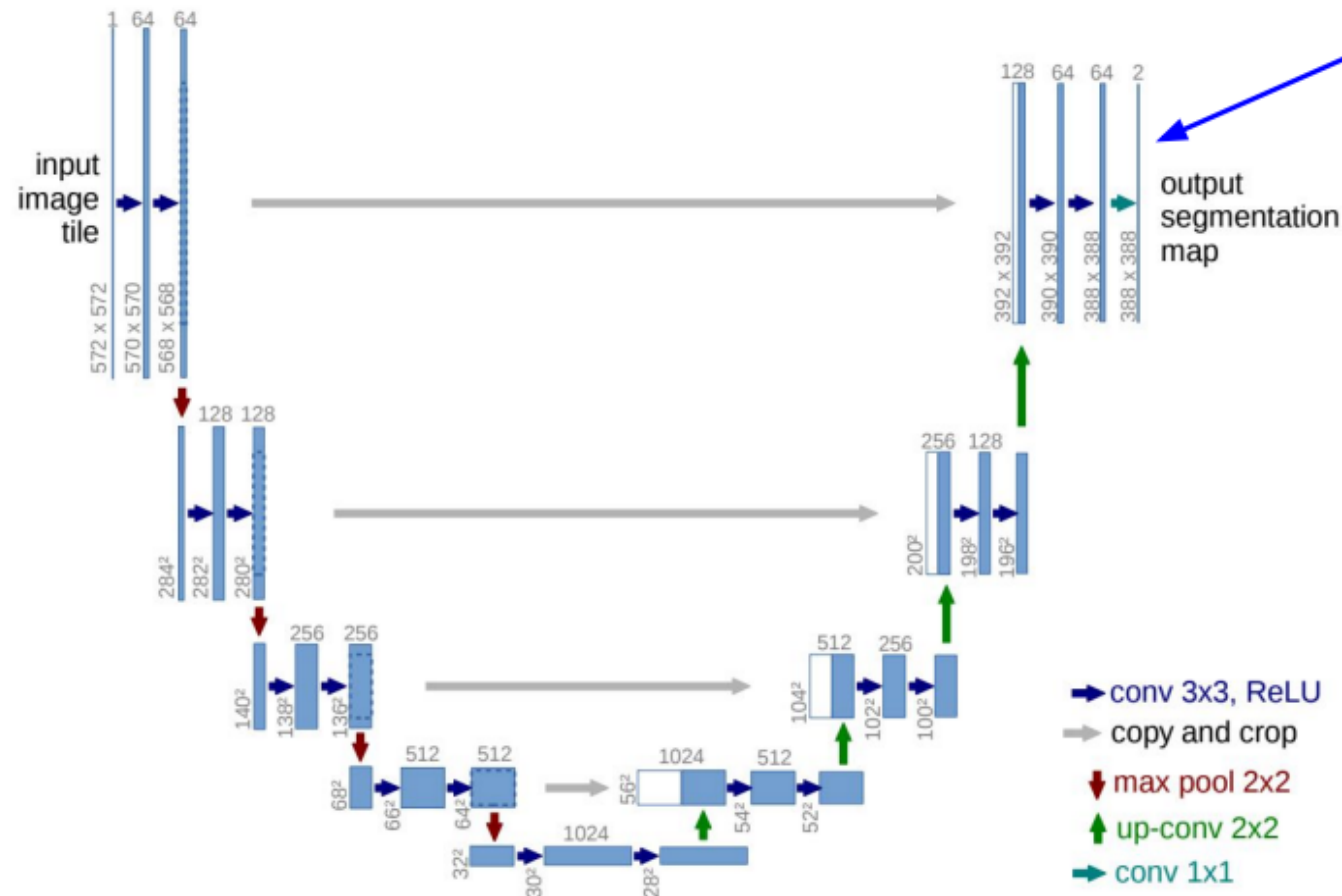


Ronneberger et al. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. 2015.

2. Basic vision models: segmentation



Semantic segmentation: U-Net



Train with classification loss
(e.g. binary cross entropy)
on every pixel, sum over all
pixels to get total loss

2. Basic vision models: segmentation



Semantic segmentation: IOU evaluation

Intersection over Union:

$$IoU = \frac{target \cap prediction}{target \cup prediction}$$

pixels included in both
target and prediction
maps

Total # pixels in the
union of both masks

2. Basic vision models: segmentation



Semantic segmentation: IOU evaluation

Intersection over Union:

$$IoU = \frac{target \cap prediction}{target \cup prediction}$$

pixels included in both
target and prediction
maps

Total # pixels in the
union of both masks

Can compute this over all masks in the
evaluation set, or at individual mask and image
levels to get finer-grained understanding of
performance.

Also known as Jaccard Index

2. Basic vision models: segmentation



Semantic segmentation: Pixel Accuracy evaluation

$$\text{Pixel Accuracy (PA)} = \frac{\text{\# correctly classified pixels}}{\text{\# total pixels}}$$

TP + TN

Total pixels in images

Q: What is a potential in image problem with this?

A: Think about what happens when there is class imbalance.

2. Basic vision models: segmentation



Semantic segmentation: Dice coefficient evaluation

$$\text{Dice Coefficient} = \frac{2 * (\text{target} \cap \text{prediction})}{\# \text{ target mask pixels} + \# \text{ prediction mask pixels}}$$

Diagram annotations:

- A blue arrow points from the text "2 * intersection" to the numerator $2 * (\text{target} \cap \text{prediction})$.
- A blue arrow points from the text "Sum of target mask size + prediction mask size" to the denominator $\# \text{ target mask pixels} + \# \text{ prediction mask pixels}$.

Very similar to IOU / Jaccard,
can derive one from the other

2. Basic vision models: segmentation



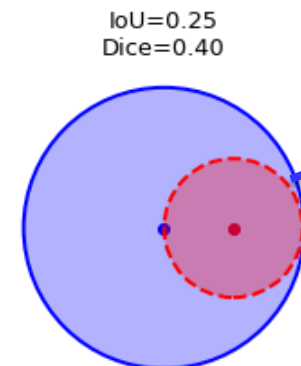
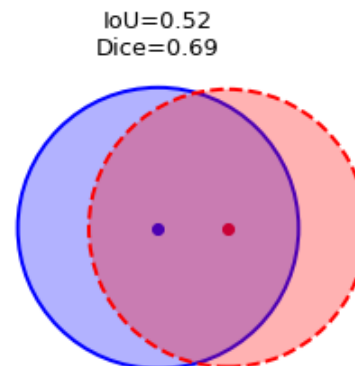
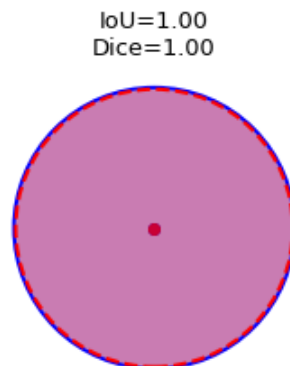
Jaccard vs Dice

$$Jaccard(A, B) = \frac{\|A \cap B\|}{\|A \cup B\|}$$

$$Dice(A, B) = \frac{2\|A \cap B\|}{\|A\| + \|B\|}$$

$$Jaccard = IoU = \frac{TP}{TP + FP + FN}$$

$$Dice = \frac{2TP}{2TP + FP + FN}$$



Jaccard: $\frac{1/4}{1} = 0.25$

Dice: $\frac{2 * (\frac{1}{4})}{1 + (\frac{1}{4})} = \frac{(\frac{1}{2})}{(\frac{5}{4})} = 0.4$

2. Basic vision models: segmentation



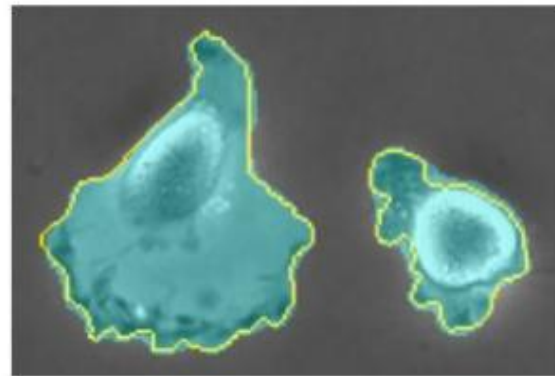
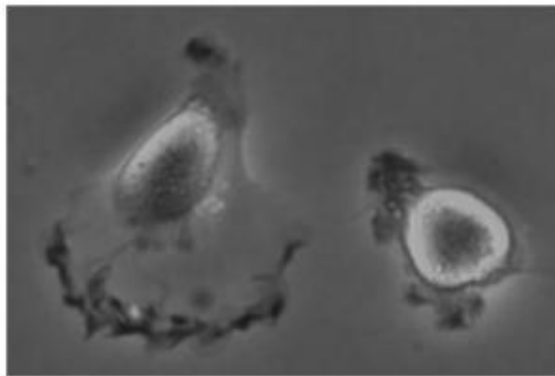
Semantic segmentation: summary of evaluation metrics

- Most commonly use IOU / Jaccard or Dice Coefficient
- Sometimes will also see pixel accuracy
- If multi-class segmentation task, typically report all these metrics per-class, and then a mean over all classes

2. Basic vision models: segmentation



Semantic segmentation: U-Net cell segmentation



Name	PhC-U373	DIC-HeLa
IMCB-SG (2014)	0.2669	0.2935
KTH-SE (2014)	0.7953	0.4607
HOUS-US (2014)	0.5323	-
second-best 2015	0.83	0.46
u-net (2015)	0.9203	0.7756

Very small dataset: 30 training images of size 512x512, in the ISBI 2012 Electron Microscopy (EM) segmentation challenge. Used excessive data augmentation to compensate.

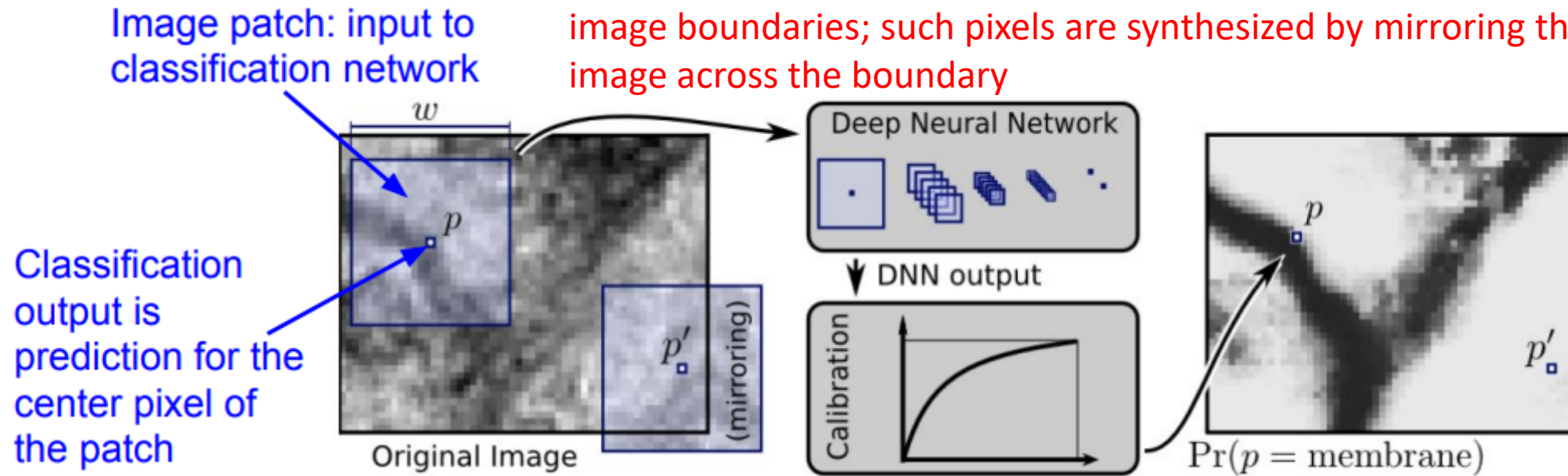
Ronneberger et al. U-Net: Convolutional Networks for Biomedical Image Segmentation. 2015.

2. Basic vision models: segmentation



Aside: segmentation through sliding-window pixel classification

When a pixel is close to the image border, its window will include pixels outside the image boundaries; such pixels are synthesized by mirroring the pixels in the actual image across the boundary



Note: a simple approach to segmentation can also be applying a classification CNN on image patches in a dense, sliding-window fashion (e.g. Ciresan et al.). But fully convolutional approaches such as U-Net generally achieve better performance.

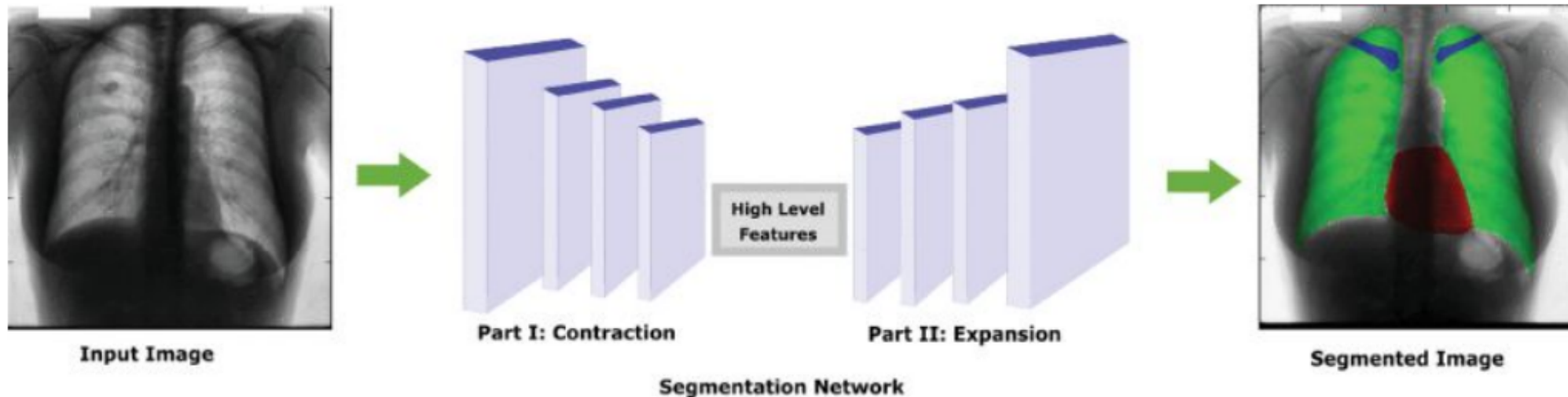
2. Basic vision models: segmentation



Novikov et al. 2018

Q: What loss function would be appropriate here?

- Chest x-ray segmentation of lungs, clavicles, and heart
- JSRT dataset of 247 chest-xrays at 2048x2048 resolution. (But downsampled to 128x128 and 256x256!)
- Used a U-Net based segmentation network with a few modifications



Novikov et al. Fully Convolutional Architectures for Multiclass Segmentation in Chest Radiographs. IEEE Trans. on Medical Imaging, 2018.

2. Basic vision models: segmentation



Novikov et al. 2018

Image ground truth class mask

$$L_{\text{dice}}(y, \hat{y}) = 1 - \frac{2 \sum_{i,j} y_{i,j} \hat{y}_{i,j}}{\sum_{i,j} y_{i,j} + \sum_{i,j} \hat{y}_{i,j}}$$

Image pixel class probabilities

- Multi-class segmentation -> tried both a per-pixel softmax loss as well as a loss based on the Dice coefficient. Note: this Dice loss is often useful to try!
- Class imbalance -> weight loss terms corresponding to each ground-truth class by inverse of class frequency: (# class pixels) / (total # pixels in data)

Body Part	Lungs		Clavicles		Heart	
Evaluation Metric	D	J	D	J	D	J
InvertedNet	0.972	0.946	0.902	0.821	0.935	0.879
All-Dropout	0.973	0.948	0.896	0.812	0.941	0.888
All-Convolutional	0.971	0.944	0.876	0.780	0.938	0.883
Original U-Net	0.971	0.944	0.880	0.785	0.938	0.883

Dice and Jaccard evaluation

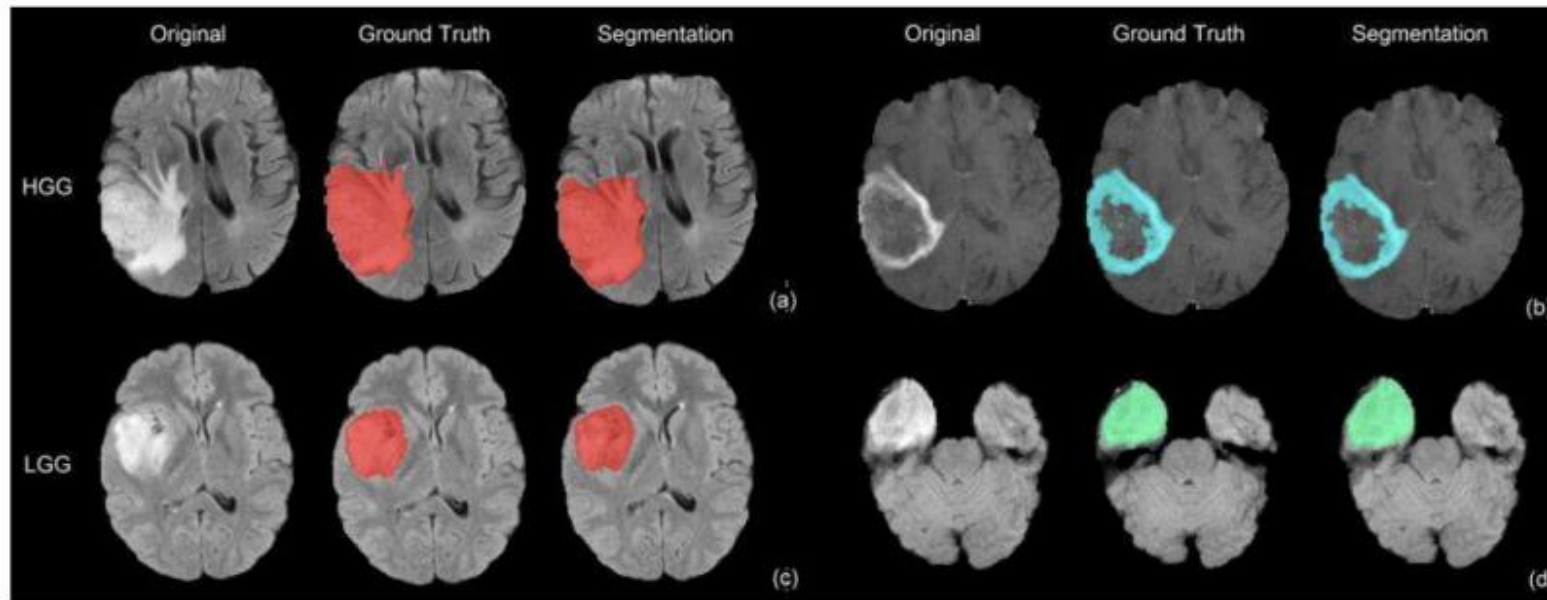
Novikov et al. Fully Convolutional Architectures for Multiclass Segmentation in Chest Radiographs. IEEE Trans. on Medical Imaging, 2018.

2. Basic vision models: segmentation



Dong et al. 2017

- Segmentation of tumors in brain MR image slices
- BRATS 2015 dataset: 220 high-grade brain tumor and 54 low-grade brain tumor MRIs
- U-Net architecture, Dice loss function



Dong et al. Automatic Brain Tumor Detection and Segmentation Using U-Net Based Fully Convolutional Networks. MIUA, 2017.

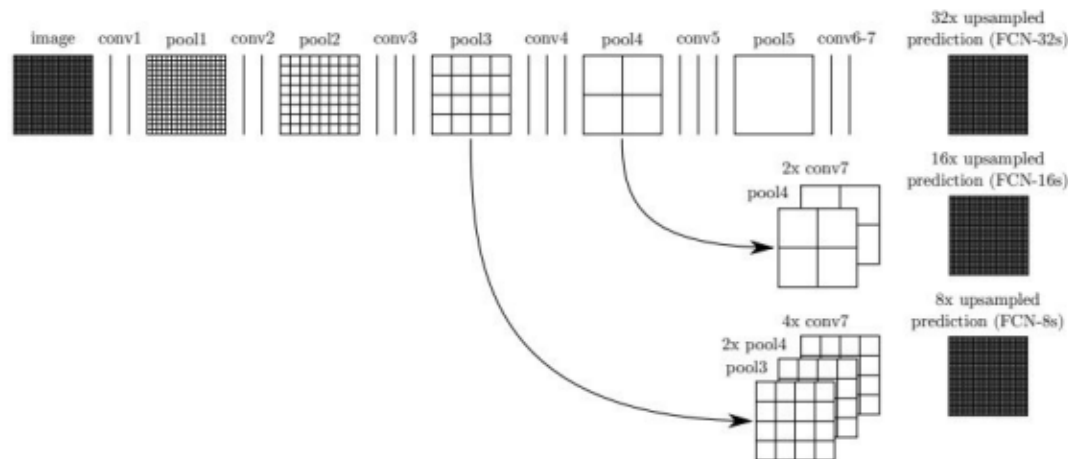
2. Basic vision models: segmentation



Can try DeepLab v3+ for segmentation projects!

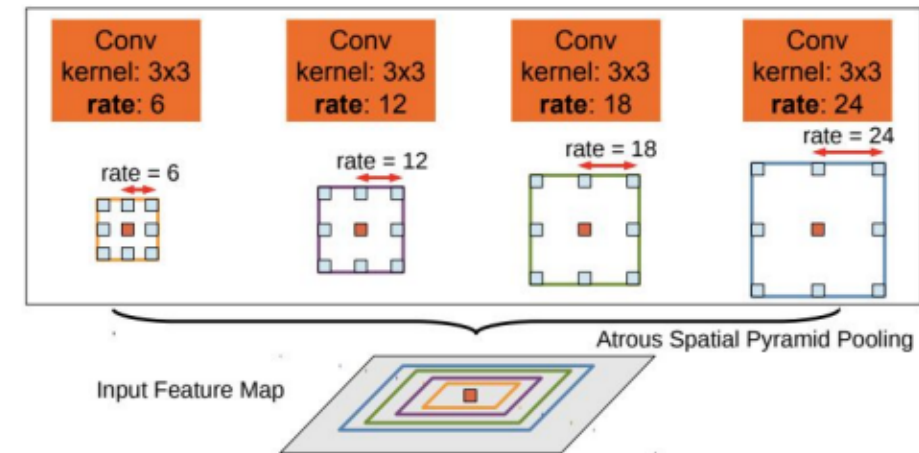
Other segmentation architectures

- **Fully convolutional networks (FCN)**
- Pre-cursor to U-Net, similar in structure but simpler upsampling pathway



Shelhamer*, Long*, et al. Fully Convolutional Networks for Semantic Segmentation. CVPR 2015.

- **DeepLab (v1-v3)**
- Uses “atrous convolutions” to control a filter’s field of view
- Parallel atrous convolutions with different rates for multi-scale features



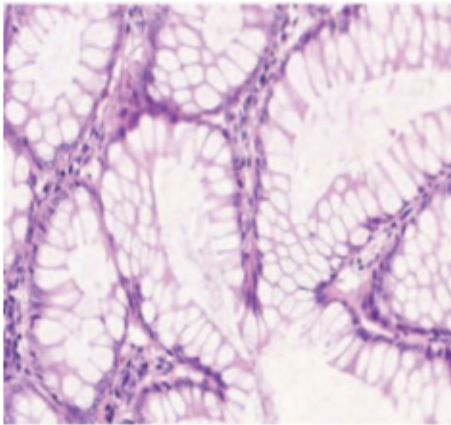
Chen et al. DeepLab: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolution, and Fully Connected CRFs. IEEE TPAMI, 2017.
Chen et al. Rethinking Atrous Convolution for Semantic Image Segmentation. 2017.

2. Basic vision models: segmentation



Richer visual recognition tasks: segmentation and detection

Classification



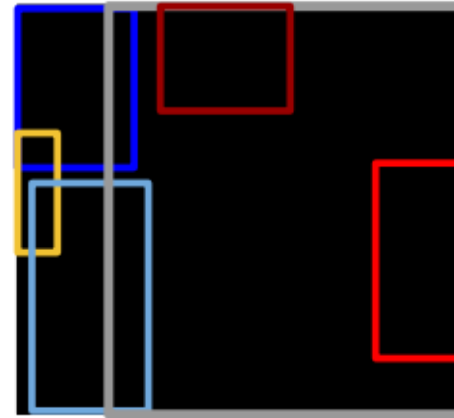
Output:
one category label for
image (e.g., colorectal
glands)

**Semantic
Segmentation**



Output:
category label for each pixel
in the image

Detection



Output:
Spatial bounding box for
each **instance** of a
category object in the
image

**Instance
Segmentation**



Output:
Category label and instance
label for each pixel in the
image

Distinguishes between different instances of an object

Figures: Chen et al. 2016. <https://arxiv.org/pdf/1604.02677.pdf>



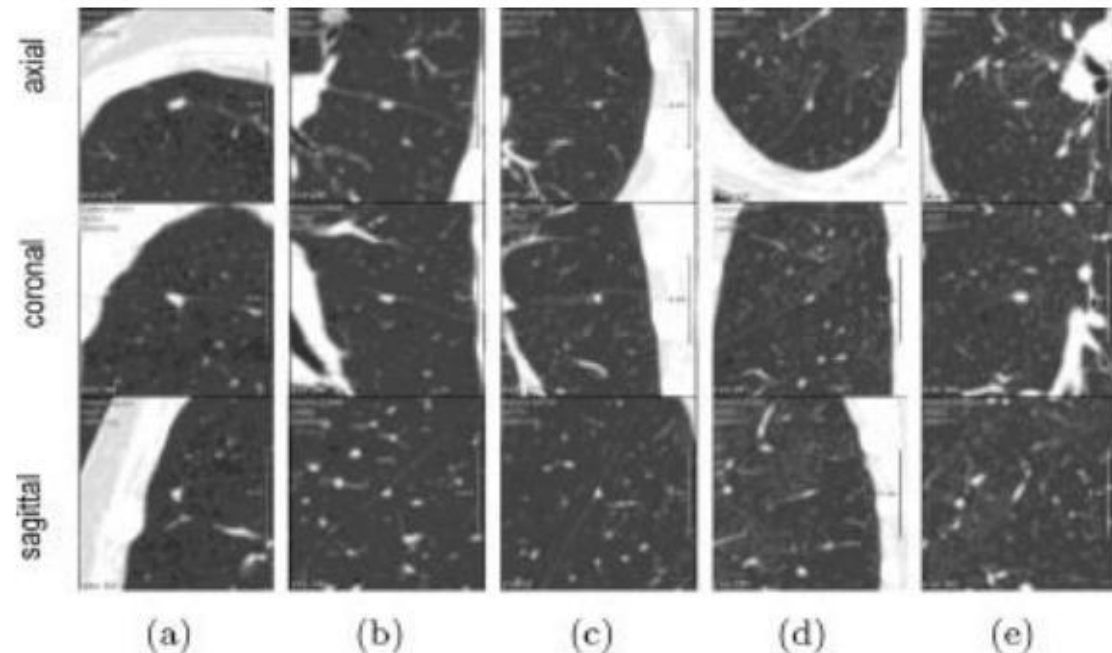
- Review what we have learned last time
- Basic vision models: segmentation
- Advanced models for higher-dimensional (3D) data
- Case studies of CNNs for medical image segmentation
- Advanced models for higher-dimensional (video) data

3. Advanced models for higher-dimensional (3D) data

How do we handle 3D data?

Recall: Ciompi et al. 2015

- Task: classification of lung nodules in 3D CT scans as peri-fissural nodules (PFN, likely to be benign) or not
- Dataset: 568 nodules from 1729 scans at a single institution. (65 typical PFNs, 19 atypical PFNs, 484 non-PFNs).
- Data pre-processing: prescaling from CT hounsfield units (HU) into [0,255]. Replicate 3x across R,G,B channels to match input dimensions of ImageNet-trained CNNs.



Ciompi et al. Automatic classification of pulmonary peri-fissural nodules in computed tomography using an ensemble of 2D views and a convolutional neural network out-of-the-box. *Medical Image Analysis*, 2015.

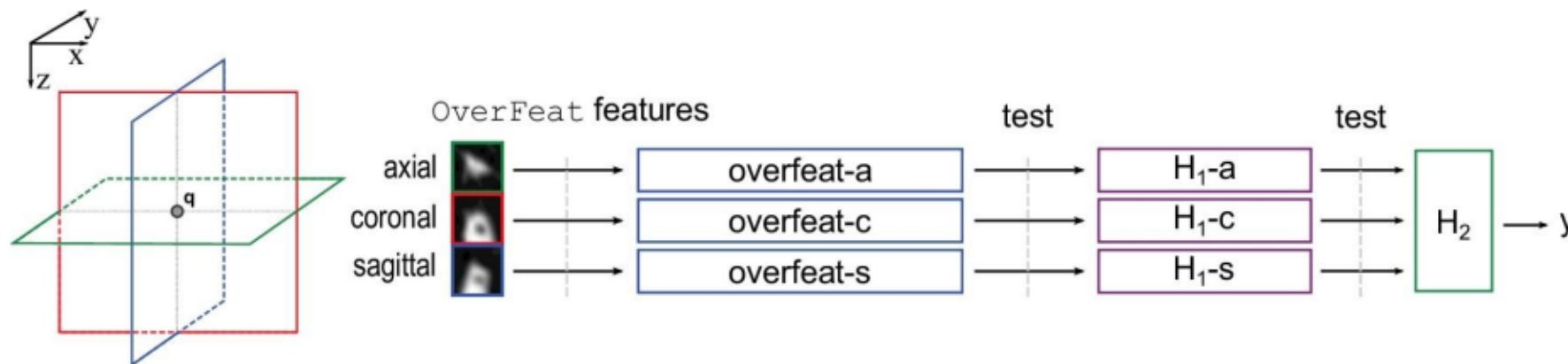
3. Advanced models for higher-dimensional (3D) data



Ciampi et al. 2015

Another approach:
3D CNNs!

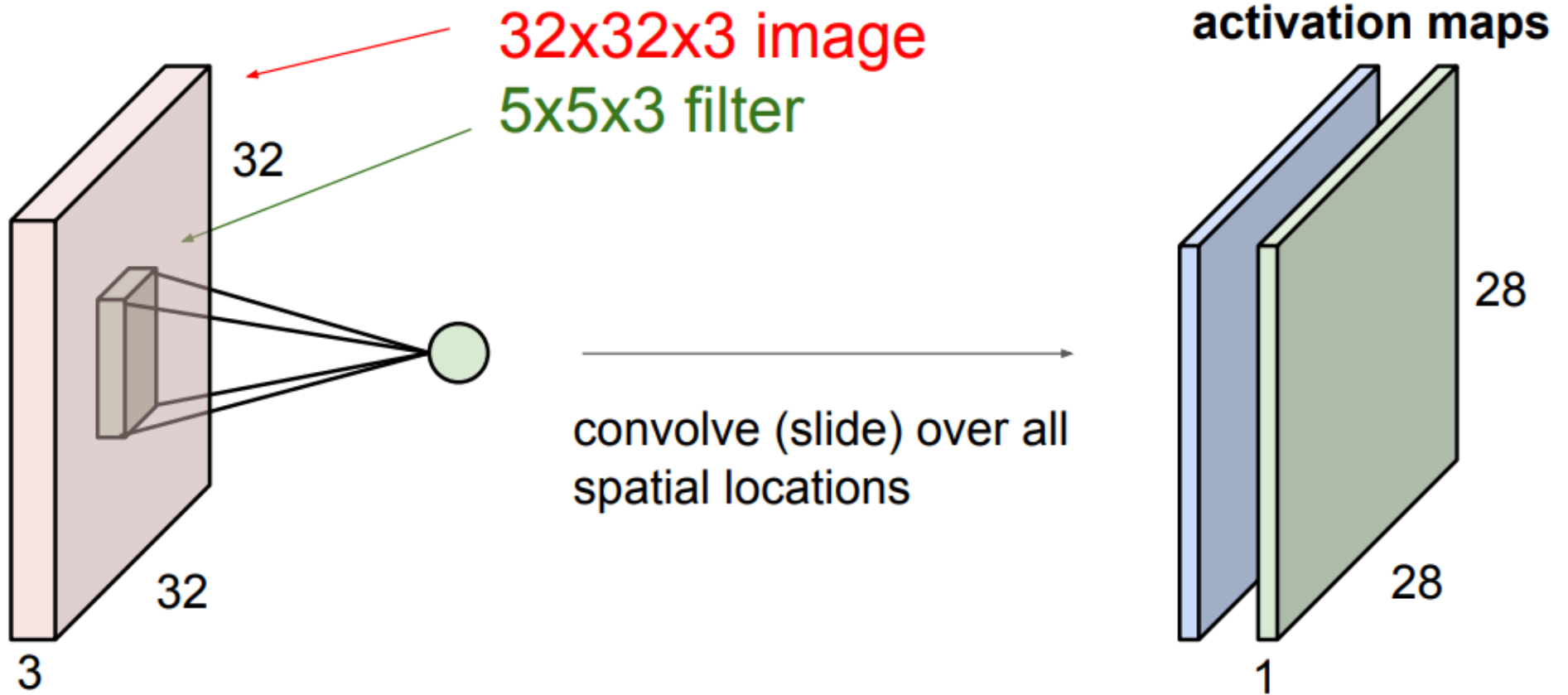
- Also extracted features from a deep learning model trained on ImageNet
 - Overfeat feature extractor (similar to AlexNet, but trained using additional losses for localization and detection)
 - To capture 3D information, extracted features from 3 different 2D views of each nodule, then input into 2-stage classifier (independent predictions on each view first, then outputs combined into second classifier).



Ciampi et al. Automatic classification of pulmonary peri-fissural nodules in computed tomography using an ensemble of 2D views and a convolutional neural network out-of-the-box. Medical Image Analysis, 2015.

3. Advanced models for higher-dimensional (3D) data

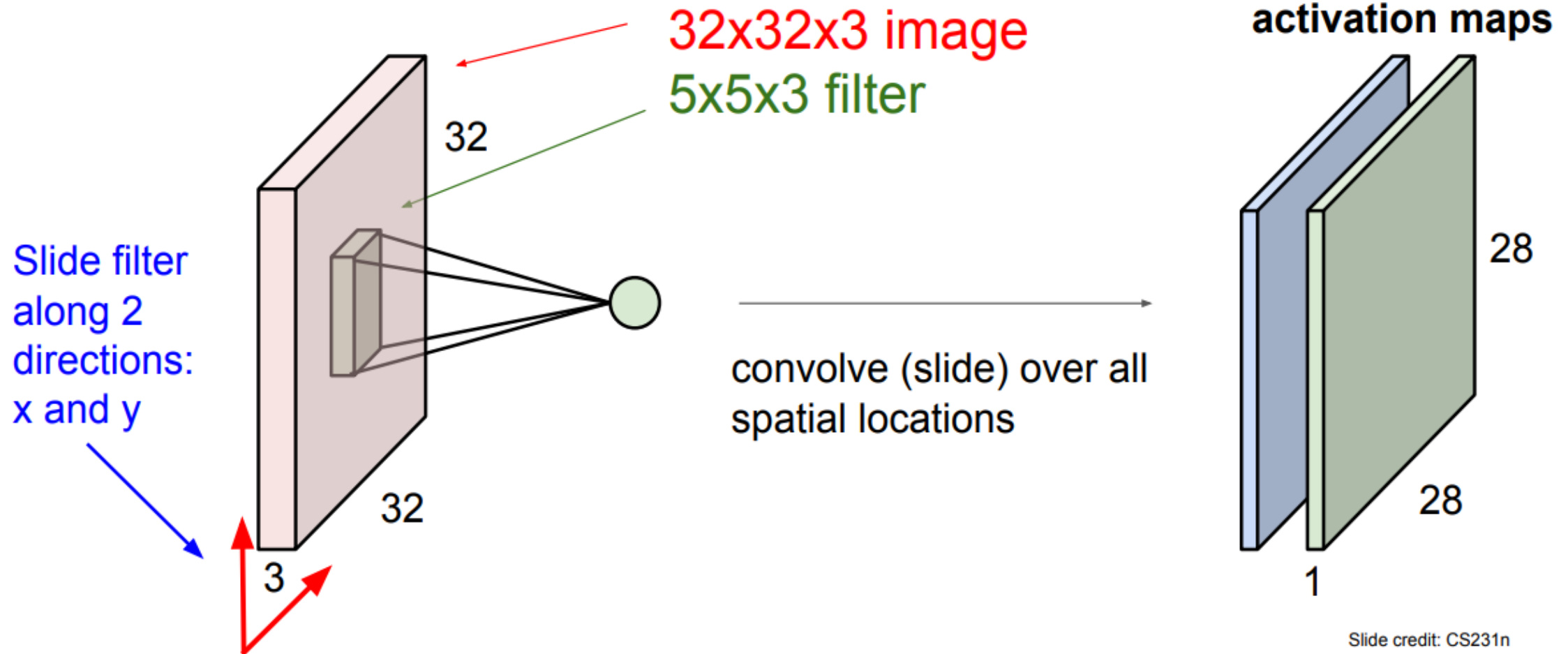
Remember 2D convolutions



Slide credit: CS231n

3. Advanced models for higher-dimensional (3D) data

Remember 2D convolutions

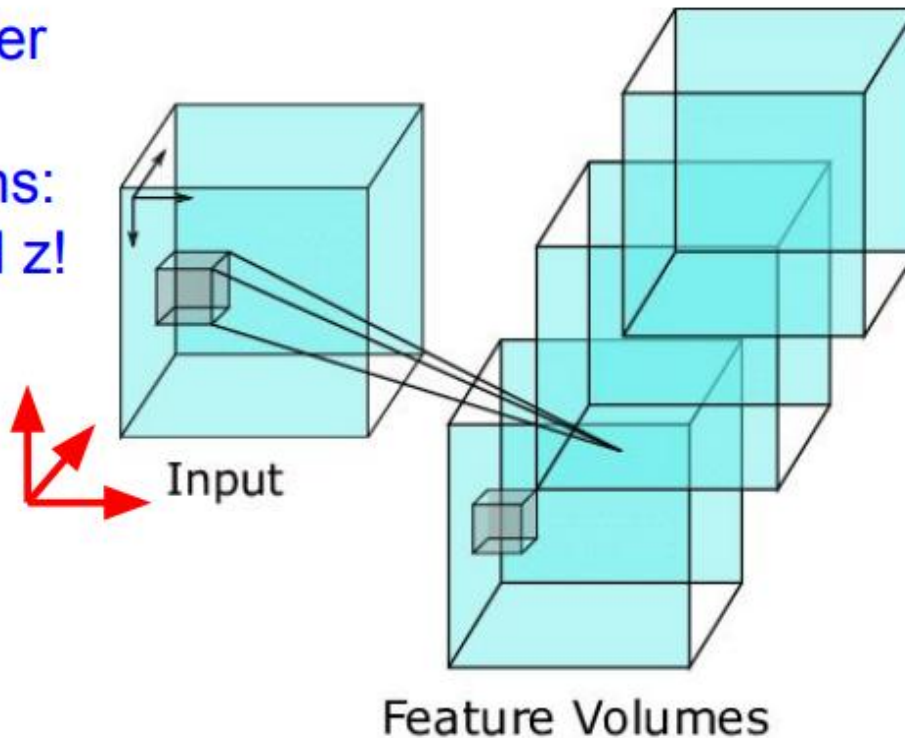


Slide credit: CS231n

3. Advanced models for higher-dimensional (3D) data

3D convolutions

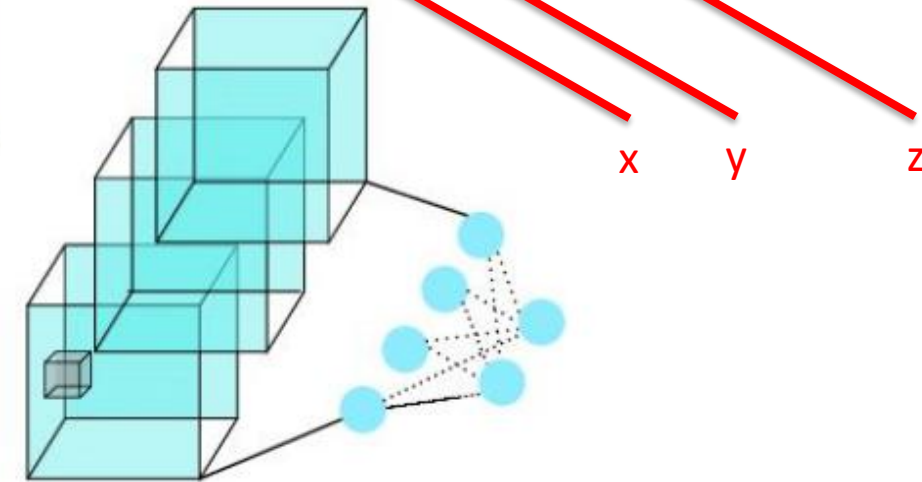
Slide filter
along 3
directions:
x, y, and z!



When might you use 3D convolutions?

Ex: 224 x 224 x 1 x 256 3D CT scan (with 256 slices)

Ex: 224 x 224 x 3 x 500 video data (with 500 temporal frames)



x,y,z are spatial and/or channel temporal dimensions.

Filter (e.g. 5 x 5 x 3 x 10 filter)
goes all the way through the
“channels” dimension as before.

Figure credit:

https://www.researchgate.net/profile/Deepak_Mishra19/publication/330912338/figure/fig1/AS:723363244810254@1549474645742/Basic-3D-CNN-architecture-the-3D-filter-is-convolved-with-the-video-in-three-dimensions.png

3. Advanced models for higher-dimensional (3D) data

Now: 3D CNNs for lung nodule classification

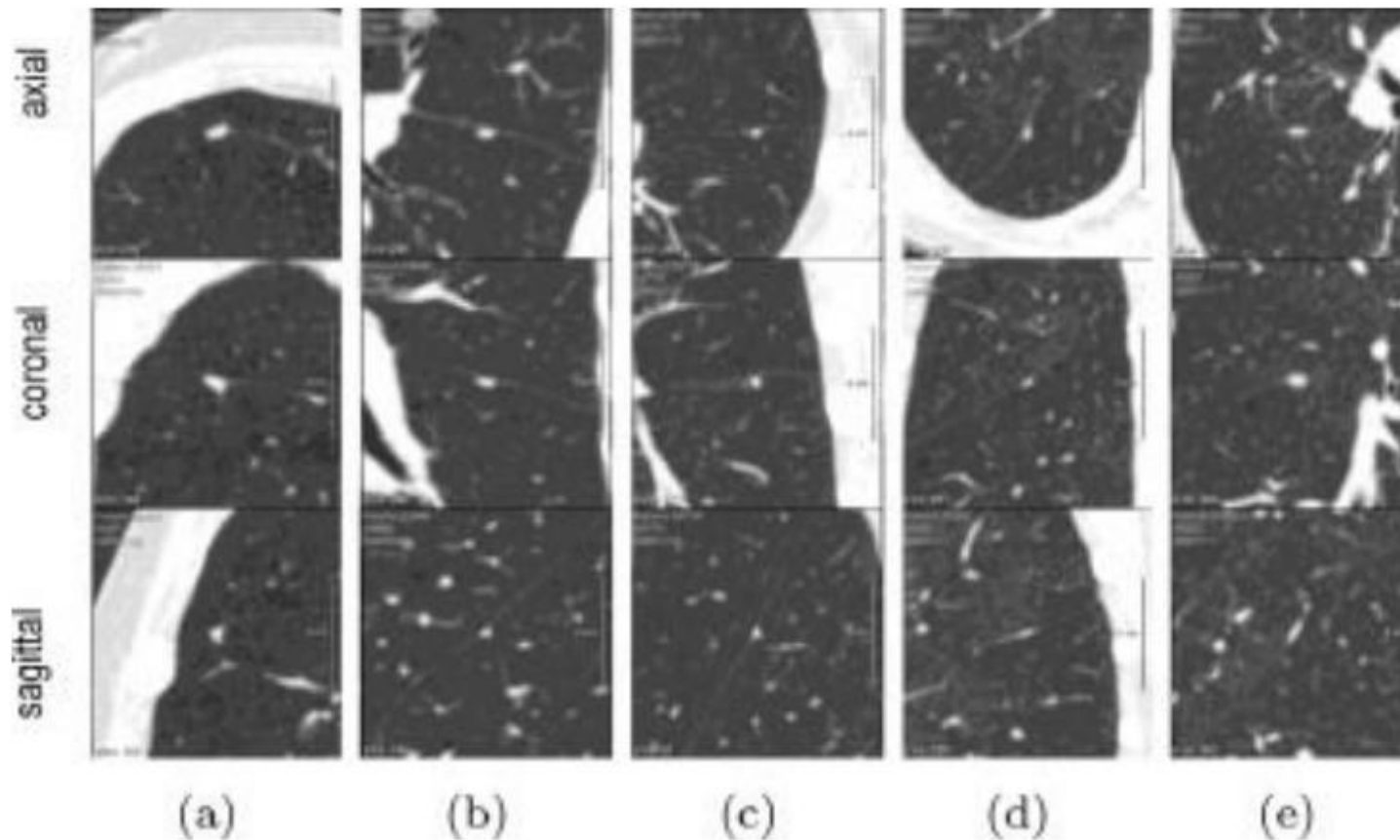
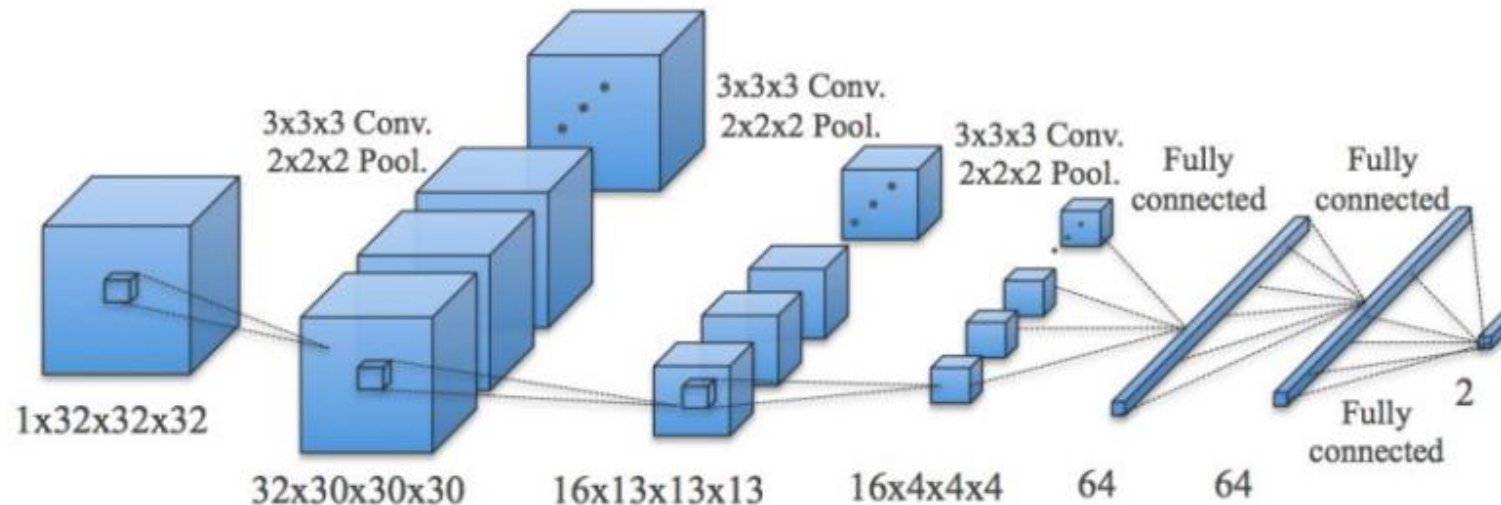


Figure credit: Ciompi et al. Automatic classification of pulmonary peri-fissural nodules in computed tomography using an ensemble of 2D views and a convolutional neural network out-of-the-box. Medical Image Analysis, 2015.

3. Advanced models for higher-dimensional (3D) data

Huang et al. 2017

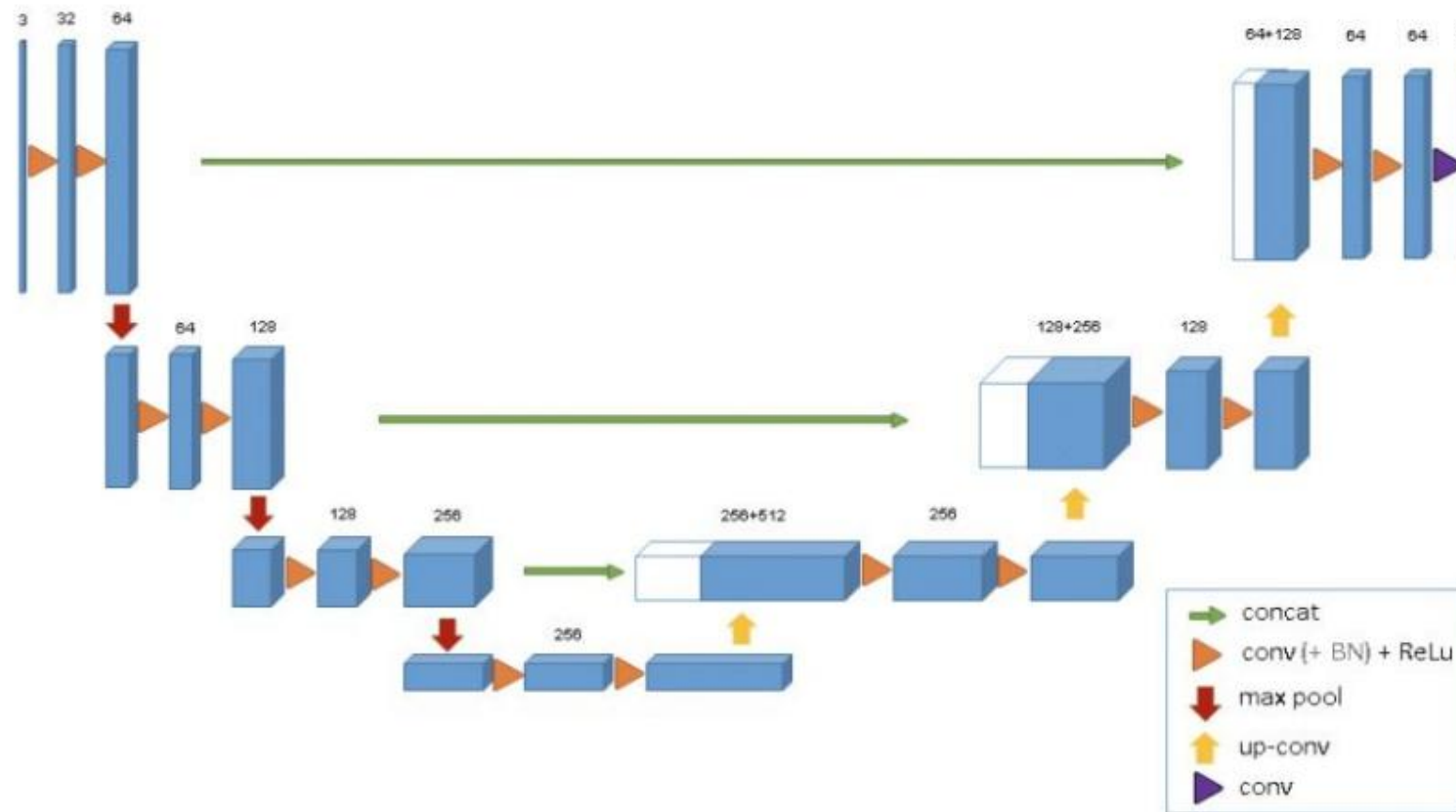
- Simple 3D CNN for lung nodule classification
- Used image processing approaches to extract candidate nodules, then 3D CNN to classify the surrounding volume
- Used the Lung Image Database Consortium (LIDC) Dataset, with 99 3D CT scans



Huang et al. Lung Nodule Detection in CT Using 3D Convolutional Neural Networks. ISBI 2017.

3. Advanced models for higher-dimensional (3D) data

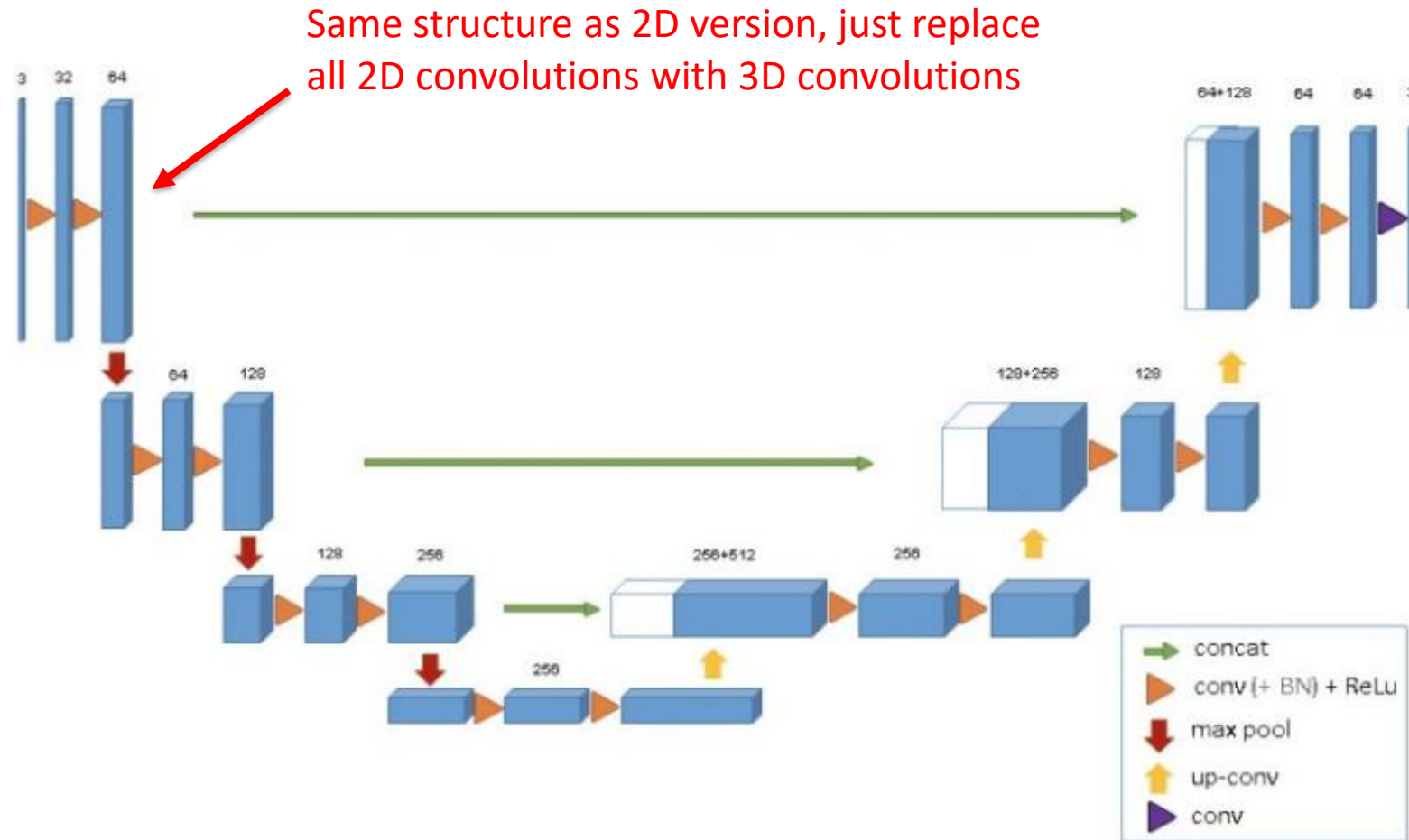
E.g. 3D U-Net



Cicek et al. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI 2016.

3. Advanced models for higher-dimensional (3D) data

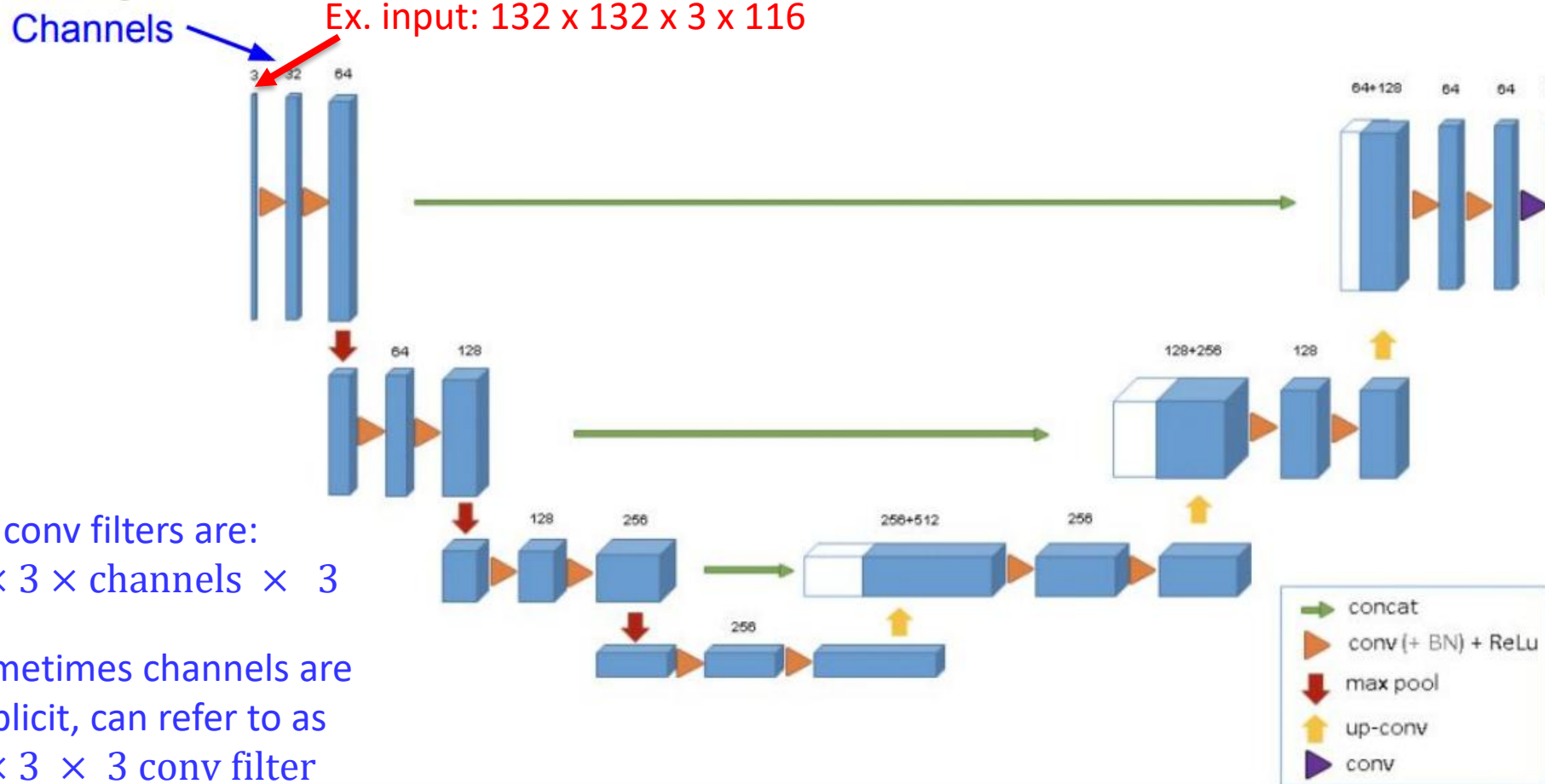
E.g. 3D U-Net



Cicek et al. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI 2016.

3. Advanced models for higher-dimensional (3D) data

E.g. 3D U-Net



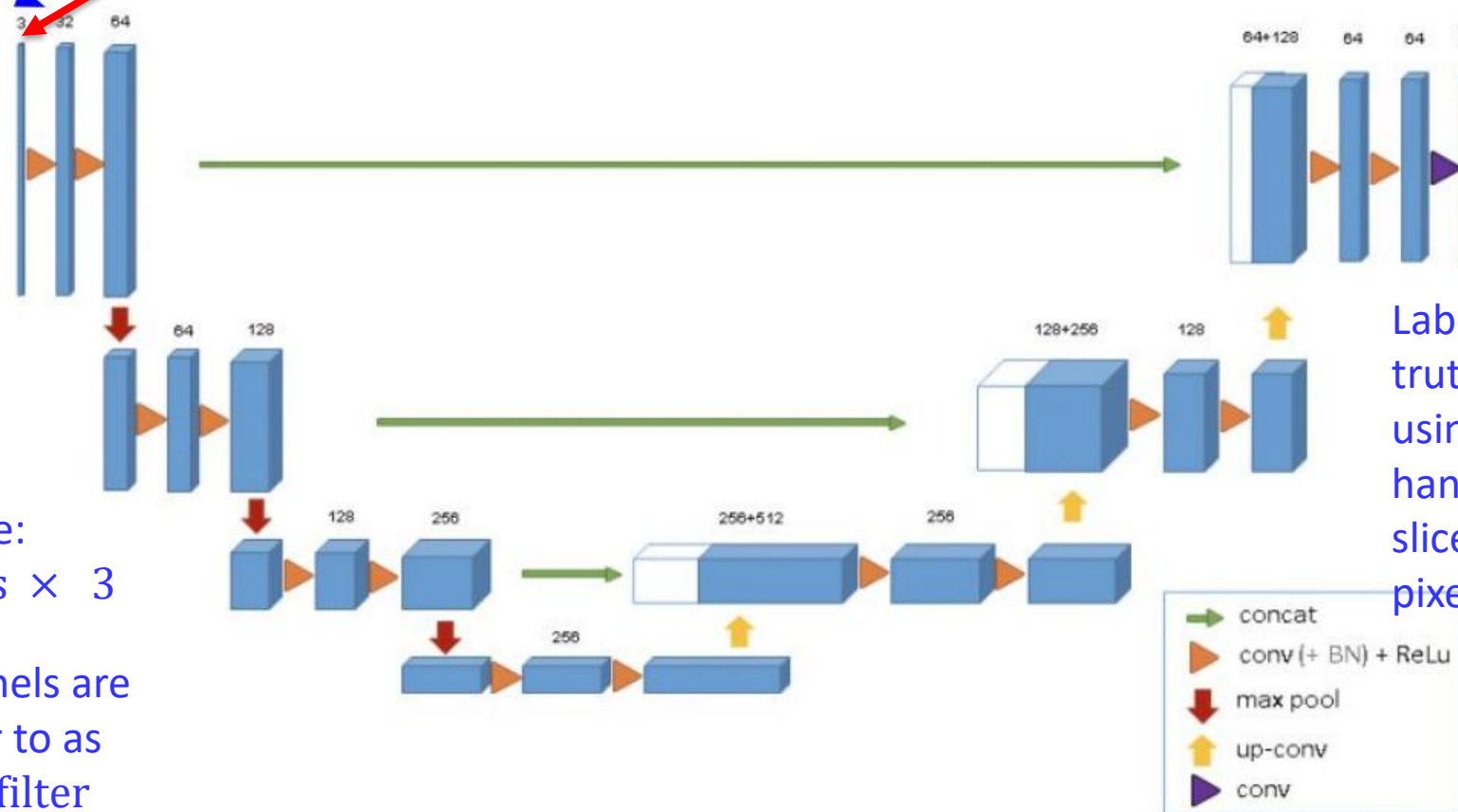
Cicek et al. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI 2016.

3. Advanced models for higher-dimensional (3D) data

E.g. 3D U-Net

Channels

Ex. input: $132 \times 132 \times 3 \times 116$



3D conv filters are:
 $3 \times 3 \times \text{channels} \times 3$

Sometimes channels are
implicit, can refer to as
 $3 \times 3 \times 3$ conv filter

Labor-intensive to provide ground truth 3D annotation. Train instead using sparse annotations: a handful of annotated xy, xz, yz 2D slices. All others are “unlabeled” pixels with no weight in the loss.

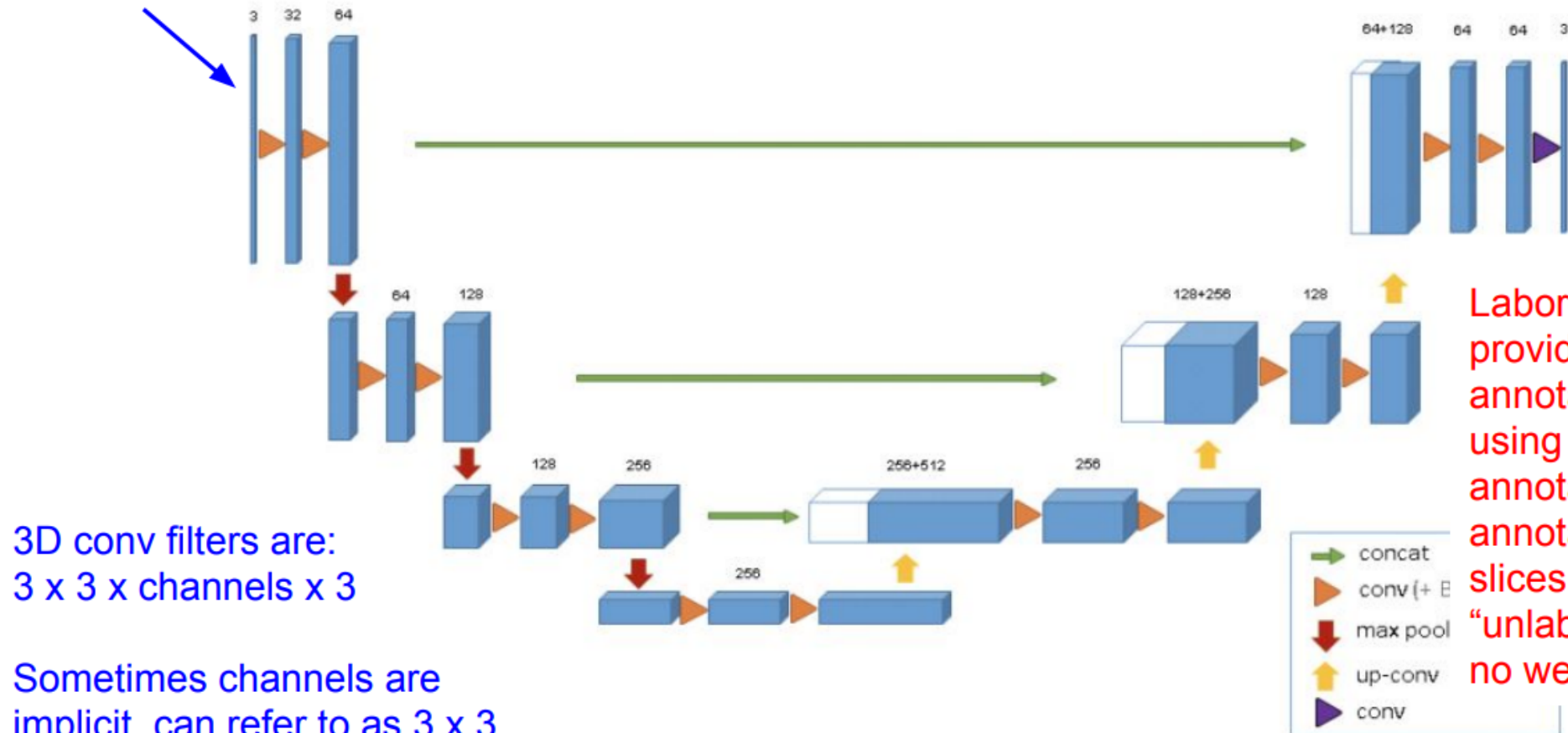
Cicek et al. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI 2016.

3. Advanced models for higher-dimensional (3D) data

E.g. 3D U-Net

Ex. input: $132 \times 132 \times 3 \times 116$

Semi-supervised learning: learning from datasets that are partially labeled (small amount of labeled data + larger amount of unlabelled data). Lots of active research on ways (e.g. loss functions which don't require manual labels) to simultaneously learn richer information from the unlabeled data.

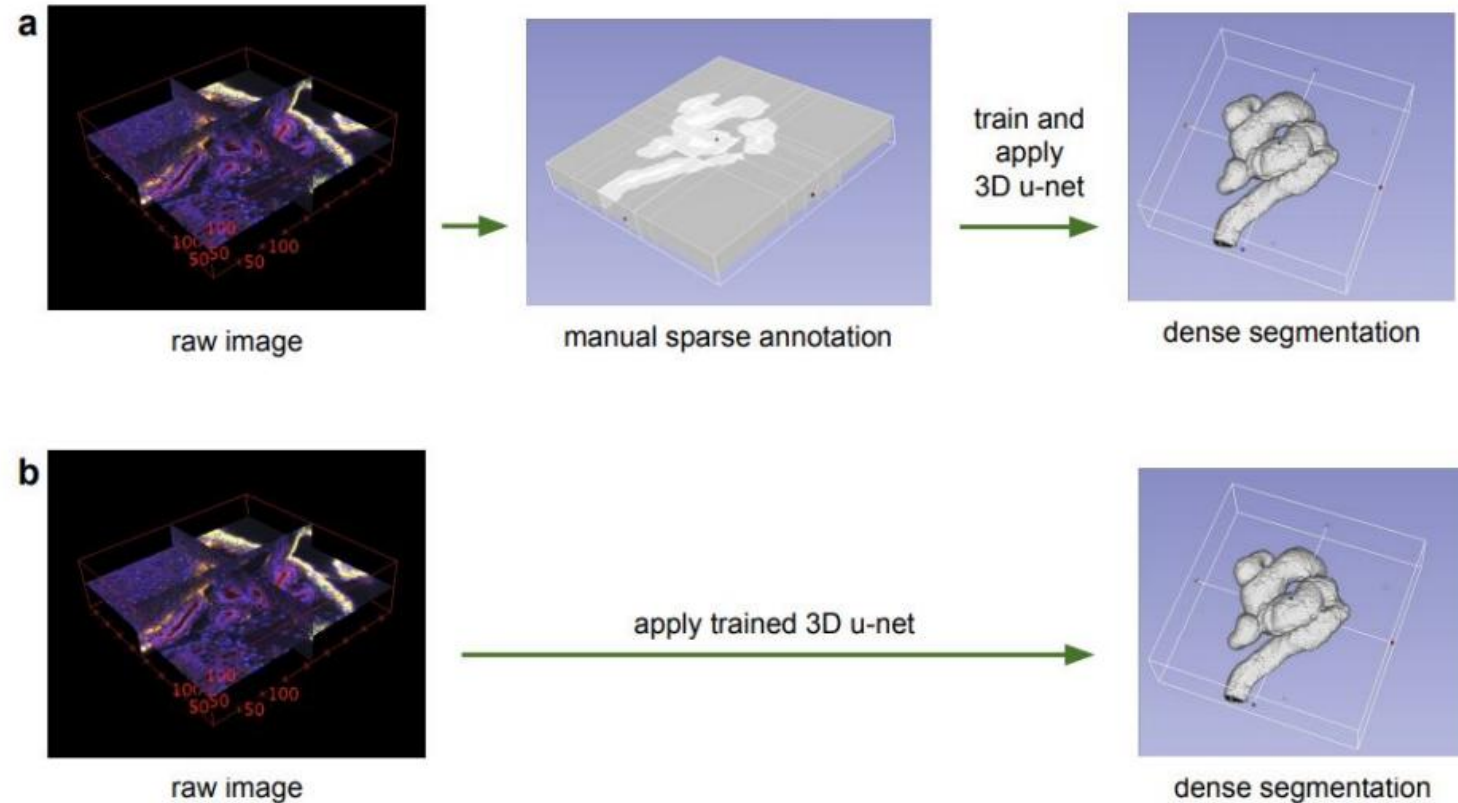


Labor-intensive to provide ground truth 3D annotation. Train instead using sparse annotations: a handful of annotated xy, xz, yz 2D slices. All others are “unlabeled” pixels with no weight in the loss.

3. Advanced models for higher-dimensional (3D) data

E.g. 3D U-Net

Ex: 3D segmentation of
Xenopus kidney in confocal
microscopic data



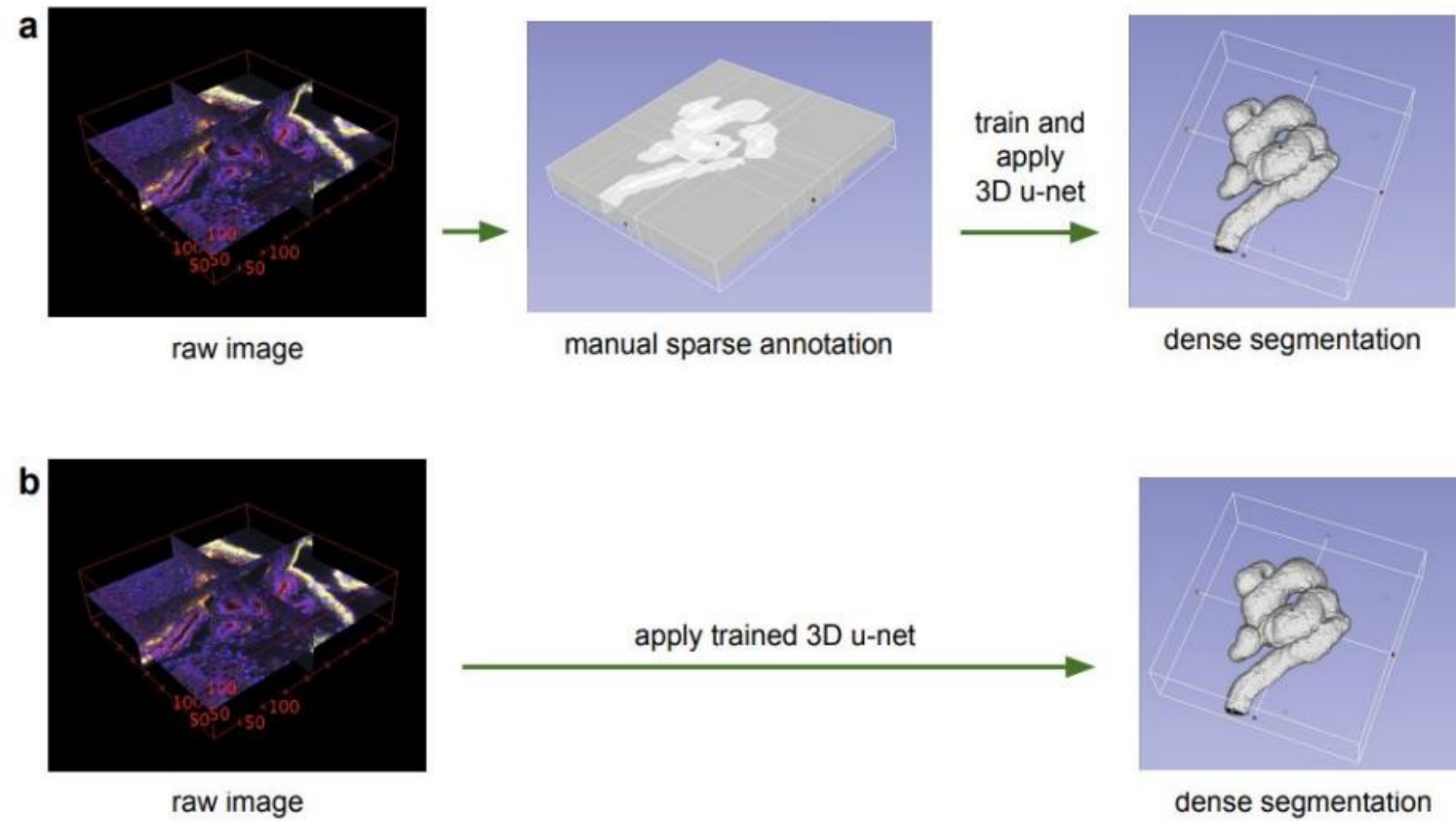
Cicek et al. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI 2016.

3. Advanced models for higher-dimensional (3D) data

E.g. 3D U-Net

Ex: 3D segmentation of
Xenopus kidney in confocal
microscopic data

Spatial dims: $\sim 250 \times 250 \times 60$.
3 channels: each channel
corresponds to a different type of
data capture



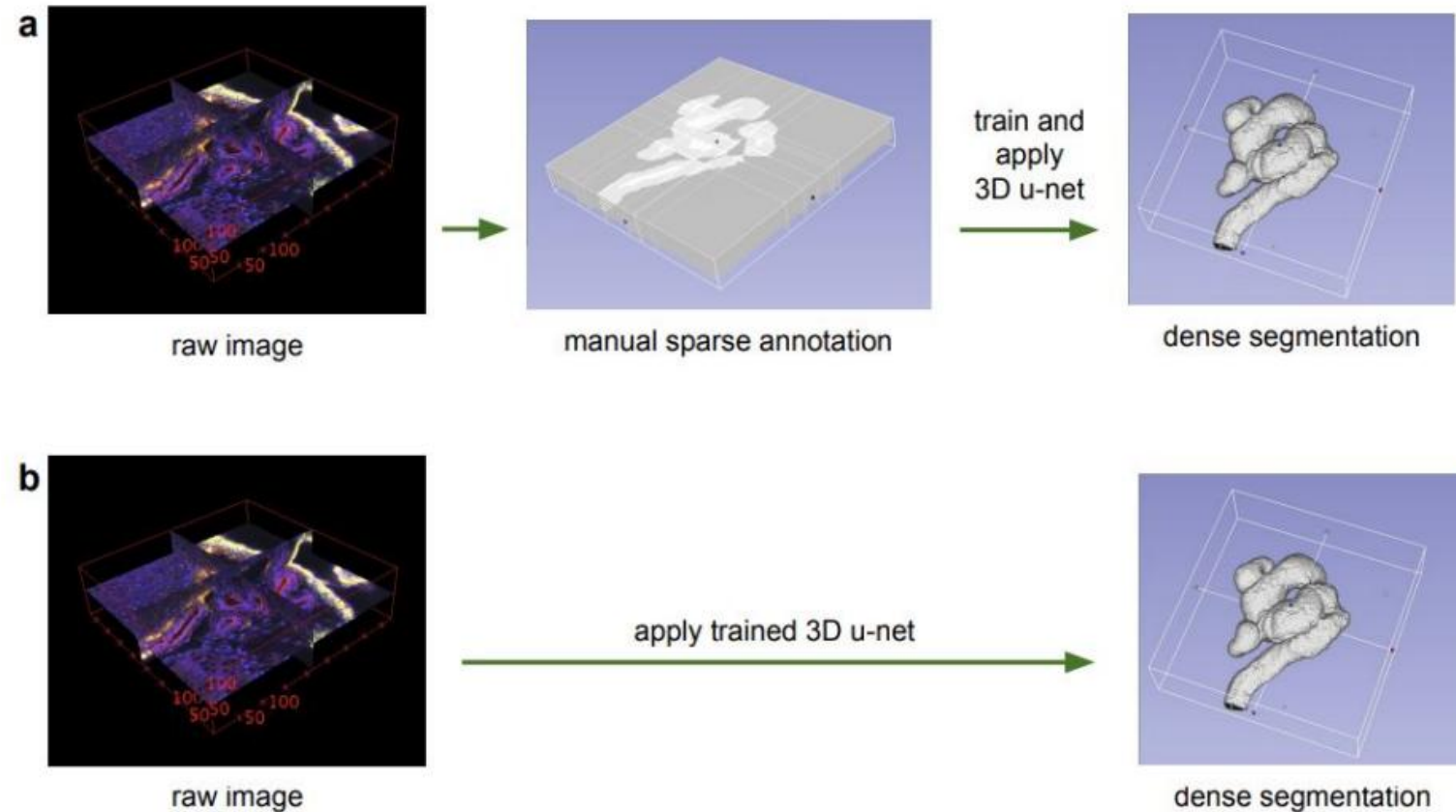
3. Advanced models for higher-dimensional (3D) data

E.g. 3D U-Net

Ex: 3D segmentation of
Xenopus kidney in confocal
microscopic data

Spatial dims: $\sim 250 \times 250 \times 60$.
3 channels: each channel
corresponds to a different type of
data capture

Used only 3 samples total! (with
total of 77 annotated 2D slices).
Leverages fact that each sample
contains many instances of same
repetitive structures w/ variation.



Cicek et al. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI 2016.

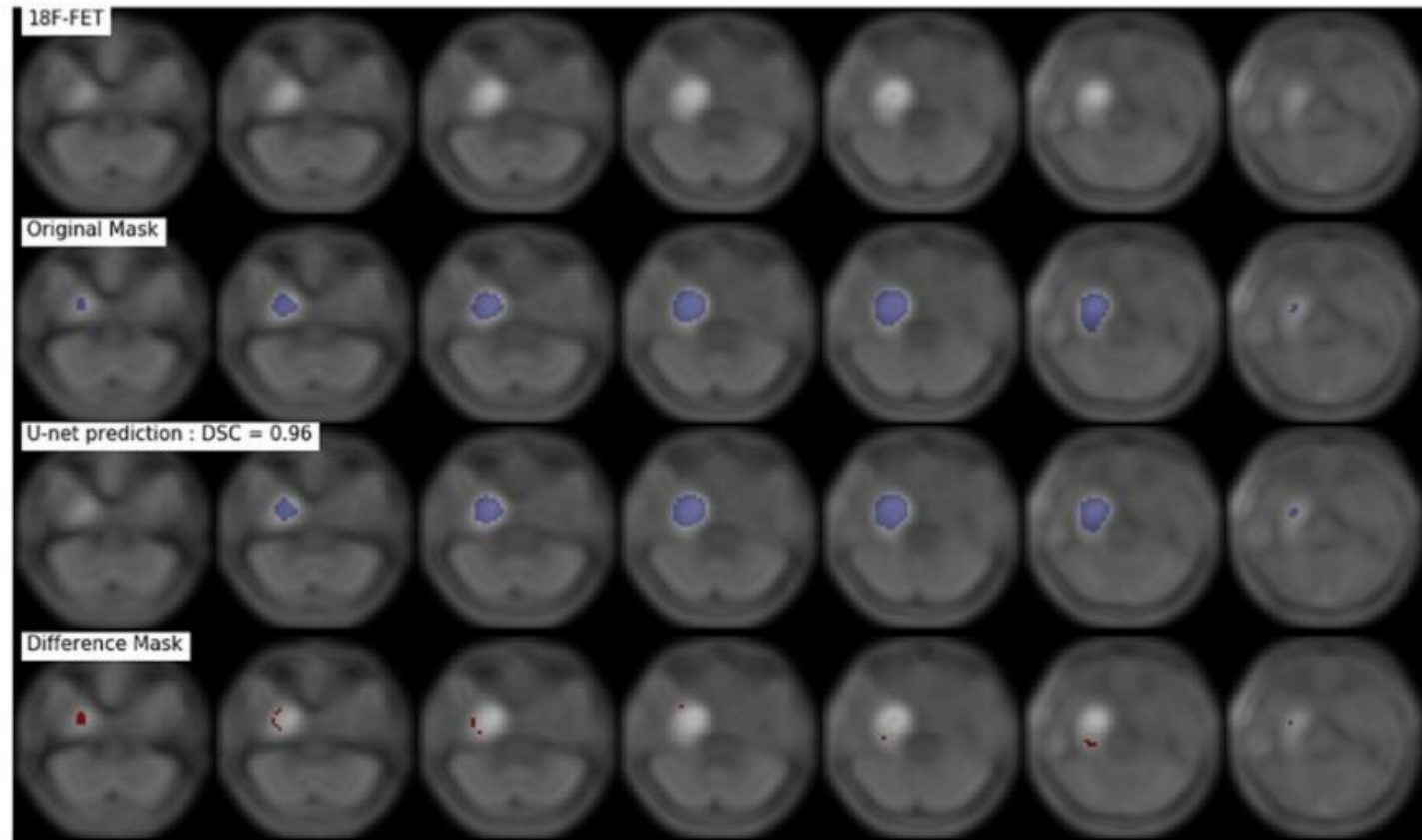
3. Advanced models for higher-dimensional (3D) data

Ex: Brain lesion segmentation

Training set: 37 PET scans
(3D volumes)

Evaluation set: 11 PET scans

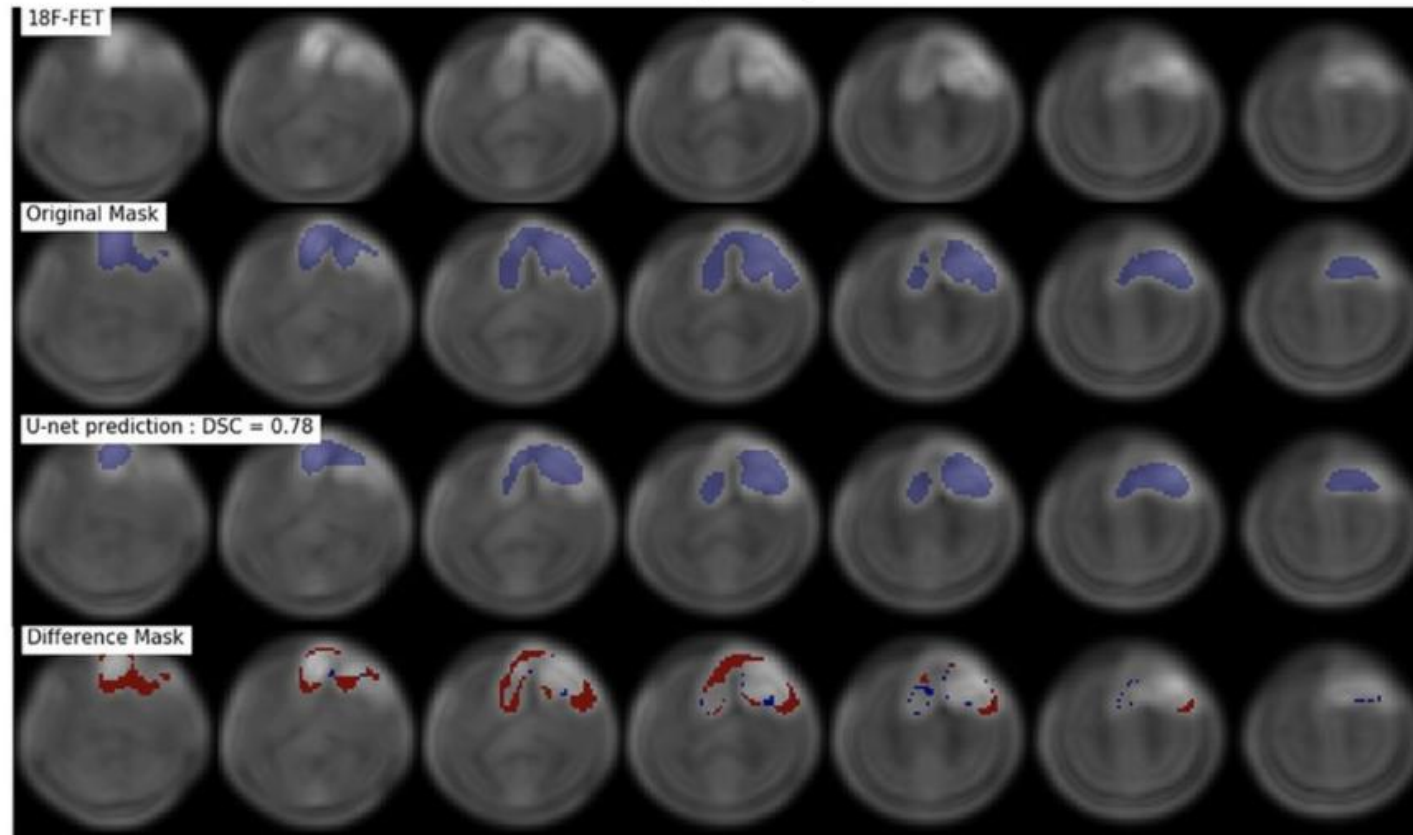
Volumes resized to 64x64x40
for computational efficiency



Blanc-Durand et al. Automatic lesion detection and segmentation of 18F-FET PET in gliomas: A full 3D U-Net convolutional neural network study. PLoS One, 2018.

3. Advanced models for higher-dimensional (3D) data

Ex: Brain lesion segmentation



Blanc-Durand et al. Automatic lesion detection and segmentation of 18F-FET PET in gliomas: A full 3D U-Net convolutional neural network study. PLoS One, 2018.



- Review what we have learned last time
- Basic vision models: segmentation
- Advanced models for higher-dimensional (3D) data
- Case studies of CNNs for medical image segmentation
- Advanced models for higher-dimensional (video) data

4. Case studies of CNNs for medical image segmentation



Automatic Detection of Cerebral Microbleeds From MR Images via 3D Convolutional Neural Networks

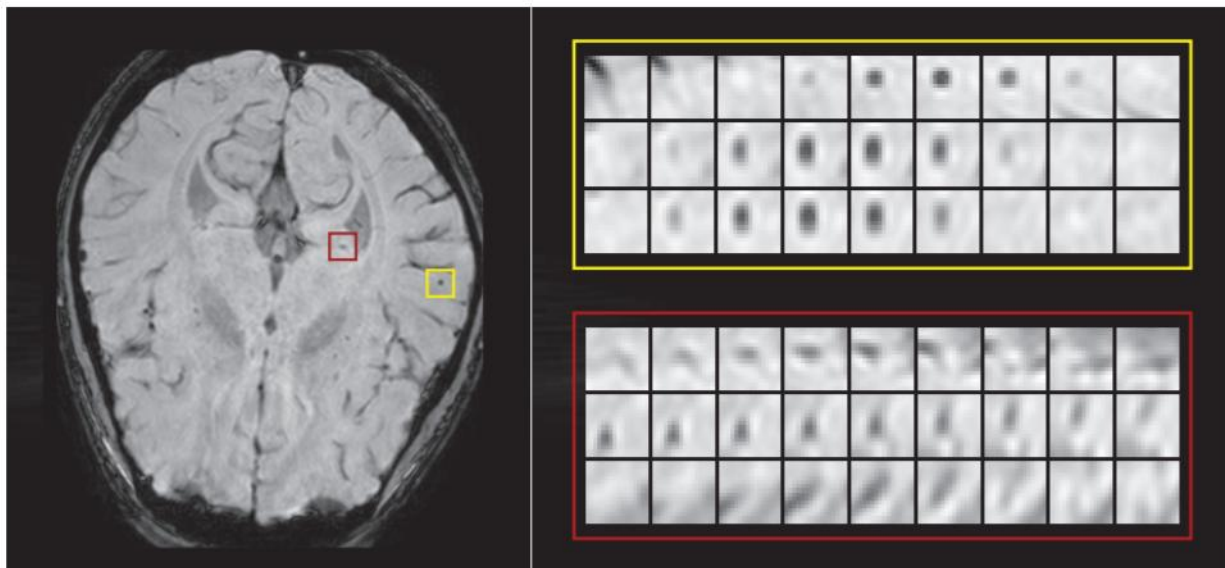


Fig. 1. Illustration of a CMB and a CMB mimic denoted with yellow and red rectangles, respectively. In each of the big rectangle, from top to down, the rows demonstrate adjacent slices in axial, sagittal and coronal planes. Best viewed in color.

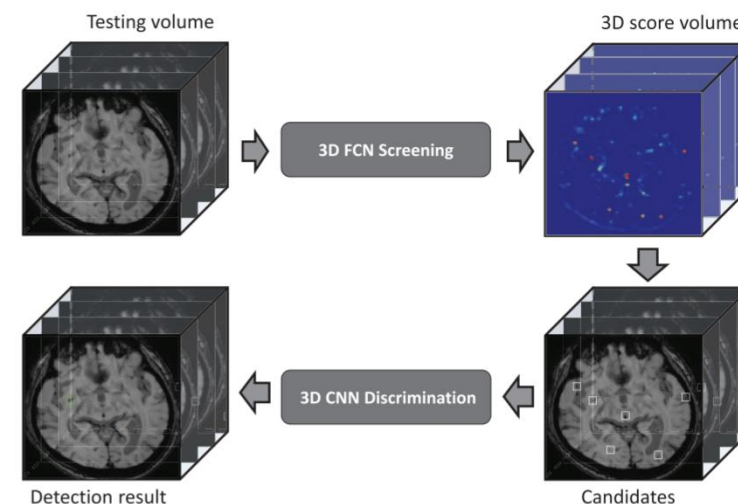


Fig. 2. An overview of the proposed cascaded framework for CMB detection.

4. Case studies of CNNs for medical image segmentation

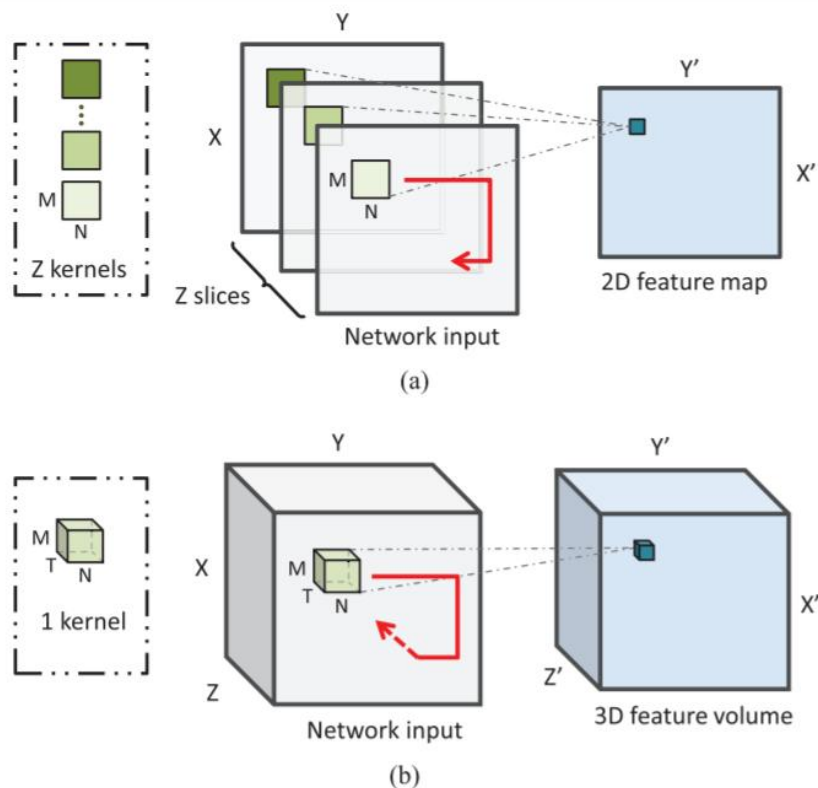


Fig. 3. Comparison of using 2D and 3D convolution kernels given volumetric image with size of $X \times Y \times Z$ in terms of network input, kernel behavior and generated feature map. Red lines represent the moving direction of kernels, i.e., sweeping over the 2D and 3D topologies, respectively. (a) With the 2D convolution (kernel size of $M \times N$), the volume is first split into Z isolated slices along the third direction and these slices are input to the network. Each generated feature map is a 2D patch. (b) With the 3D convolution (kernel size of $M \times N \times T$), the entire volume is input to the network. Each generated feature map is a 3D volume. (Note that kernel sizes M , N and T need not to be equal. Best viewed in color.)

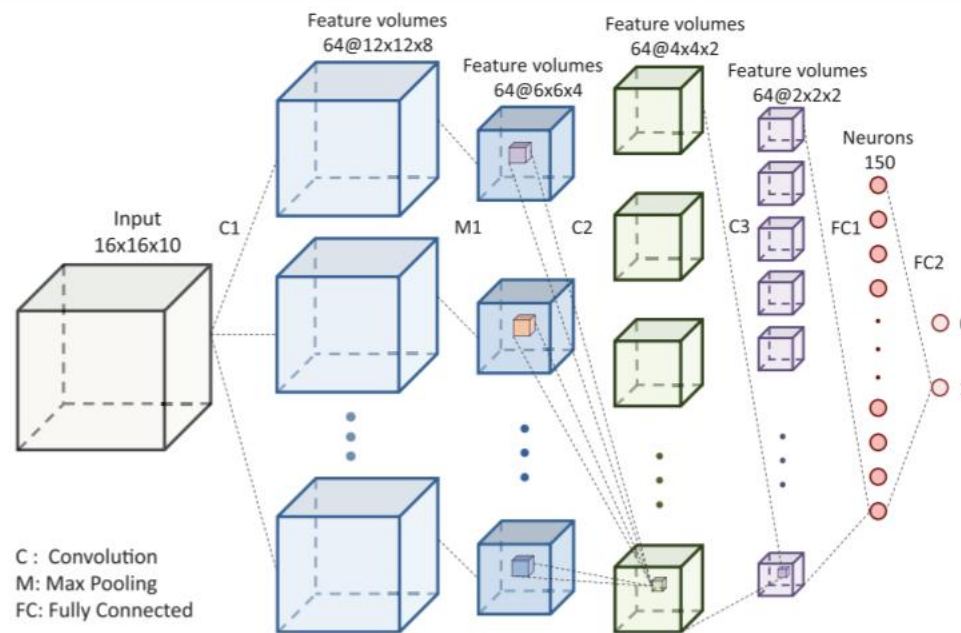


Fig. 4. The hierarchical architecture of the 3D CNN model.

4. Case studies of CNNs for medical image segmentation



TABLE I
THE ARCHITECTURE OF 3D FCN SCREENING MODEL

Layer	Kernel size	Stride	Output size	Feature volumes
Input	-	-	$16 \times 16 \times 10$	1
C1	$5 \times 5 \times 3$	1	$12 \times 12 \times 8$	64
M1	$2 \times 2 \times 2$	2	$6 \times 6 \times 4$	64
C2	$3 \times 3 \times 3$	1	$4 \times 4 \times 2$	64
C3	$3 \times 3 \times 1$	1	$2 \times 2 \times 2$	64
FC1	$2 \times 2 \times 2$	1	$1 \times 1 \times 1$	150
FC2	$1 \times 1 \times 1$	1	$1 \times 1 \times 1$	2

Note: FC layers are converted into the convolutional fashion.

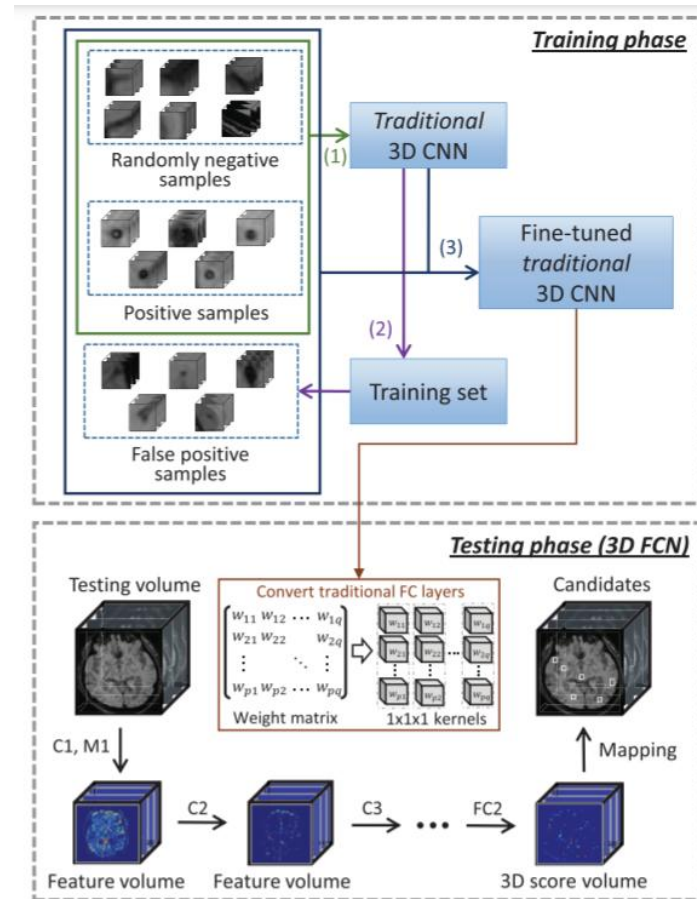
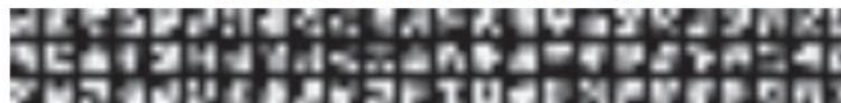


Fig. 6. Illustration of the workflow of the screening stage. The training phase is conducted in three sub-steps: (1) train an initial *traditional* 3D CNN with positive samples and randomly selected negative samples; (2) apply the initial model on training set and obtain false positive samples to enlarge the training database; (3) fine-tune the initial *traditional* 3D CNN model with the enlarged database to strengthen its discrimination capability. Once training is done, the traditional FC layers are converted into the convolutional fashion (as shown in the brown box). During the testing phase, the 3D FCN takes a whole volume as input, extracts representative feature volumes and finally produces a 3D score volume to retrieve candidates.

4. Case studies of CNNs for medical image segmentation



(a) Visualization of 3D kernels in the 1st C layer



(b) Visualization of 3D kernels in the 2nd C layer

Fig. 7. Visualization of typical learned filters in the screening 3D FCN model: (a) visualization of the C1 layer kernels, where each column represents a 3D kernel of size $5 \times 5 \times 3$, which is visualized as three 5×5 maps; (b) visualization of the C2 layer kernels, where each column represents a 3D kernel of size $3 \times 3 \times 3$, which is visualized as three 3×3 maps.

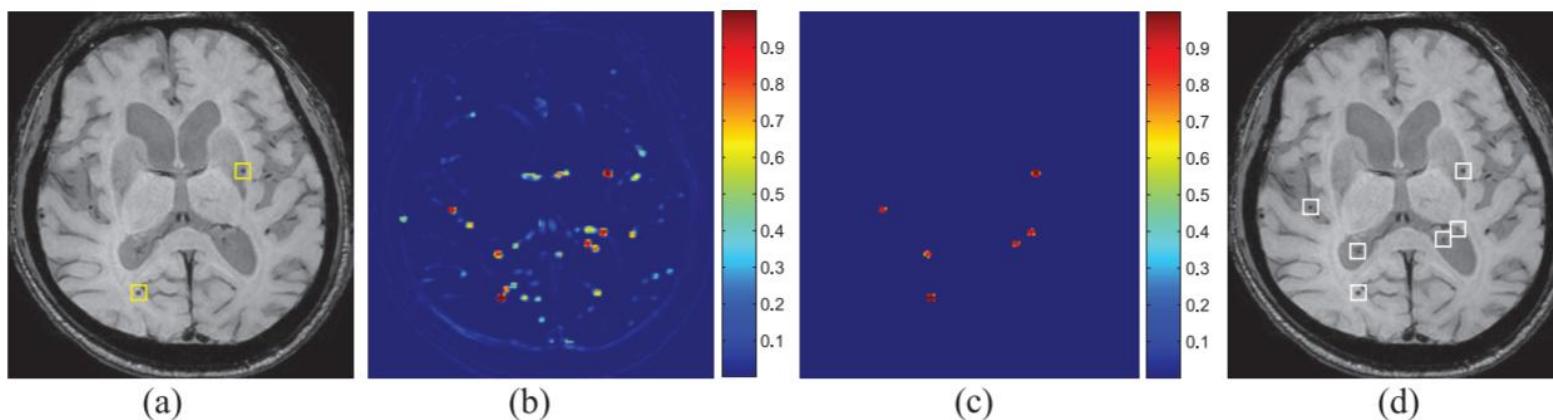


Fig. 8. Typical results of the 3D FCN screening model with score volume projection onto the axial plane. (a) Raw data with true CMBs (yellow rectangles). (b) 2D projection of the score volume generated with FCN. (c) 2D projection of the post-processed score volume. (d) Retrieved candidates (white rectangles). Best viewed in color.

4. Case studies of CNNs for medical image segmentation



TABLE VI
EVALUATION OF DETECTION RESULTS

Methods	Sensitivity	Precision	FP _{avg}
Barnes et al. [15]	64.96%	5.13%	28.10
Random forest [40]	85.47%	17.24%	9.60
2D-CNN-SVM [18]	88.03%	22.69%	7.02
Ours (single)	92.31%	42.69%	2.90
Ours (average)	93.16%	44.31%	2.74

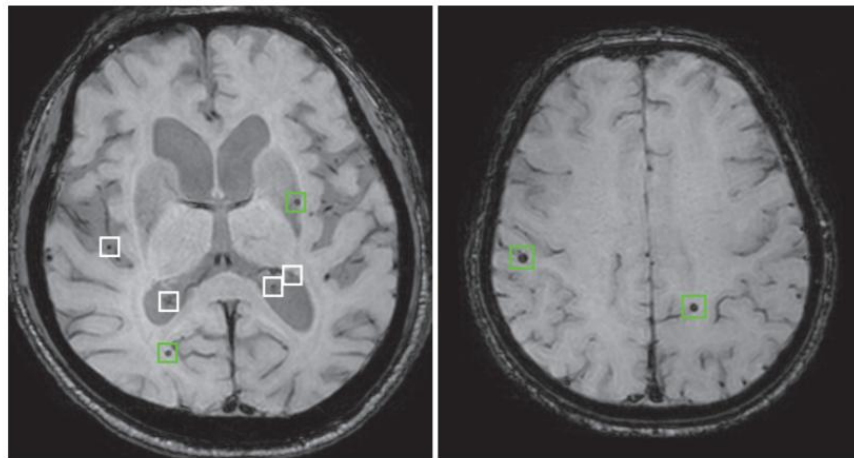


Fig. 12. Examples of CMB detection results (viewed in axial planes). Green rectangles denote the correctly detected CMBs and white rectangles denote the removed false positive candidates by our method.

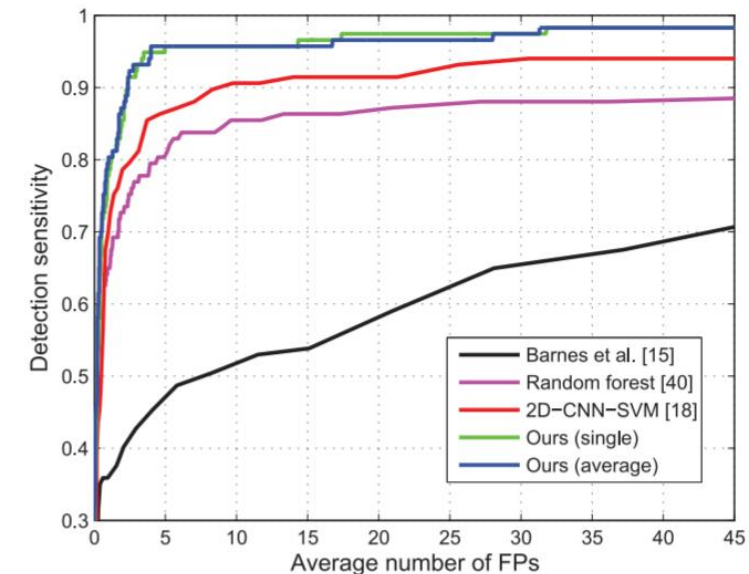


Fig. 11. Comparison of FROC curves of different methods. The top two lines are results produced by our 3D CNN based cascaded frameworks.

4. Case studies of CNNs for medical image segmentation



3D Deeply Supervised Network for Automatic Liver Segmentation from CT Volumes

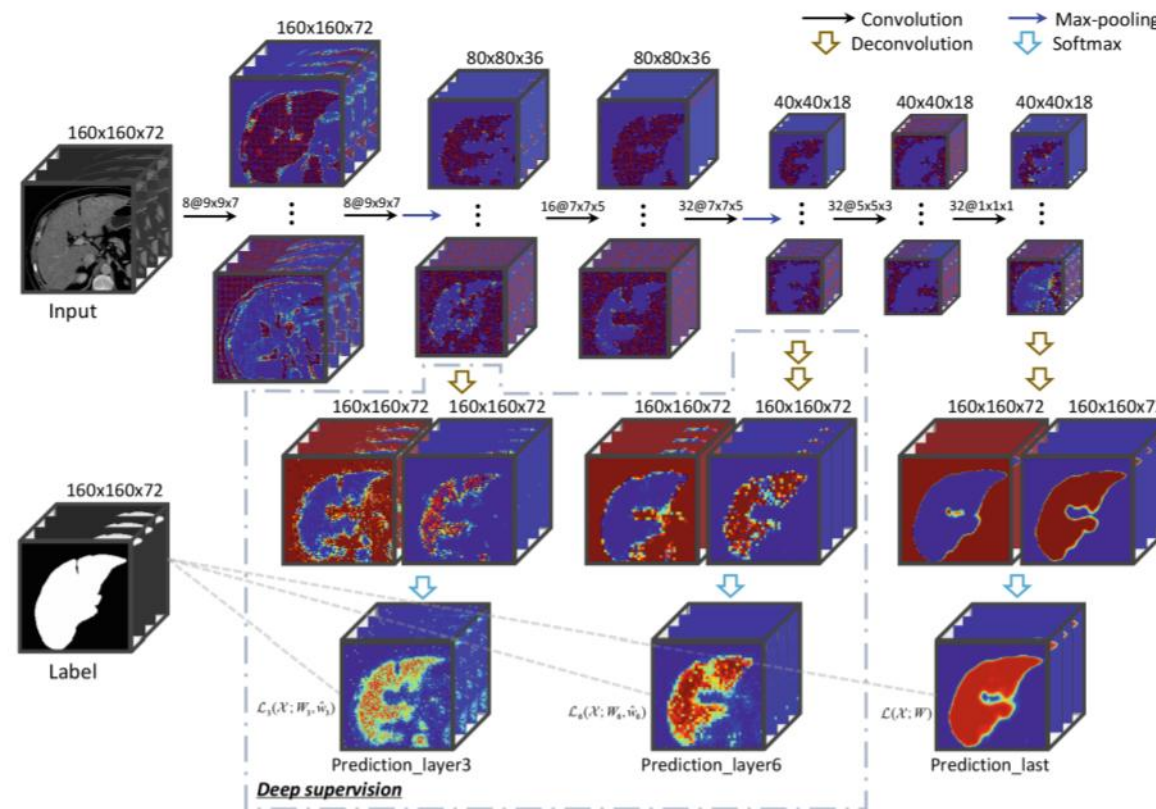


Fig. 1. Architecture of the proposed 3D DSN, with intermediate feature volumes, deep supervision layer predictions and last layer predictions visualized in colormap. The sizes of input and feature volumes, and the numbers and sizes of 3D kernels are indicated.

4. Case studies of CNNs for medical image segmentation

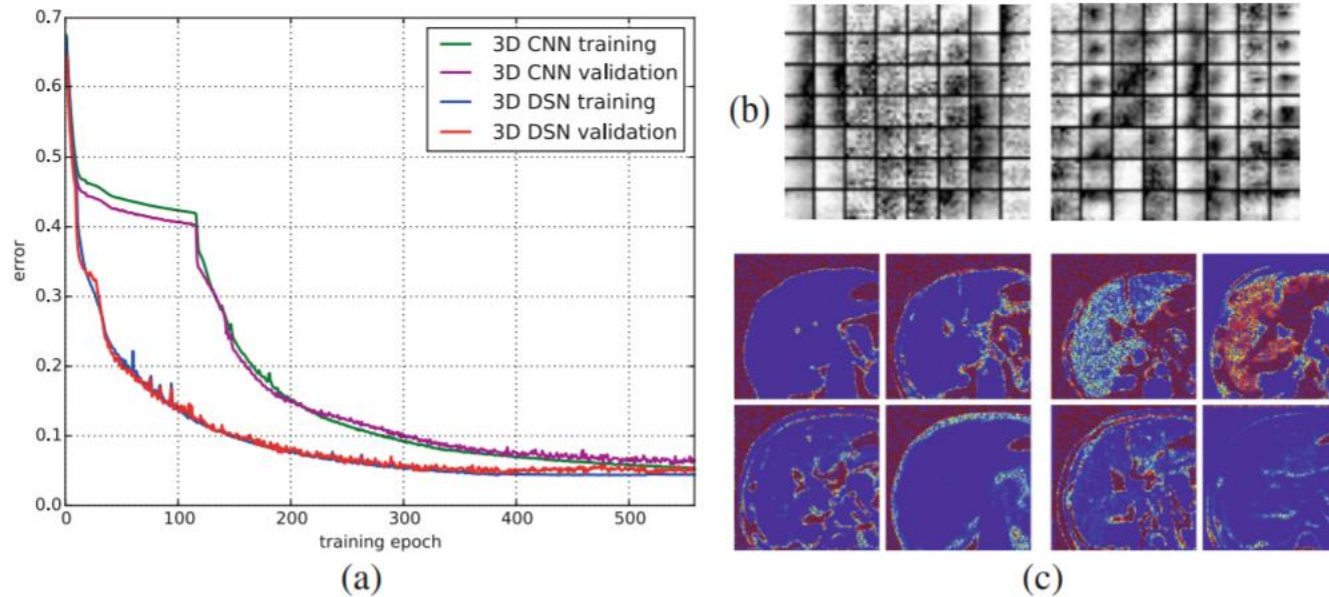


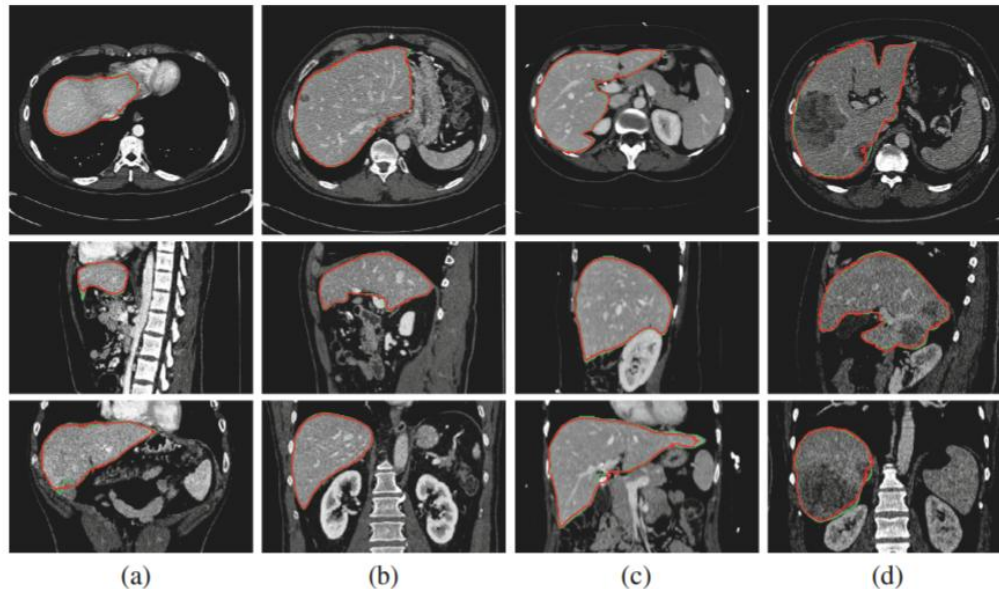
Fig. 2. (a) Comparison of the learning curves of 3D CNN and 3D DSN. (b) Visualization of the learned 3D kernels in the 1st layer of 3D CNN (left) and 3D DSN (right), each column presents a single kernel of size $9 \times 9 \times 7$ expanded along the third dimension as seven 9×9 maps. (c) Visualization of typical features in the 1st layer of 3D CNN (left) and 3D DSN (right).

4. Case studies of CNNs for medical image segmentation



Table 1. Quantitative evaluations of our methods on the training set.

Dataset	Methods	VOE	VD	AvgD	RMSD	MaxD
Training Set	3D-CNN	7.68	1.98	1.56	4.09	45.99
	3D-DSN	6.27	1.46	1.32	3.38	36.49
	3D-CNN+CRF	5.64	1.72	0.89	1.73	34.42
	3D-DSN+CRF	5.37	1.32	0.67	1.48	29.63



3D U-net, 3D V-net better?

4. Case studies of CNNs for medical image segmentation

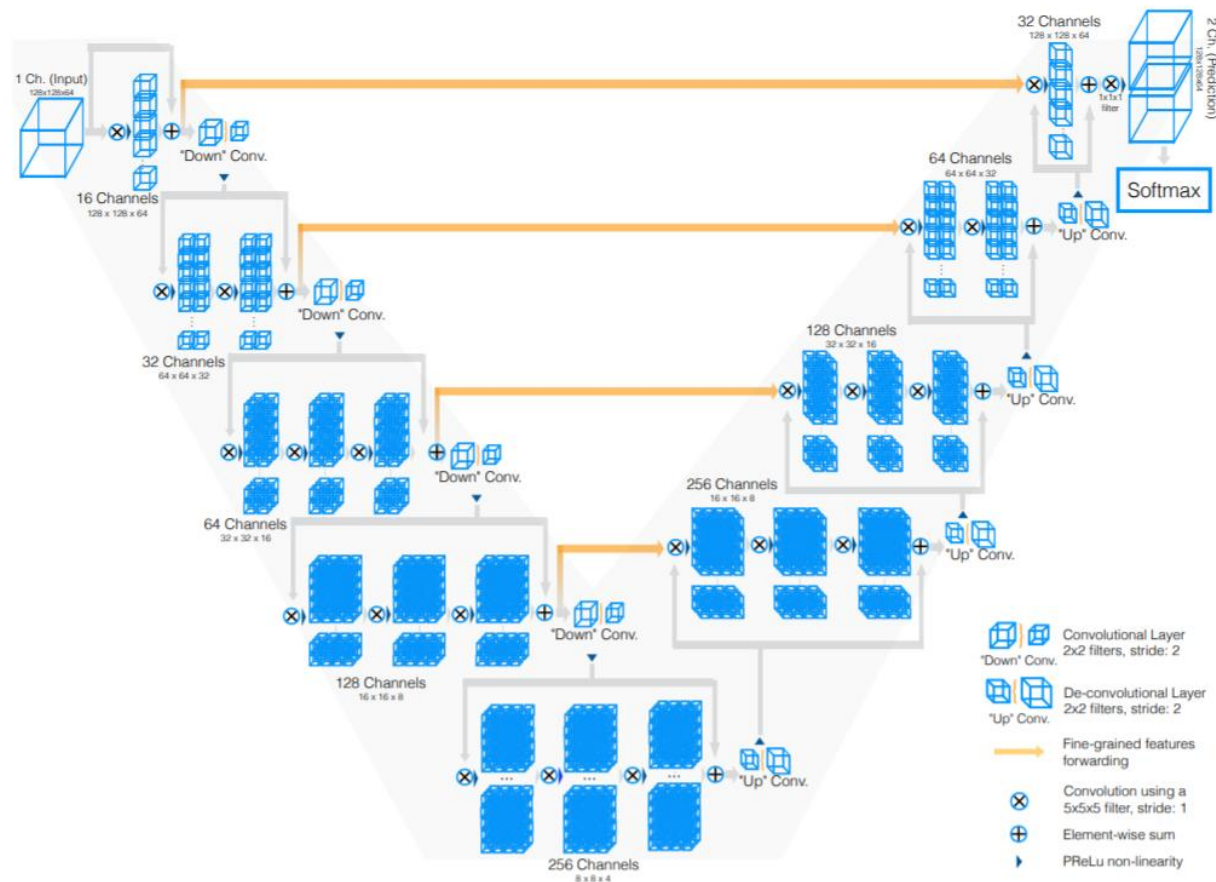


Fig. 2. Schematic representation of our network architecture. Our custom implementation of Caffe [5] processes 3D data by performing volumetric convolutions. Best viewed in electronic format.

4. Case studies of CNNs for medical image segmentation

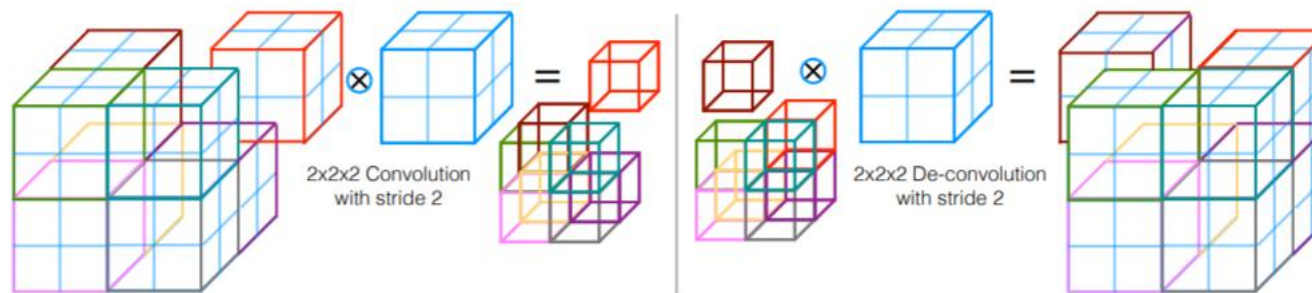


Fig. 3. Convolutions with appropriate stride can be used to reduce the size of the data. Conversely, de-convolutions increase the data size by projecting each input voxel to a bigger region through the kernel.

Table 1. Theoretical receptive field of the $3 \times 3 \times 3$ convolutional layers of the network.

Layer	Input Size	Receptive Field	Layer	Input Size	Receptive Field
L-Stage 1	128	$5 \times 5 \times 5$	R-Stage 4	16	$476 \times 476 \times 476$
L-Stage 2	64	$22 \times 22 \times 22$	R-Stage 3	32	$528 \times 528 \times 528$
L-Stage 3	32	$72 \times 72 \times 72$	R-Stage 2	64	$546 \times 546 \times 546$
L-Stage 4	16	$172 \times 172 \times 172$	R-Stage 1	128	$551 \times 551 \times 551$
L-Stage 5	8	$372 \times 372 \times 372$	Output	128	$551 \times 551 \times 551$

4. Case studies of CNNs for medical image segmentation

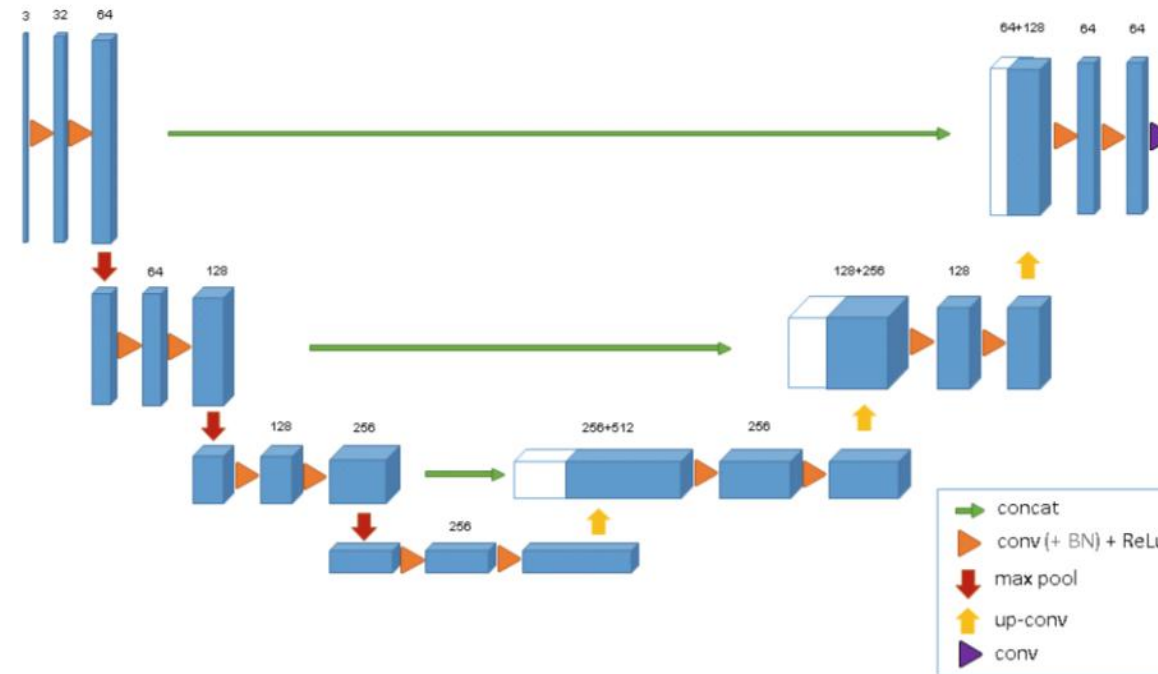


Fig. 2. The 3D u-net architecture. Blue boxes represent feature maps. The number of channels is denoted above each feature map.

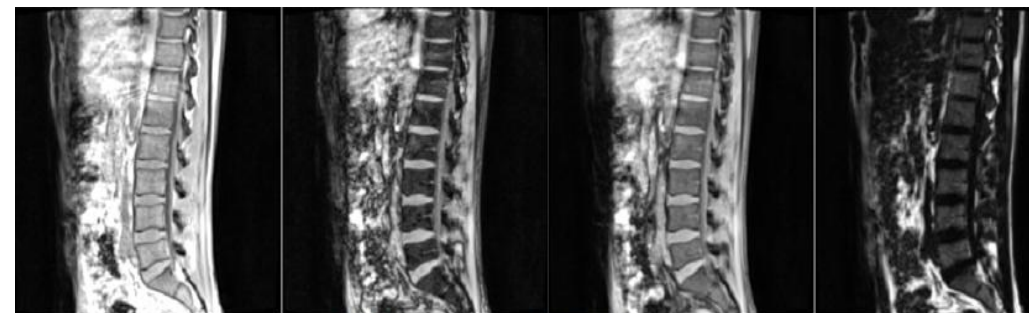
4. Case studies of CNNs for medical image segmentation



Intervertebral disc segmentation from multi-modality MRI

What is intervertebral disc (IVD) ?

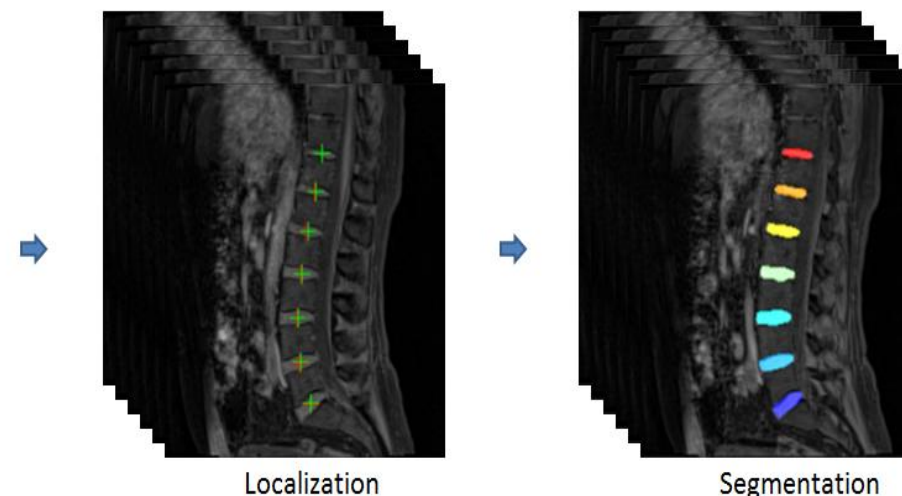
- small spine components
- lie between each pair of adjacent vertebrae
- disc degeneration can occur in all people



Multi-modality MR (M^3) Images

Characteristics

- very small structure
- limited data
- better utilize multi-modality images



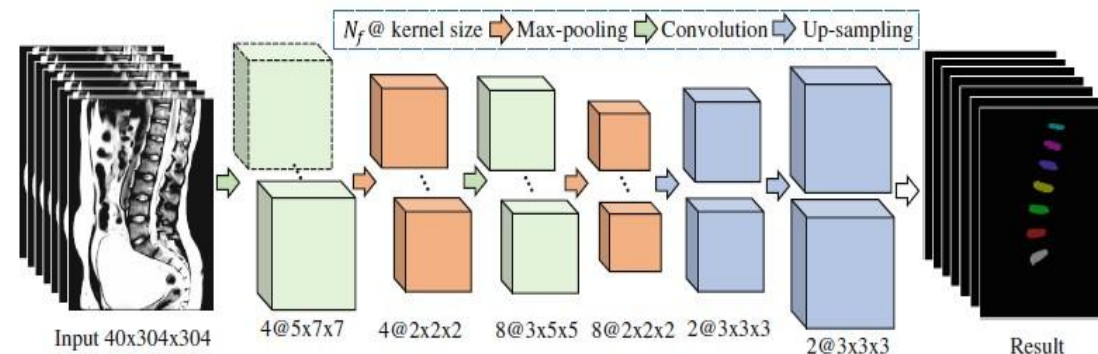
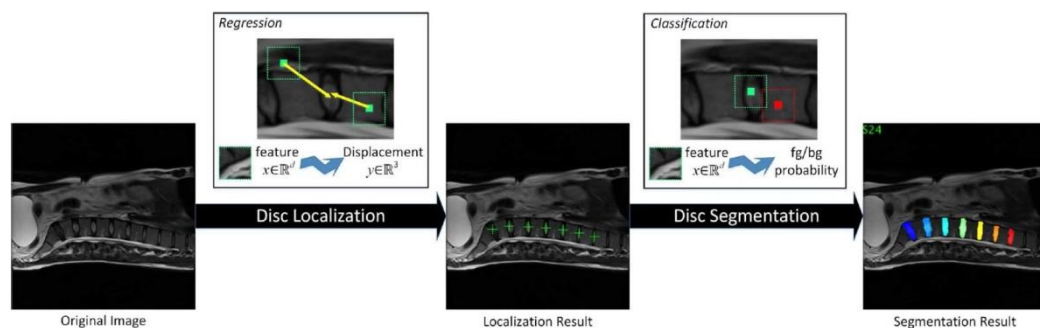
4. Case studies of CNNs for medical image segmentation



Feature engineering

hand-crafted: shape, size, geometry, intensity, radial symmetry transformation, etc.

learning-based: deep convolution neural networks.



Limitations

hand-crafted methods have limited representation capacity.

learning-based methods ignore **different scales of anatomical structures**.

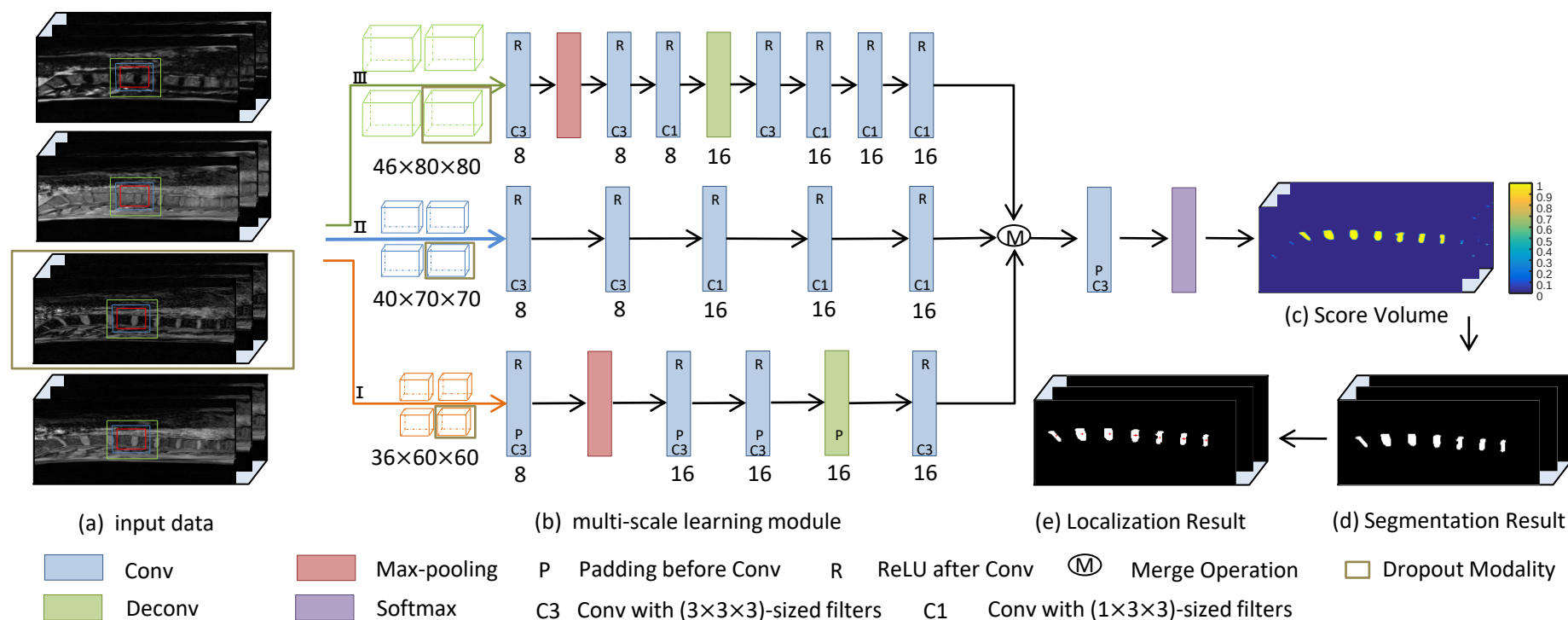
4. Case studies of CNNs for medical image segmentation



Our motivation: richer context information is important for recognition.

Multi-scale 3D fully convolutional neural network (MsFCN)

multi-scale contextual information

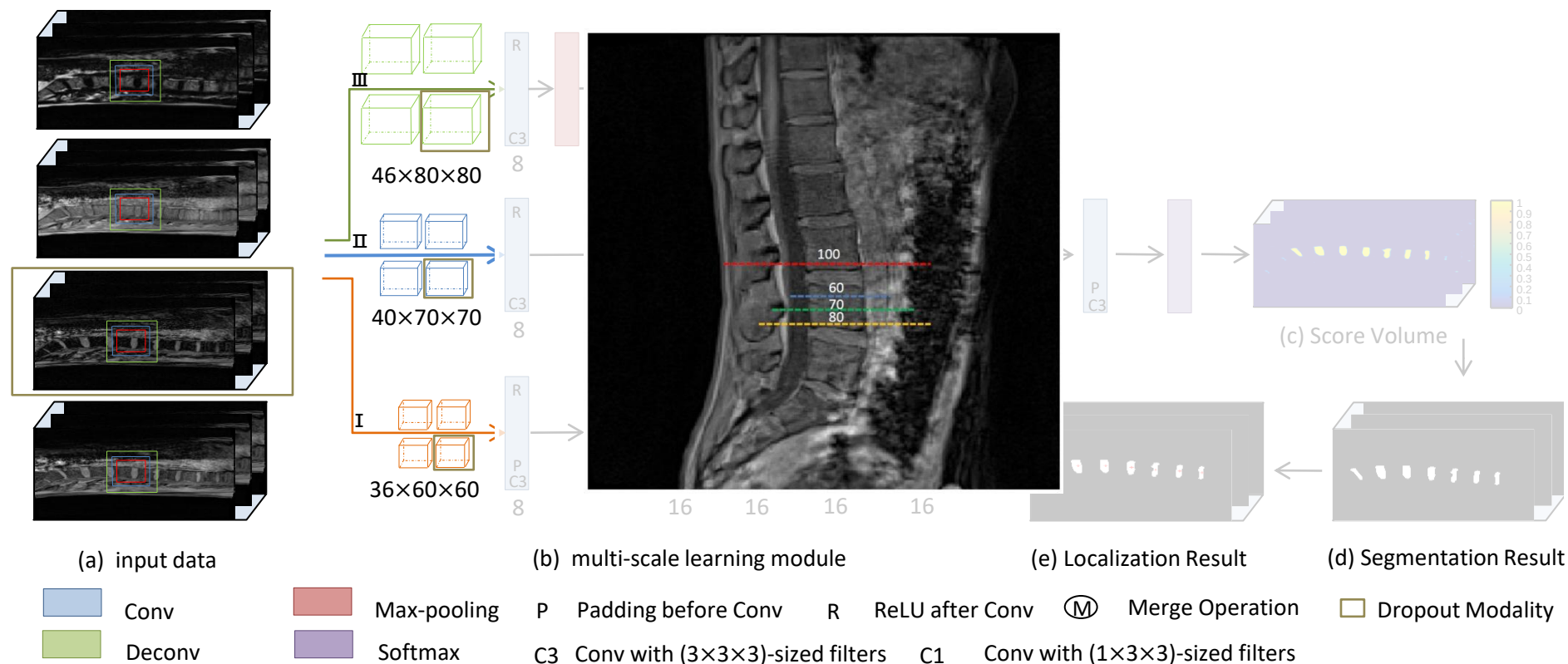


4. Case studies of CNNs for medical image segmentation



Multi-scale 3D fully convolutional neural network (MsFCN)

multi-scale contextual information

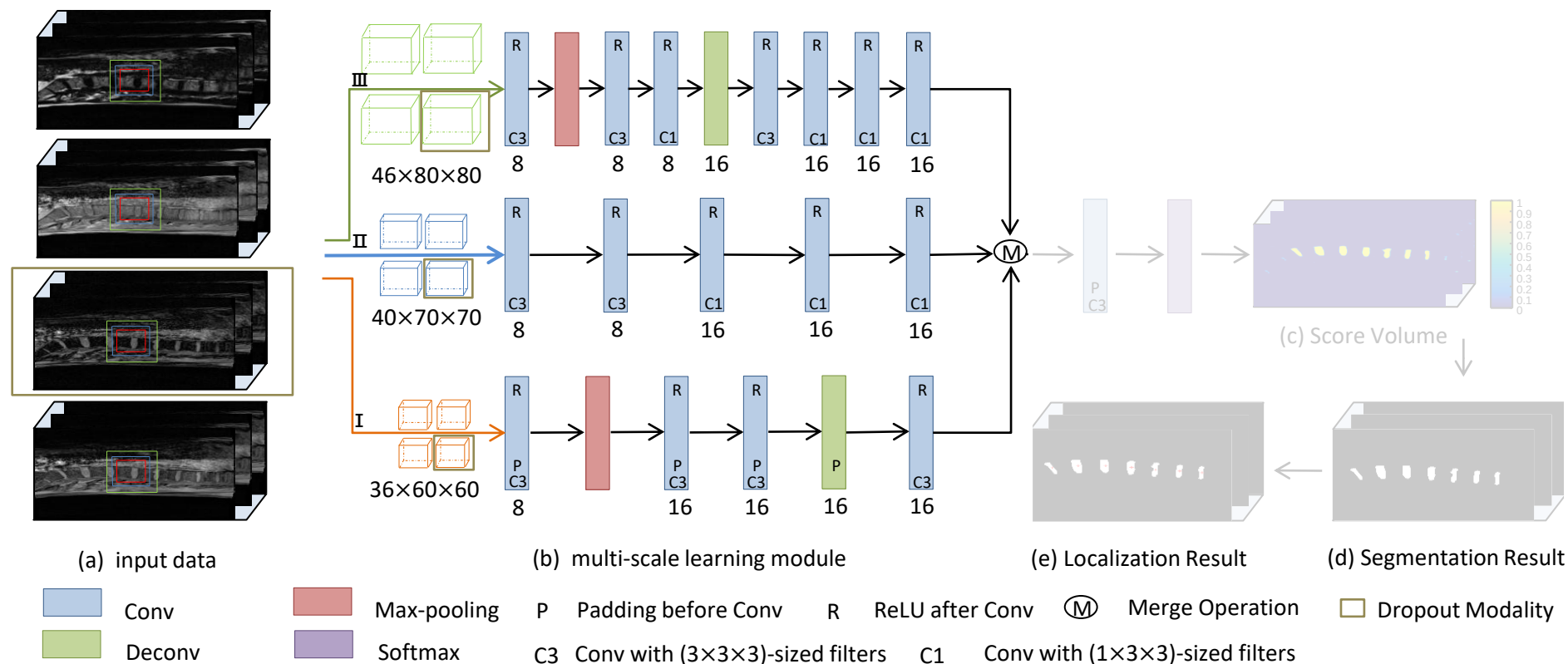


4. Case studies of CNNs for medical image segmentation



Multi-scale 3D fully convolutional neural network (MsFCN)

multi-scale contextual information

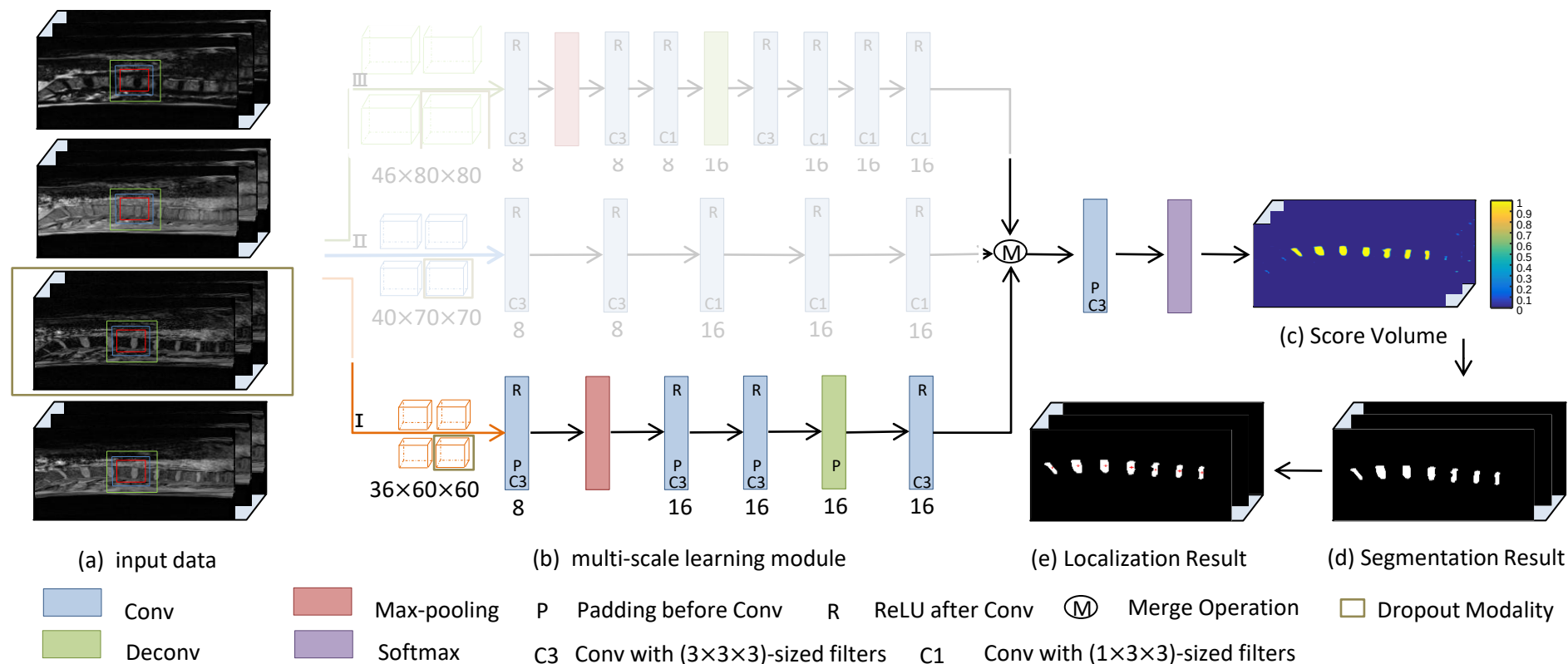


4. Case studies of CNNs for medical image segmentation



Multi-scale 3D fully convolutional neural network (MsFCN)

multi-scale contextual information

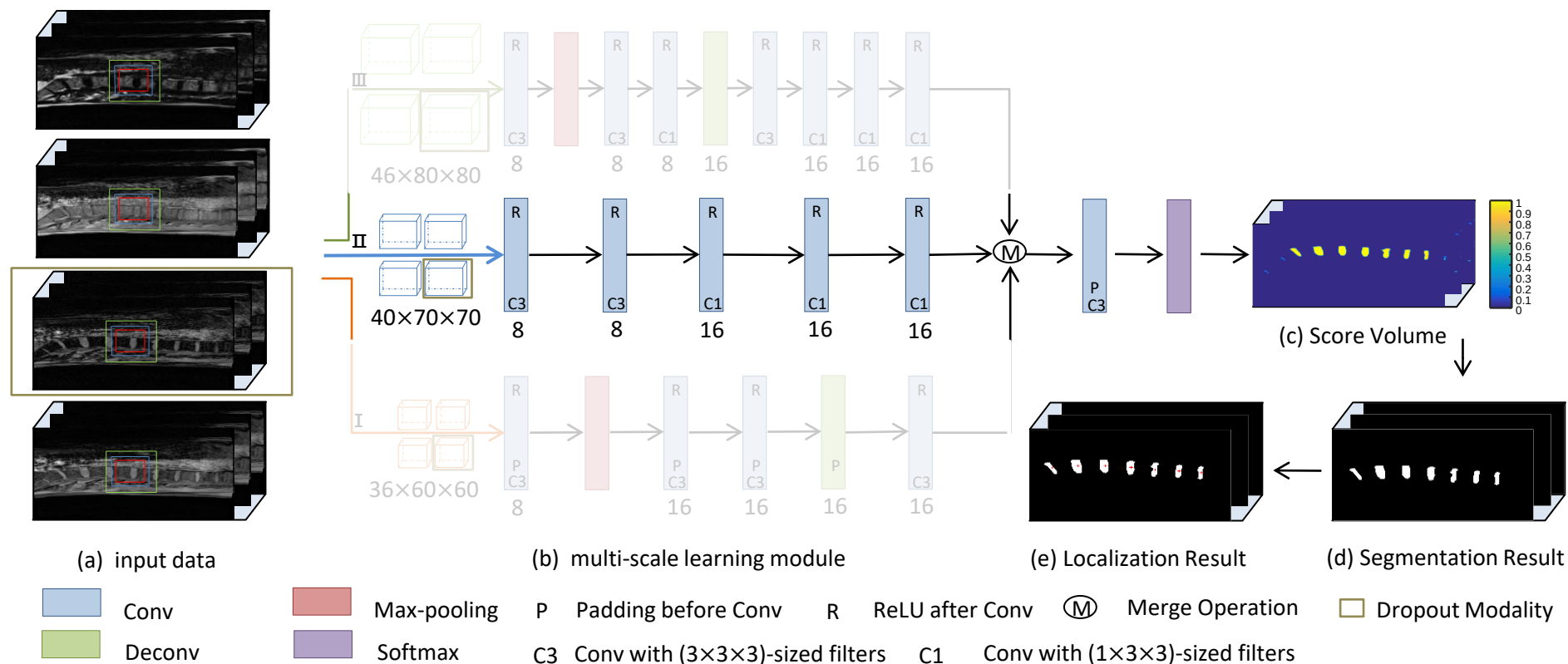


4. Case studies of CNNs for medical image segmentation



Multi-scale 3D fully convolutional neural network (MsFCN)

multi-scale contextual information

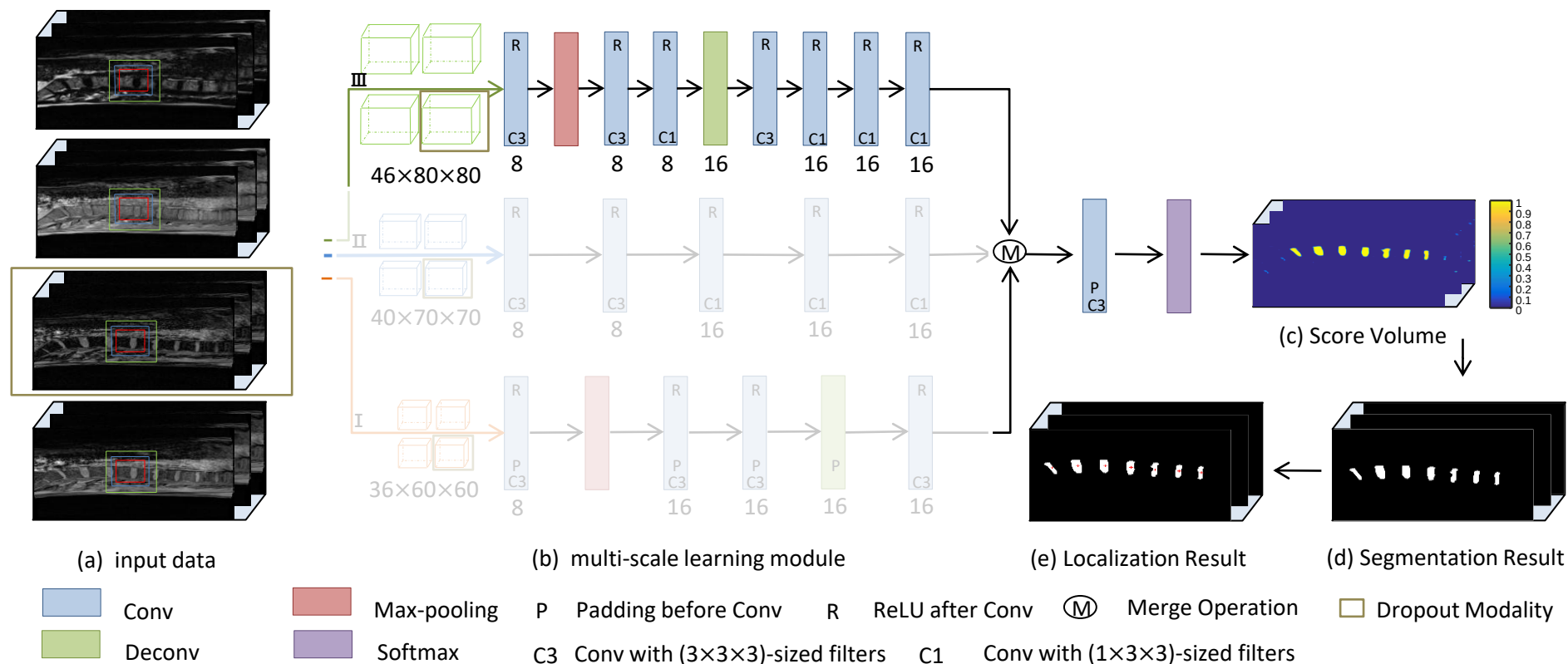


4. Case studies of CNNs for medical image segmentation



Multi-scale 3D fully convolutional neural network (MsFCN)

multi-scale contextual information

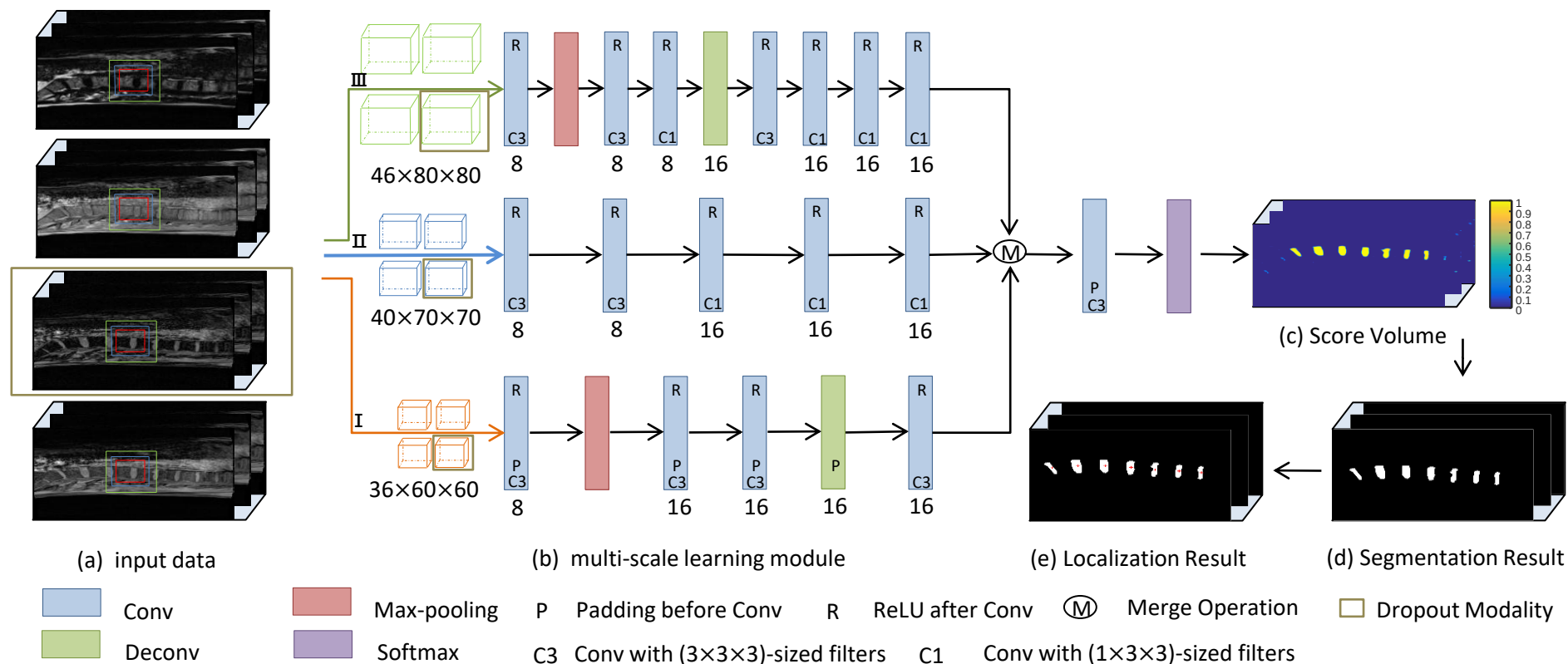


4. Case studies of CNNs for medical image segmentation



Multi-scale 3D fully convolutional neural network (MsFCN)

- multi-scale contextual information



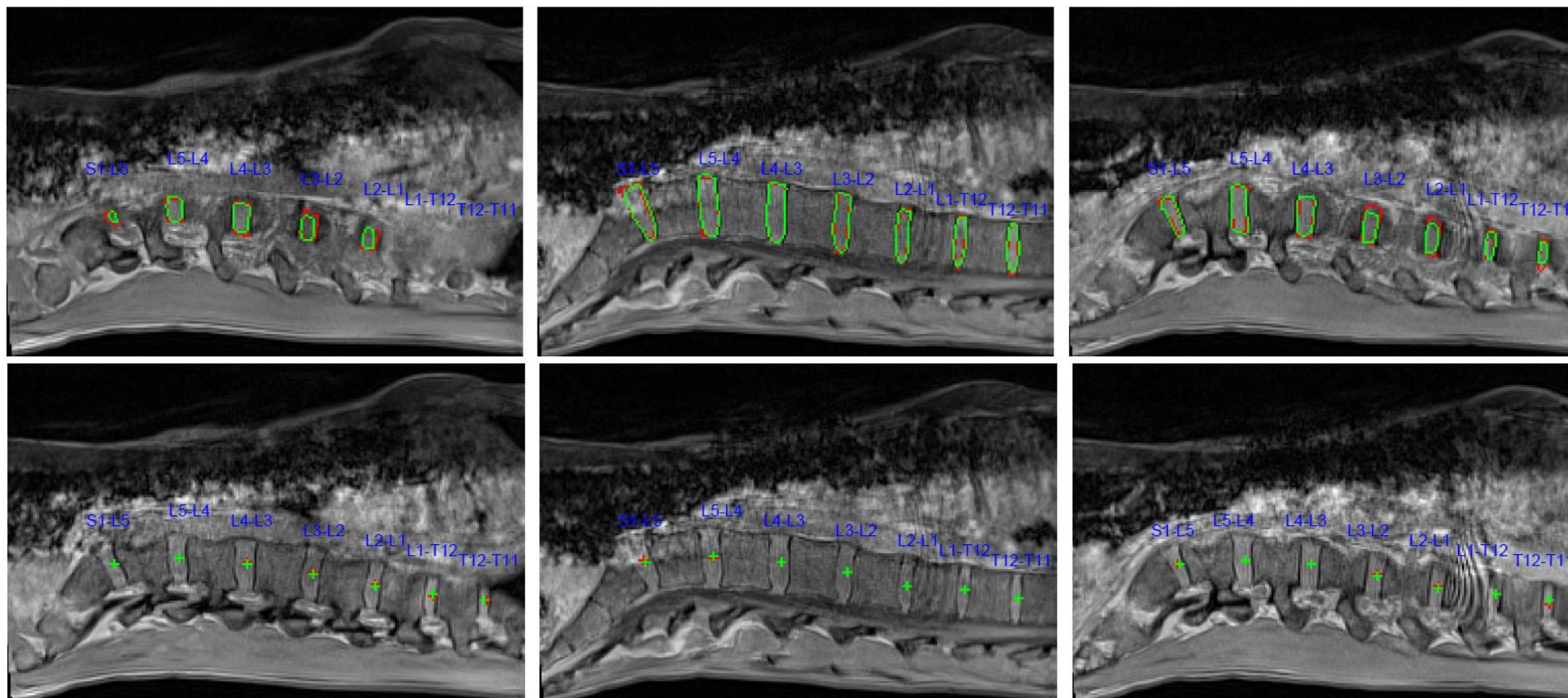
4. Case studies of CNNs for medical image segmentation



Evaluation of IVD localization and segmentation on datasets acquired from two different time points

Training Data	Test Data	Localization	Segmentation
		MLD(mm) $\pm$ SD(mm)	MDOC(%) $\pm$ SDDOC(%)
Time point 1	Time point 2	0.47 $\pm$ 0.45	90.56 $\pm$ 2.48
Time point 2	Time point 1	0.46 $\pm$ 0.34	90.48 $\pm$ 0.02

Robustness at different time points.



Localization results

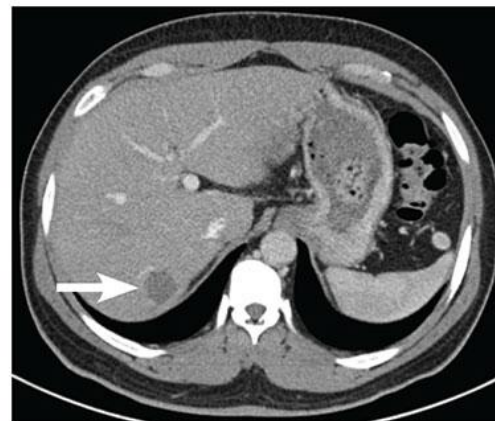
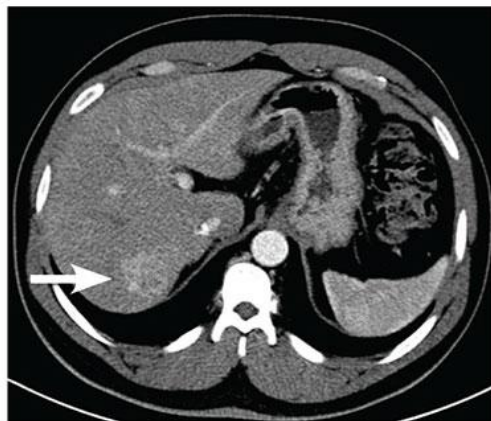
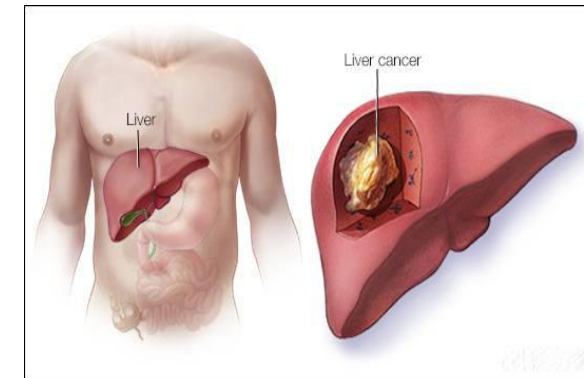
4. Case studies of CNNs for medical image segmentation



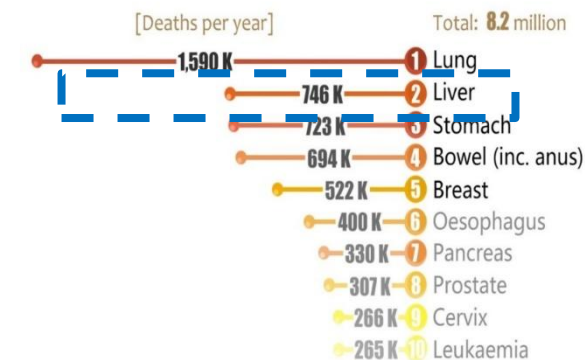
HDenseUNet - Liver tumor segmentation from CT volumes

Clinical Importance

- the second most deadly cancer
- causes about 700,000 deaths per year
- assist doctors in making accurate cancer evaluation and treatment planning.



Most Common Causes of Cancer Death



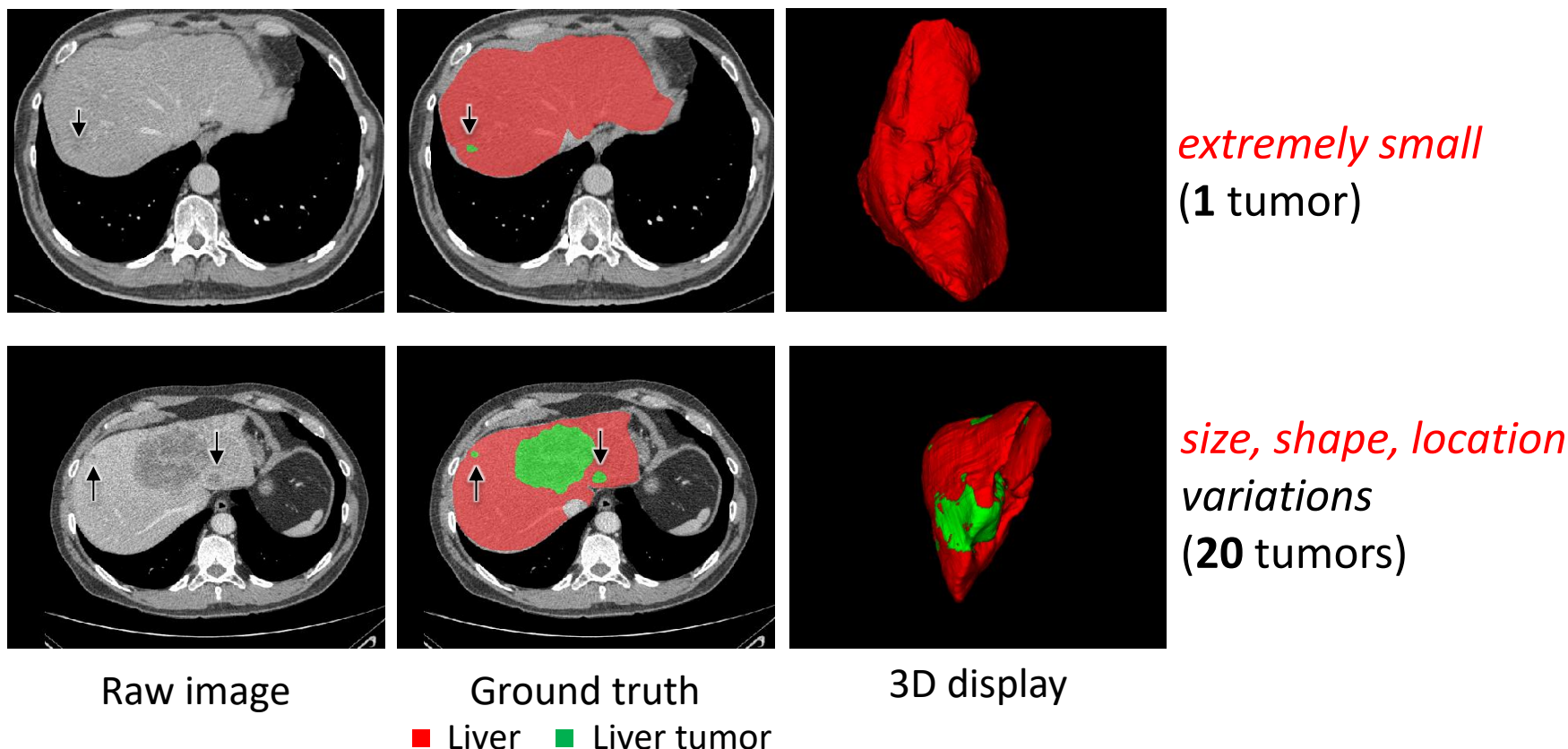
[2018 Liver Cancer: Statistics]

[pictures from Thailand medical news]

4. Case studies of CNNs for medical image segmentation



- CT is 3-dimensional and usually has a large volume size. (heavy computational cost)
- Liver tumors have large variations in **shape, size, location, intensity**, and the **number of lesions** in different patients. (3D contexts, effective encoder, small tumors)



4. Case studies of CNNs for medical image segmentation



Hand-crafted features:

thresholding
region-based
active-contours models
graph cut, etc.

Deep learning-based features:

- cascaded network based on Unet (2D network)
- multi-channel 2D FCN (2D network)
- Resnet, VGG net (2D network)
- 3D Unet, 3D Vnet (3D network)

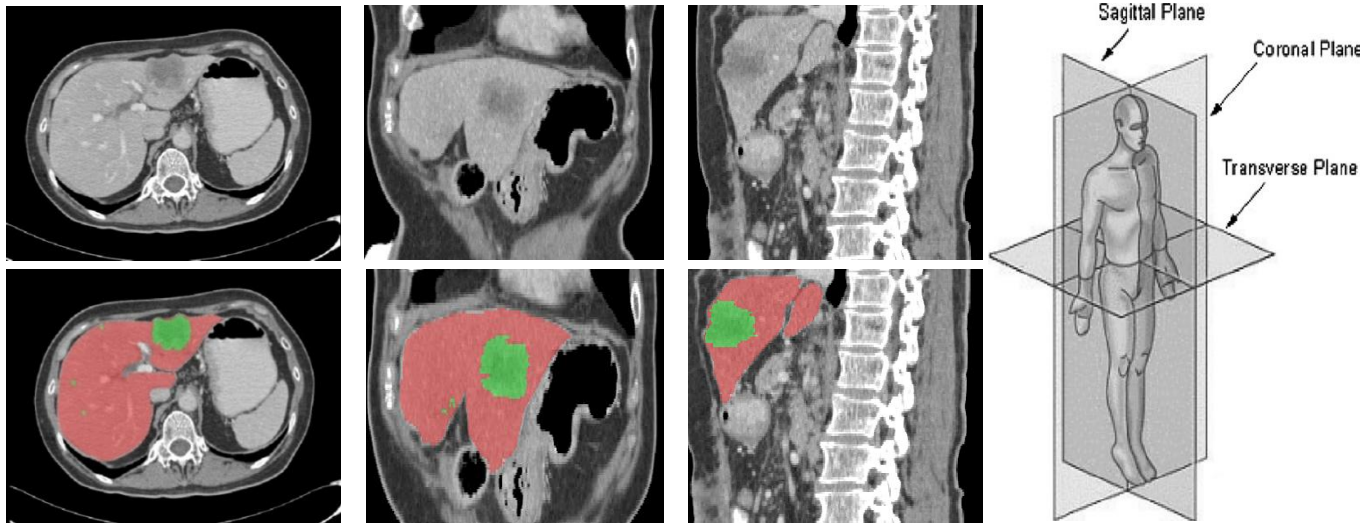
Limitations:

	2D Networks	3D Networks
3D context	No, ✗	Yes, ✓
Computational cost	Low, ✓	High, ✗
Memory Consumption	Low, ✓	High, ✗

4. Case studies of CNNs for medical image segmentation



- large-scale CT -> heavy computational cost -> **a deep and effective network.**
- 3D context is important for recognition in 3D CT scans -> **include 3D context for effective encoder design.**



Liver and tumors shown in three orthogonal planes

Take advantages of 2D and 3D networks

	2D Network	3D Network
3D context	No, ×	Yes, ✓
Computational cost	Low, ✓	High, ×
Memory Consumption	Low, ✓	High, ×

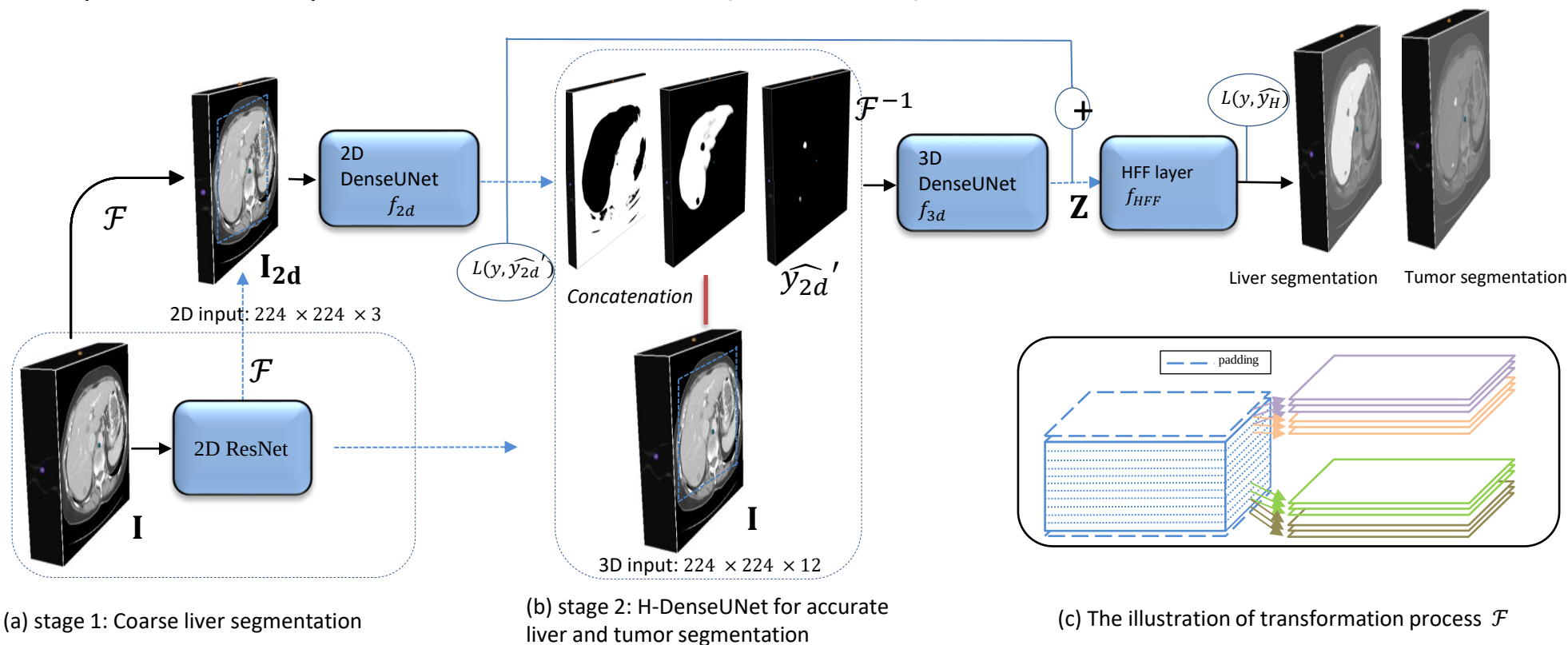
4. Case studies of CNNs for medical image segmentation



Hybrid Densely Connected Unet (H-DenseUNet): a 2D DenseUNet and a 3D counterpart.

2D DenseUNet aims to extract **intra-slice hierarchical features** (*very deep and effective*)

3D counterpart is to incorporate inter-slice contexts (*3D contexts*)

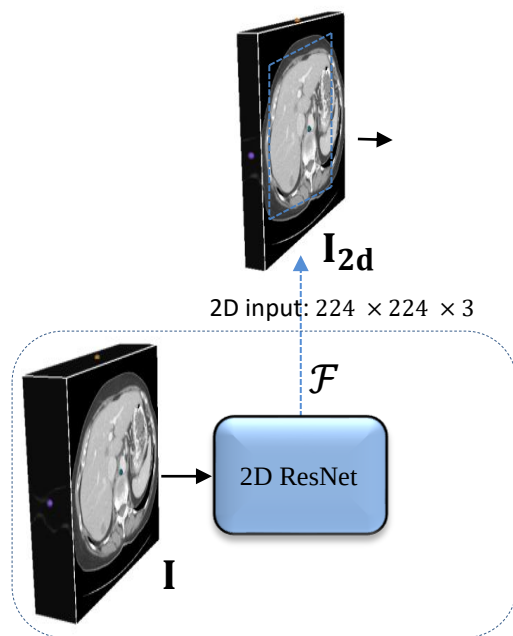


4. Case studies of CNNs for medical image segmentation



Hybrid Densely Connected Unet (H-DenseUNet): a 2D DenseUNet and a 3D counterpart.

Stage 1: Get the liver ROI to save *the following computational cost*.



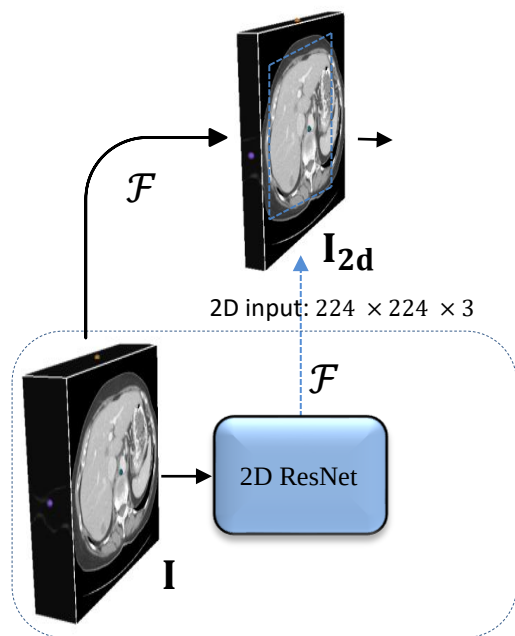
(a) stage 1: Coarse liver segmentation

4. Case studies of CNNs for medical image segmentation

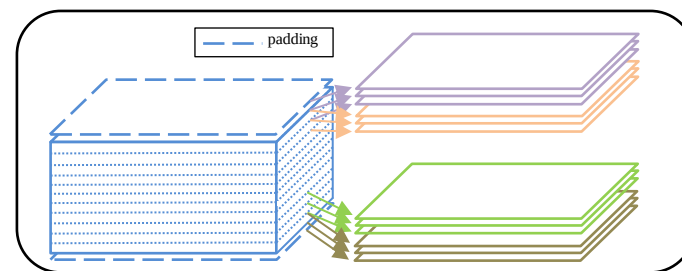


Hybrid Densely Connected Unet (H-DenseUNet): a 2D DenseUNet and a 3D counterpart.

Stage 2: 3D input is sliced into 2d adjacent slices.



(a) stage 1: Coarse liver segmentation



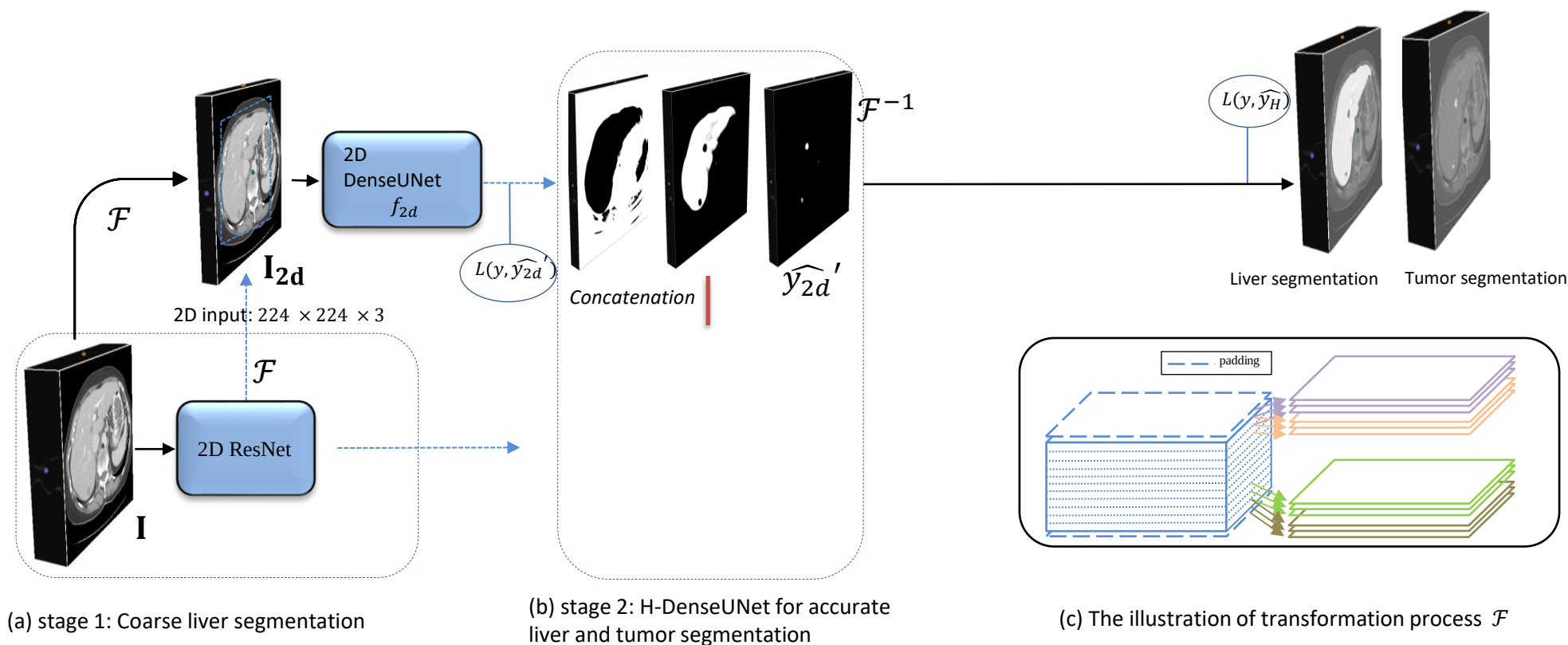
(c) The illustration of transformation process $\mathcal{F}$

4. Case studies of CNNs for medical image segmentation



Hybrid Densely Connected Unet (H-DenseUNet): a 2D DenseUNet and a 3D counterpart.

Stage 2: 2D slices are fed into the 2D DenseUNet.

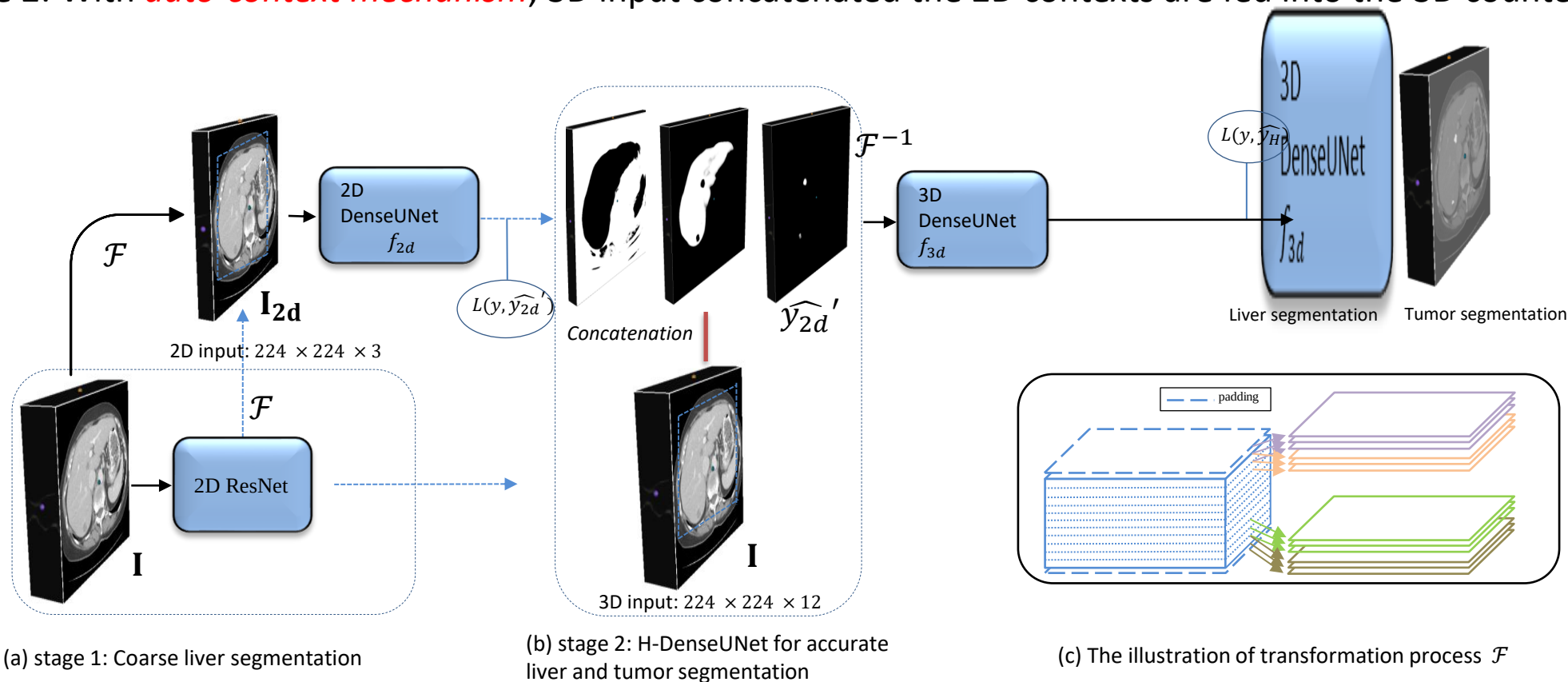


4. Case studies of CNNs for medical image segmentation



Hybrid Densely Connected Unet (H-DenseUNet): a 2D DenseUNet and a 3D counterpart.

Stage 2: With *auto-context mechanism*, 3D input concatenated the 2D contexts are fed into the 3D counterpart.

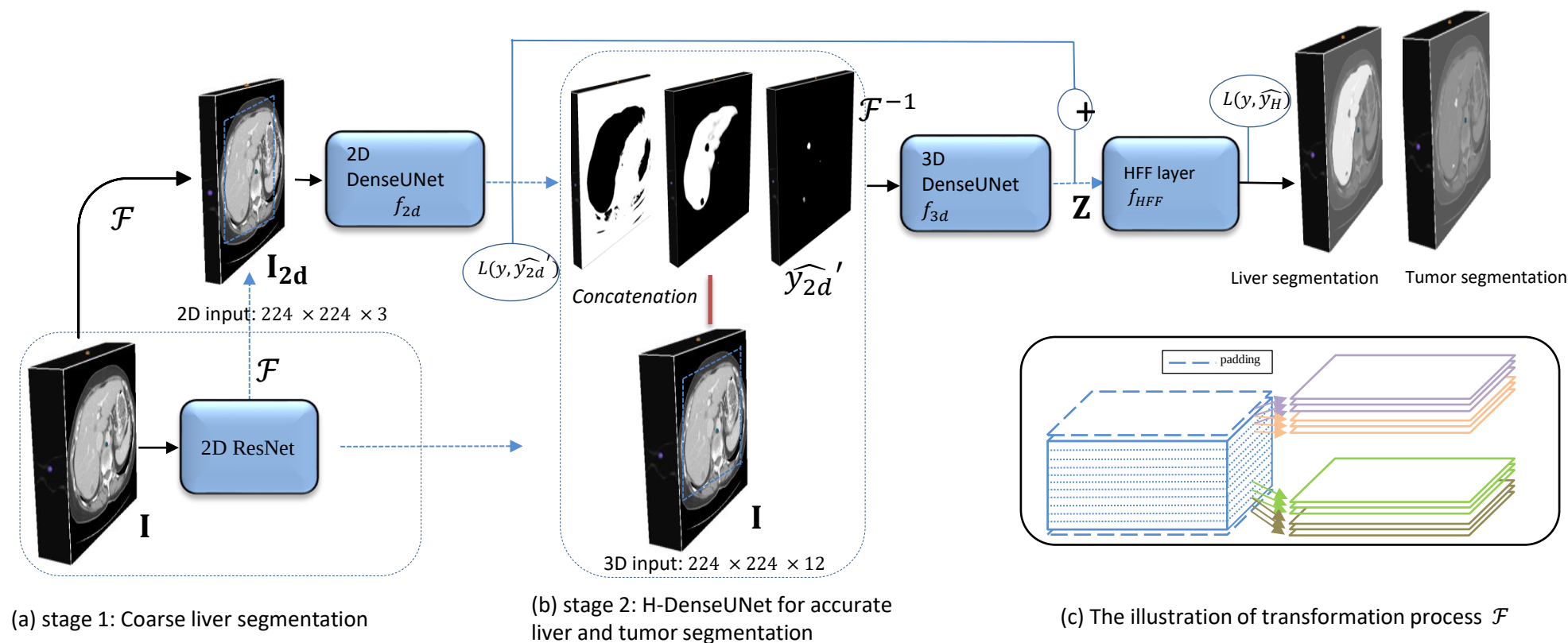


4. Case studies of CNNs for medical image segmentation



Hybrid Densely Connected Unet (H-DenseUNet): a 2D DenseUNet and a 3D counterpart.

Stage 3: *intra-slices* and inter-slices features are fused together.



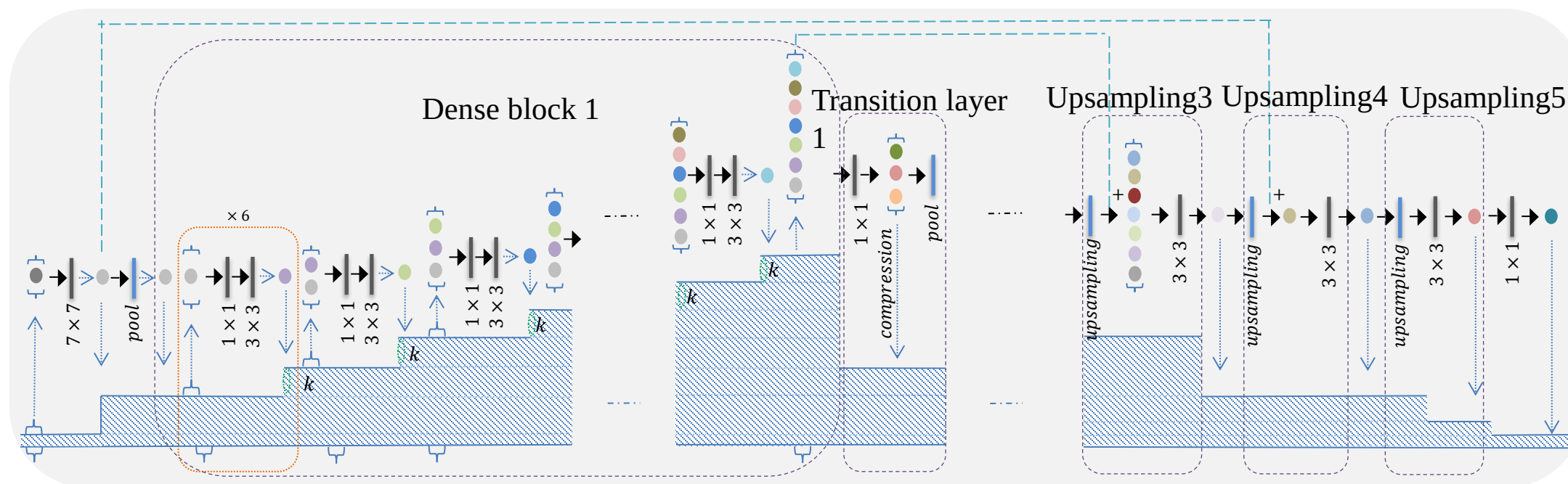
4. Case studies of CNNs for medical image segmentation



2D DenseUNet -> *Deep and Effective*

Dense connected path improves the information flow and parameters efficiency

Long range UNet connections enable the preservation of low-level features



The illustration of the 2D DenseUNet.

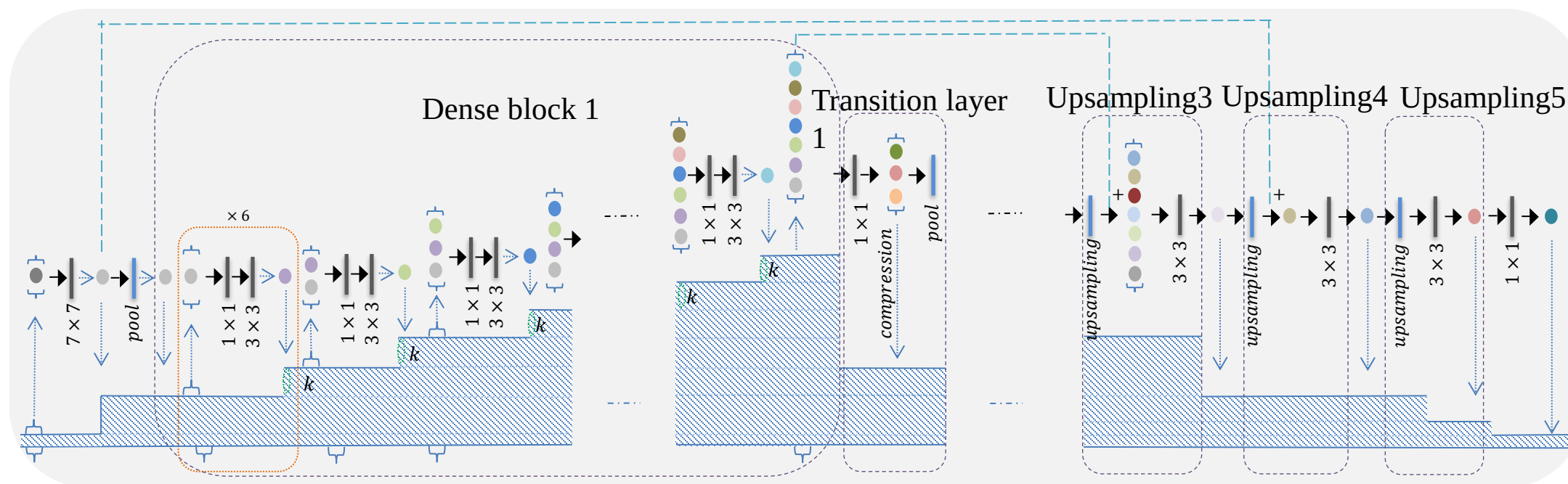
4. Case studies of CNNs for medical image segmentation



2D DenseUNet -> *Deep and Effective*

Dense connected path improves the information flow and parameters efficiency

Long range UNet connections enable the preservation of low-level features



The illustration of the 2D DenseUNet.

4. Case studies of CNNs for medical image segmentation



Advantages of H-DenseUNet

- deep and effective
 - the design of densely connected network with the Unet connections.
 - with **auto-context**, the optimization burden in the 3D counterpart is well alleviated.
- 3D contexts
 - incorporate voxel information by 3D counterpart.

	2D Network	3D Network
3D context	No, ×	Yes, √
Computational cost	Low, √	High, ×
Memory Consumption	Low, √	High, ×

4. Case studies of CNNs for medical image segmentation



Datasets

2017 MICCAI Liver Tumor Segmentation (LiTS) Challenge

- training Data / test Data (**131 / 70, CT Scans**)
 - **varying spatial** resolution (slice spacing differs between 0.45-6.0 mm)
 - number of slices varies from **42 to 1024**
-
- **3Dircadb Dataset – public clinical dataset**
 - number of slices varies from **64 to 502**

Evaluation metrics

- Dice per case; Dice global
- Volumetric overlap error (VOE)
- Relative volume difference (RVD)
- Average symmetric surface distance (ASD)
- Root mean square symmetric surface distance (RMSD)

4. Case studies of CNNs for medical image segmentation



Ablation study on each component of our method

Model	Lesion		Liver	
	Dice per case	Dice global	Dice per case	Dice global
3D DenseUNet without pre-trained model	59.4	78.8	93.6	92.9
UNet Chlebus et al.	65.0	-	-	-
ResNet Han et al.	67.0	-	-	-
2D DenseUNet without pre-trained model	67.7	80.1	94.7	94.7
2D DenseNet with pre-trained model	68.3	81.8	95.3	95.9
2D DenseUNet with pre-trained model	70.2	82.1	95.8	96.3
H-DenseUNet	72.2	82.4	96.1	96.5

- **2D DenseUNet outperforms other 2D networks such as ResNet and UNet.**

4. Case studies of CNNs for medical image segmentation



Ablation study on each component of our method

Model	Lesion		Liver	
	Dice per case	Dice global	Dice per case	Dice global
3D DenseUNet without pre-trained model	59.4	78.8	93.6	92.9
UNet Chlebus et al.	65.0	-	-	-
ResNet Han et al.	67.0	-	-	-
2D DenseUNet without pre-trained model	67.7	80.1	94.7	94.7
2D DenseNet with pre-trained model	68.3	81.8	95.3	95.9
2D DenseUNet with pre-trained model	70.2	82.1	95.8	96.3
H-DenseUNet	72.2	82.4	96.1	96.5

- **2D DenseUNet outperforms other 2D networks such as ResNet and UNet.**
- **2D DenseUNet outperforms 3D DenseUNet.**

4. Case studies of CNNs for medical image segmentation



Ablation study on each component of our method

Model	Lesion		Liver	
	Dice per case	Dice global	Dice per case	Dice global
3D DenseUNet without pre-trained model	59.4	78.8	93.6	92.9
UNet Chlebus et al.	65.0	-	-	-
ResNet Han et al.	67.0	-	-	-
2D DenseUNet without pre-trained model	67.7	80.1	94.7	94.7
2D DenseNet with pre-trained model	68.3	81.8	95.3	95.9
2D DenseUNet with pre-trained model	70.2	82.1	95.8	96.3
H-DenseUNet	72.2	82.4	96.1	96.5

- 2D DenseUNet outperforms other 2D networks such as ResNet and UNet.
- 2D DenseUNet outperforms 3D DenseUNet.
- 2D DenseUNet outperforms 2D DenseNet. (*effectiveness of unet connections*)
- H-DenseUNet outperforms 2D DenseUNet. (*effectiveness of the hybrid network*)

4. Case studies of CNNs for medical image segmentation



Results on the 2017 MICCAI LiTS challenge leaderboard (only list **top 15** results)

Team	Lesion		Liver		Tumor Burden
	Dice per case	Dice global	Dice per case	Dice global	RMSE
our	72.2	82.4	96.1	96.5	0.015
IeHealth	70.2	79.4	96.1	96.4	0.017
hans.meine	67.6	79.6	96.0	96.5	0.020
superAI	67.4	81.4	0.0	0.0	1251.447
Elehanx Han et al.	67.0	-	-	-	-
medical	66.1	78.3	95.1	95.1	0.023
deepX Yuan et al.	65.7	82.0	96.3	96.7	0.017
Njust768	65.5	76.8	4.10	13.5	0.920
Medical Vorontsov et al.	65.0	-	-	-	-
Gchlebus Chlebus et al.	65.0	-	-	-	-
predible	64.0	77.0	95.0	95.0	0.020
Lei Bi et al.	64.0	-	-	-	-
ed10b047	63.0	77.0	94.0	94.0	0.020
chunliang	62.5	78.8	95.8	96.2	0.016
yaya	62.4	79.2	95.9	96.3	0.016

- **Higher than over 50** participated teams.
- Include teams **from big companies** such as siemens.

Note: - denotes that the team participated in ISBI competition and the measurement was not evaluated.

4. Case studies of CNNs for medical image segmentation



Liver segmentation results on the clinical dataset.

Model	Year	VOE(%)	RVD(%)	ASD(mm)	RMSD(mm)	DICE
Unet Chlebus et al.	2017	62.55 ± 22.36	0.380 ± 1.95	11.11 ± 12.02	16.71 ± 13.81	0.51 ± 0.25
Christ et al.	2017	-	-	-	-	0.56 ± 0.26
ResNet Han et al.	2017	56.47 ± 13.62	-0.41 ± 0.21	6.36 ± 3.77	11.69 ± 7.60	0.60 ± 0.12
ours		49.72 ± 5.2	-0.33 ± 0.10	5.293 ± 6.15	11.11 ± 29.14	0.65 ± 0.02
Foruzan et al.*	2016	30.61 ± 10.44	15.97 ± 12.04	4.18 ± 9.60	5.09 ± 10.71	0.82 ± 0.07
Wu et al.*	2017	29.04 ± 8.16	-2.20 ± 15.88	0.72 ± 0.33	1.10 ± 0.49	0.83 ± 0.06
Li et al. †	2013	14.4 ± 5.3	-8.1 ± 2.1	2.4 ± 0.8	2.9 ± 0.7	-
Moghbel et al. †	2016	22.78 ± 12.15	8.59 ± 18.78	-	-	0.75 ± 0.15
Sun et al. †	2017	15.6 ± 4.3	5.8 ± 3.5	2.0 ± 0.9	2.9 ± 1.5	-
ours †		11.68 ± 4.33	-0.01 ± 0.05	0.58 ± 0.46	1.87 ± 2.33	0.937 ± 0.02

Note: * denotes the semi-automatic methods; † denotes the method use additional datasets; - denotes the result is not reported.

93.7% Dice on **liver** segmentation

Over 10% higher than others on Dice

Tumor segmentation results on the clinical dataset.

Model	Year	VOE(%)	RVD(%)	ASD(mm)	RMSD(mm)	DICE
Unet Chlebus et al.	2017	14.21 ± 5.71	-0.05 ± 0.10	4.33 ± 3.39	8.35 ± 7.54	0.923 ± 0.03
ResNet Han et al.	2017	11.65 ± 4.06	-0.03 ± 0.06	3.91 ± 3.95	8.11 ± 9.68	0.938 ± 0.02
Christ et al.	2017	10.7	-1.4	1.5	24.0	0.943
ours		10.02 ± 3.44	-0.01 ± 0.05	4.06 ± 3.85	9.63 ± 10.41	0.947 ± 0.01
Li et al. †	2015	9.15 ± 1.44	-0.07 ± 3.64	1.55 ± 0.39	3.15 ± 0.98	-
Moghbel et al. †	2016	5.95	7.49	-	-	0.911
Lu et al. †	2017	9.36 ± 3.34	0.97 ± 3.26	1.89 ± 1.08	4.15 ± 3.16	-
ours †		3.57 ± 1.66	0.01 ± 0.02	1.28 ± 2.02	3.58 ± 6.58	0.982 ± 0.01

Note: † denotes the method use additional datasets. - denotes the result is not reported.

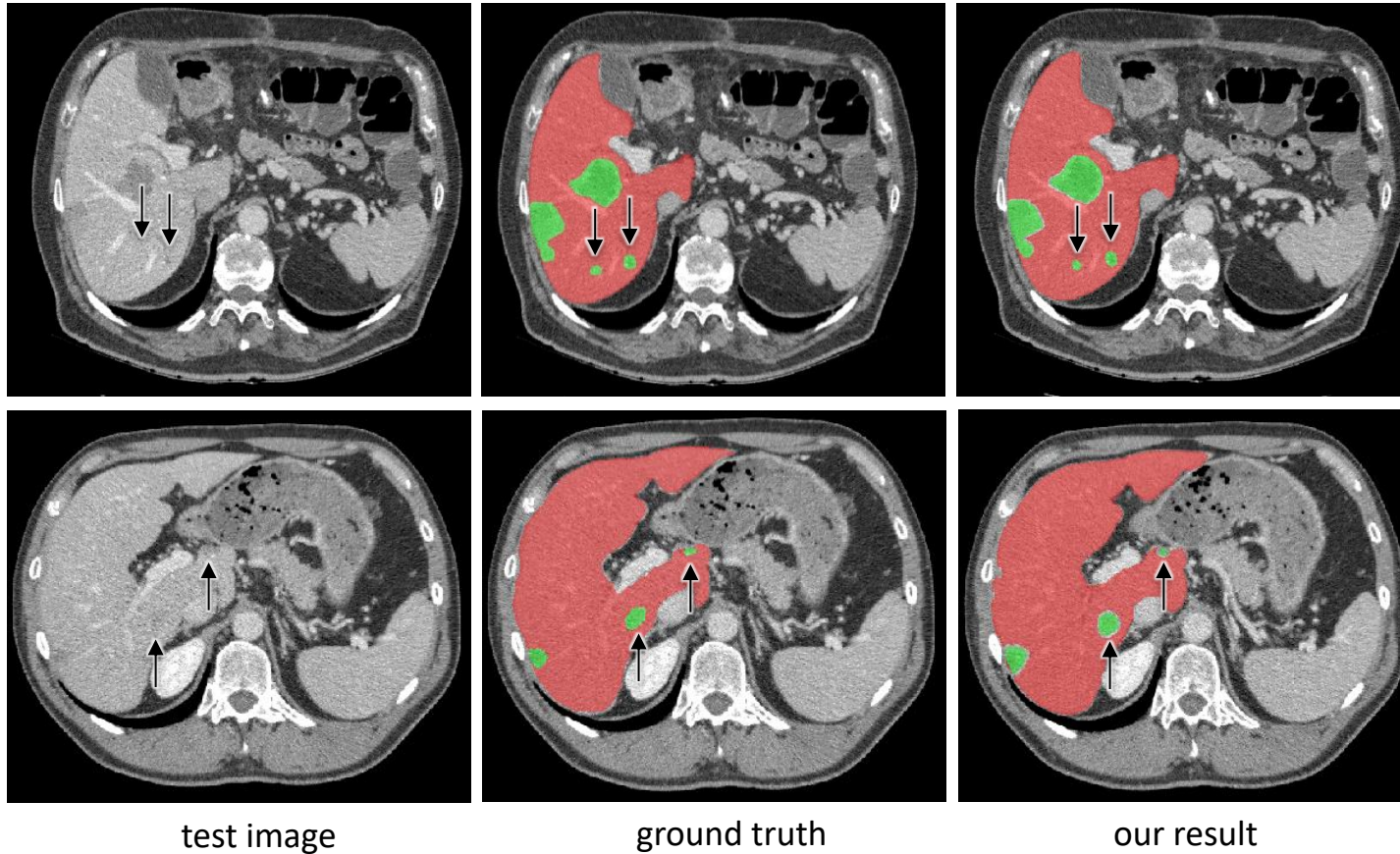
98.2% Dice on **tumor** segmentation

Over 7% higher than others on Dice

4. Case studies of CNNs for medical image segmentation



The visualization of liver tumor segmentation results.

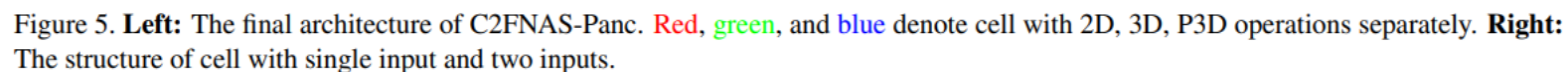


4. Case studies of CNNs for medical image segmentation



Hybrid network is more complex but has higher performance for 3D segmentation in some tasks.

Neural architecture search



4. Case studies of CNNs for medical image segmentation



Neural architecture search

Task	Brain				Liver			Pancreas			Prostate		
Class	1	2	3	Avg	1	2	Avg	1	2	Avg	1	2	Avg
CerebriuDIKU [22]	69.52	43.11	66.74	59.79	94.27	57.25	75.76	71.23	24.98	48.11	69.11	86.34	77.73
Lupin	66.15	41.63	64.15	57.31	94.79	61.40	78.10	75.99	21.24	48.62	72.73	87.62	80.18
NVDLMED [34]	67.52	45.00	68.01	60.18	95.06	71.40	83.23	78.42	38.48	58.45	69.36	86.66	78.01
K.A.V.athlon	66.63	46.62	67.46	60.24	94.74	61.65	78.20	74.97	43.20	59.09	73.42	87.80	80.61
nnU-Net [12]	67.71	47.73	68.16	61.20	95.24	73.71	84.48	79.53	52.27	65.90	75.81	89.59	82.70
C2FNAS-Panc	67.62	48.56	69.09	61.76	94.91	71.63	83.27	80.59	52.87	66.73	73.11	87.43	80.27
C2FNAS-Panc*	67.62	48.60	69.72	61.98	94.98	72.89	83.94	80.76	54.41	67.59	74.88	88.75	81.82

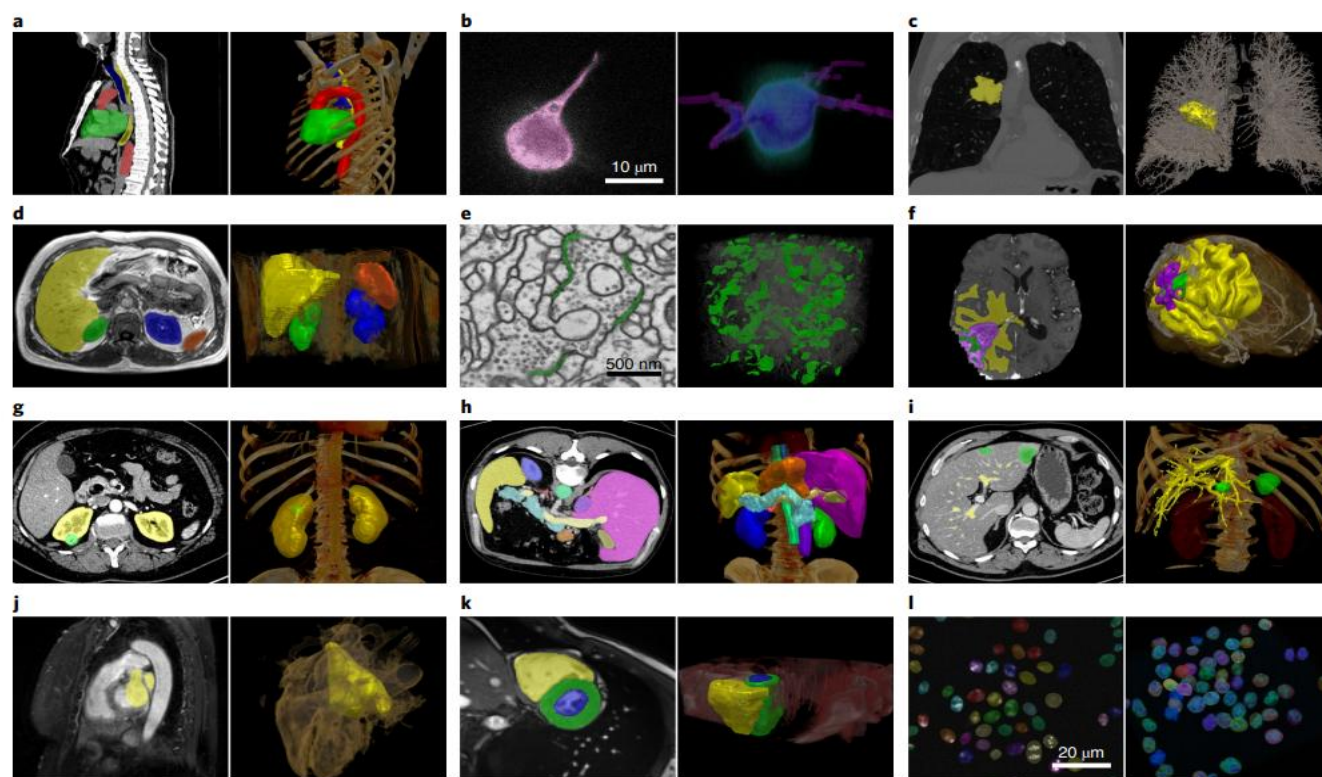
Task	Lung	Heart	Hippocampus			HepaticVessel			Spleen	Colon	Avg (Task)	Avg (Class)
Class	1	1	1	2	Avg	1	2	Avg	1	1		
CerebriuDIKU [22]	58.71	89.47	89.68	88.31	89.00	59.00	38.00	48.50	95.00	28.00	67.01	66.40
Lupin	54.61	91.86	89.66	88.26	88.96	60.00	47.00	53.50	94.00	9.00	65.61	65.89
NVDLMED [34]	52.15	92.46	87.97	86.71	87.34	63.00	64.00	63.50	96.00	56.00	72.73	71.66
K.A.V.athlon	60.56	91.72	89.83	88.52	89.18	62.00	63.00	62.50	97.00	36.00	71.51	70.89
nnU-Net [12]	69.20	92.77	90.37	88.95	89.66	63.00	69.00	66.00	96.00	56.00	76.39	75.00
C2FNAS-Panc	69.47	92.13	86.87	85.44	86.16	63.78	69.41	66.60	96.60	55.68	75.87	74.42
C2FNAS-Panc*	70.44	92.49	89.37	87.96	88.67	64.30	71.00	67.65	96.28	58.90	76.97	75.49

Table 1. Comparison with state-of-the-art methods on MSD challenge test set (number from MSD leaderboard) measured by **Dice-Sørensen coefficient (DSC)**. * denotes the 5-fold model ensemble. The numbers of tasks hepatic vessel, spleen, and colon from other teams are rounded. We also report the average on tasks and on targets respectively for an overall comparison across all tasks/targets.

4. Case studies of CNNs for medical image segmentation



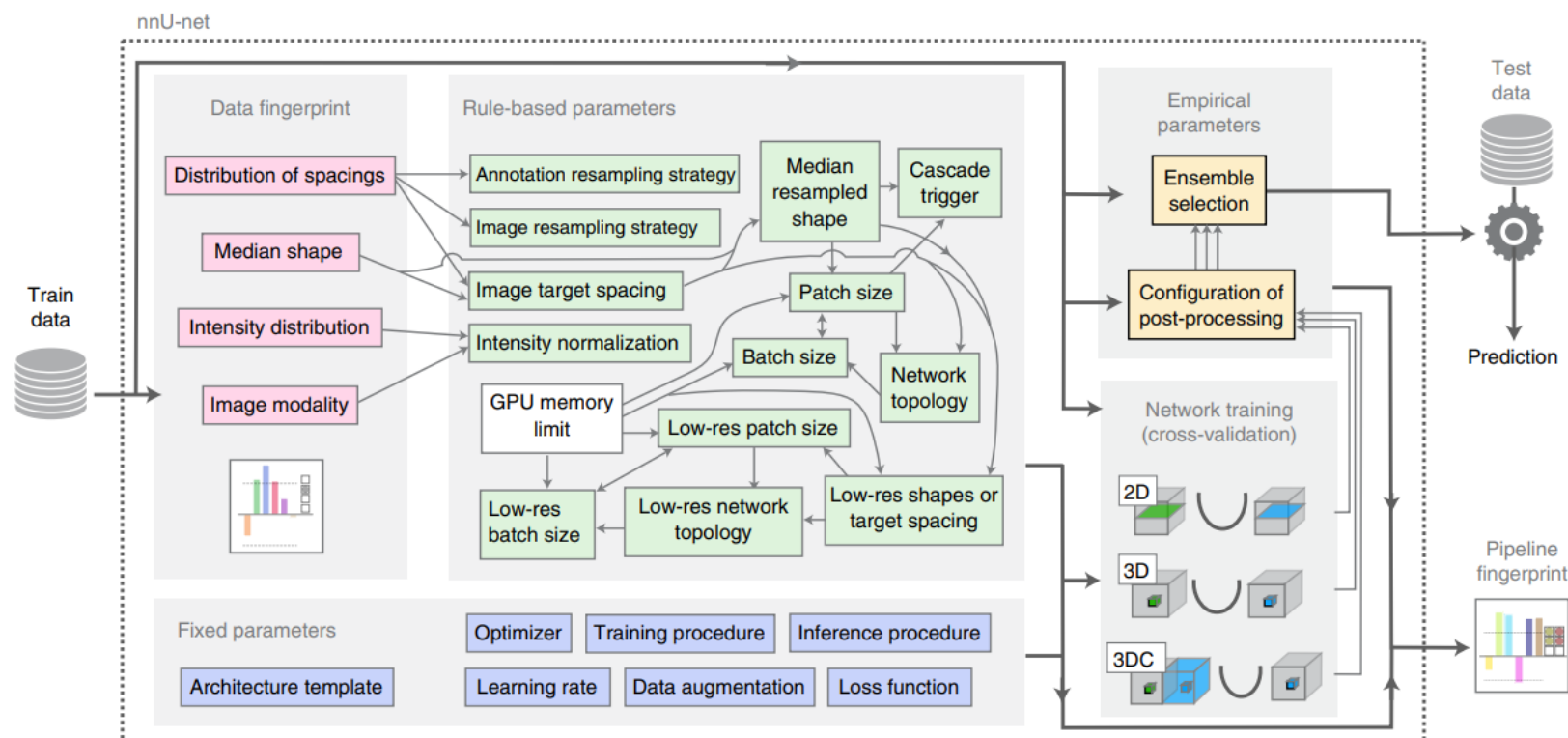
nnUNet: a deep learning-based segmentation method that automatically configures itself, including preprocessing, network architecture, training and post-processing for any new task.



4. Case studies of CNNs for medical image segmentation



nnUNet:



4. Case studies of CNNs for medical image segmentation



Transformer architecture for medical image segmentation

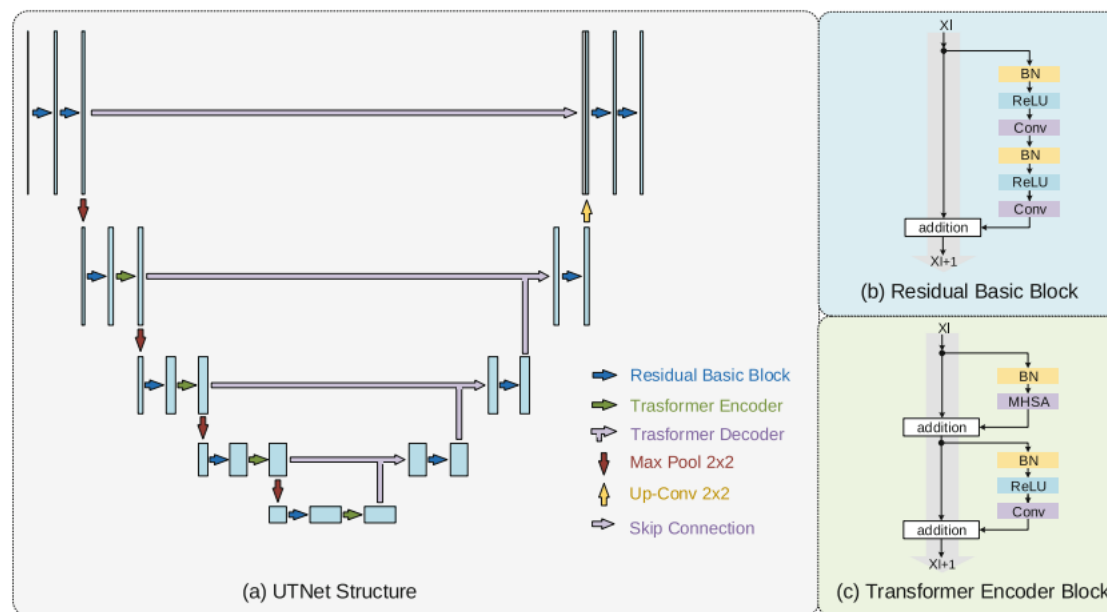


Fig. 1. (a) The hybrid architecture of the proposed UTNet. The proposed efficient self-attention mechanism and relative positional encoding allow us to apply Transformer to aggregate global context information from multiple scales in both encoder and decoder. (b) Pre-activation residual basic block. (c) The structure of Transformer encoder block.

4. Case studies of CNNs for medical image segmentation



Transformer architecture for medical image segmentation

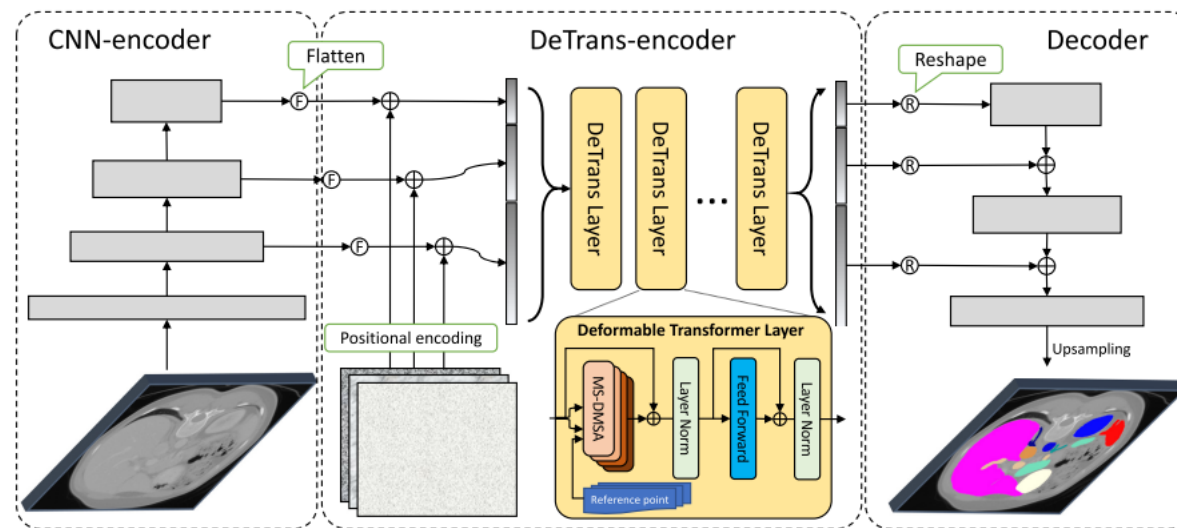


Fig. 2. Diagram of CoTr: A CNN-encoder, a DeTrans-encoder, and a decoder. Gray rectangles: CNN blocks. Yellow rectangles: 3D deformable Transformer layers. The CNN-encoder extracts multi-scale feature maps from an input image. The DeTrans-encoder processes the flattened multi-scale feature maps that embedded with the positional encoding in a sequence-to-sequence manner. The features with long-range dependency are generated by the DeTrans-encoder and fed to the decoder for segmentation. (Color figure online)

4. Case studies of CNNs for medical image segmentation



Read more papers in MICCAI, IEEE Transactions on Medical Imaging, Medical Image Analysis, CVPR, ECCV, ICCV, AAAI, IJCAI

Google Scholar

3D network medical image segmentation

Articles

About 391,000 results (0.12 sec)

Any time
Since 2022
Since 2021
Since 2018
Custom range...

Sort by relevance
Sort by date

Any type
Review articles

☐ include patents
☒ include citations

Create alert

Unetr: Transformers for 3d medical image segmentation
[A Hatamizadeh, Y. Tang, V. Nath...](#) - Proceedings of the ..., 2022 - openaccess.thecvf.com
... Despite their success, a limitation of these **networks** is their poor ... us- ing it as an attention layer within the **segmentation network** (3) UNETR ... A 3D input volume (eg C=4 channels for MRI ...
☆ Save 99 Cite Cited by 37 Related articles All 9 versions 88

An application of cascaded 3D fully convolutional networks for medical image segmentation
[H. Roth, H. Oda, X. Zhou, N. Shimizu, Y. Yang...](#) - ... **Medical Imaging** and ..., 2018 - Elsevier
Recent advances in 3D fully convolutional **networks** (FCN) have made it feasible to produce dense voxel-wise predictions of volumetric images. In this work, we show that a multi-class ...
☆ Save 99 Cite Cited by 184 Related articles All 12 versions Web of Science: 120

One network to segment them all: A general, lightweight system for accurate 3d medical image segmentation
[M. Perslev, E. B. Dam, A. Paj, C. Igel](#) - ... Conference on **Medical Image** ..., 2019 - Springer
Many recent **medical segmentation** systems rely on powerful deep learning models to solve highly specific tasks. To maximize performance, it is standard practice to evaluate numerous ...
☆ Save 99 Cite Cited by 41 Related articles All 6 versions 88

Dints: Differentiable neural network topology search for 3d medical image segmentation
[Y. He, D. Yang, H. Roth, C. Zhao...](#) - Proceedings of the IEEE ..., 2021 - openaccess.thecvf.com
Recently, neural architecture search (NAS) has been applied to automatically search high-performance **networks** for **medical image segmentation**. The NAS search space usually ...
☆ Save 99 Cite Cited by 7 Related articles All 5 versions 88

Efficient 3D medical image segmentation algorithm over a secured multimedia network

Google Scholar

Hybrid dense network medical image segmentation IEEE TMI

Articles

About 9,180 results (0.03 sec)

Any time
Since 2022
Since 2021
Since 2018
Custom range...

Sort by relevance
Sort by date

Any type
Review articles

☐ include patents
☒ include citations

Create alert

H-DenseUNet: hybrid densely connected UNet for liver and tumor segmentation from CT volumes
[X. Li, H. Chen, X. Qi, Q. Dou, C. W. Fu...](#) - ... **transactions on medical** ..., 2018 - ieeexplore. **ieee.org**
... Li et al.: **HYBRID DENSELY CONNECTED UNET FOR LIVER AND TUMOR** ... where H denotes the optimized **hybrid** features and *yH ... the number of micro-blocks in each **dense** block is ...
☆ Save 99 Cite Cited by 993 Related articles All 8 versions Web of Science: 494

SESV: Accurate medical image segmentation by predicting and correcting errors
[Y. Xie, J. Zhang, H. Lu, C. Shen...](#) - ... **Transactions on Medical** ..., 2020 - ieeexplore. **ieee.org**
... In [14], [38] and used the cross-entropy loss and **hybrid** loss as ... the convolutional-deconvolutional neural **network** (CDNN) [59], **dense** deconvolutional **network** (DDN) [7 ...
☆ Save 99 Cite Cited by 17 Related articles All 3 versions Web of Science: 10

Hi-net: hybrid-fusion network for multi-modal MR image synthesis
[T. Zhou, H. Fu, G. Chen, J. Shen...](#) - ... **transactions on medical** ..., 2020 - ieeexplore. **ieee.org**
... a multi-modal synthesis **network** is designed to **densely** combine the ... To this end, we propose a novel **Hybrid-fusion Network** ... the modality-specific **networks** and fusion **network** to learn a ...
☆ Save 99 Cite Cited by 73 Related articles All 6 versions Web of Science: 31

Efficient medical image segmentation based on knowledge distillation
[D. Qin, J. J. Bu, Z. Liu, X. Shen, S. Zhou...](#) - ... **on Medical Imaging**, 2021 - ieeexplore. **ieee.org**
... in **medical image segmentation** methods, such as adding **dense** connections, replacing ... performance in the LITS challenge with a **hybrid** use of ... that the size of this student **network** is 21 ...
☆ Save 99 Cite Cited by 2 Related articles All 6 versions Web of Science: 1

Ce-net: Context encoder network for 2d medical image segmentation
[Z. Gu, J. Cheng, H. Fu, K. Zhou, H. Hao...](#) - **IEEE transactions on** ..., 2019 - ieeexplore. **ieee.org**
Since the proposed block looks like a **denselv** connected block we name it **dense** atrous con

4. Case studies of CNNs for medical image segmentation



Practice by participating in medical image challenges

<https://grand-challenge.org/challenges/>

Winner can get bonus or be invited to submit a paper

Grand Challenge Challenges Algorithms ... Help Sign In Register

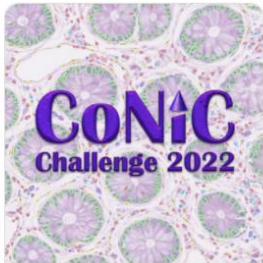
Challenges

Here is an overview over the medical image analysis challenges that have been hosted on Grand Challenge.
Please [contact us](#) if you would like to host your own challenge.

+ Add a new challenge View all challenges

Filter Challenges

121 challenges found



CoNIC 2022
Accepting submissions



AIROGS
Accepting submissions



STOIC 2021
COVID-19 AI Challenge
with 10,000 CT scans
Accepting submissions for qualification until Mar 23 2022 at 16:00



TIGER
Accepting submissions for Segmentation and Detection

Grand Challenge Challenges Algorithms ... Help Sign In Register

+ Add a new challenge View all challenges

Filter Challenges

Title or Description

Modality

CT

Anatomical Structure

Anatomical Region

Organization

Challenge Series

Task Type

Segmentation

Educational

Unknown



- Review what we have learned last time
- Case studies of CNNs for object detection
- Advanced models for higher-dimensional (3D) data
- Case studies of CNNs for medical image segmentation
- Advanced models for higher-dimensional (video) data

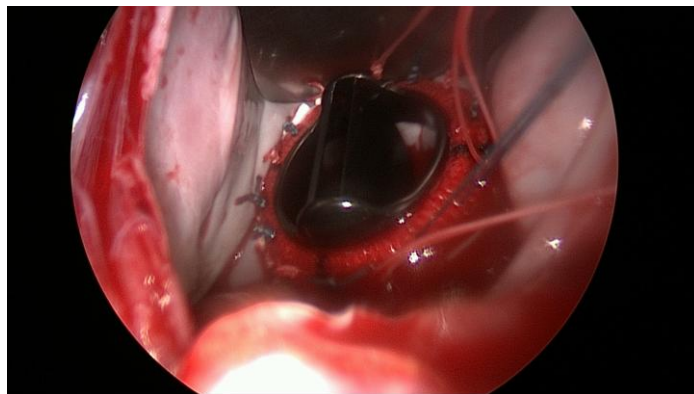
5. Advanced models for higher-dimensional (video) data



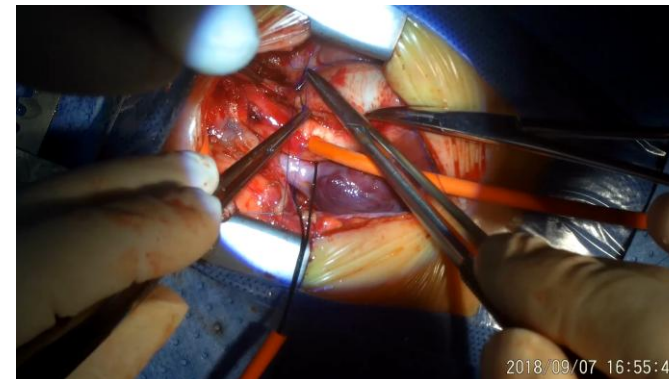
Predict heart disease from echocardiogram videos



cholecystectomy surgeries



Thoracoscopy-assisted mitral valve replacement



thoracotomy

Surgical skill assessment, workflow recognition, instrument segmentation, future frame prediction

5. Advanced models for higher-dimensional (video) data



Another approach: 3D convolutions

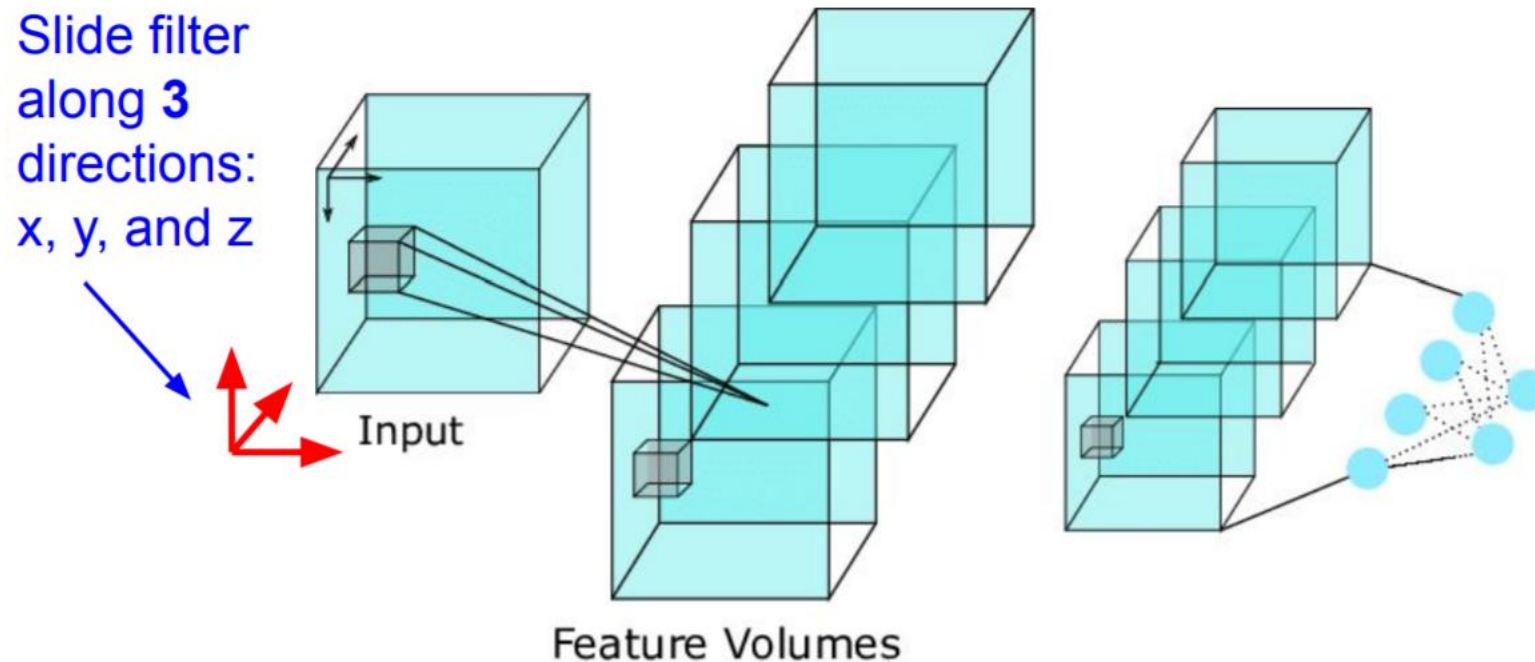


Figure credit:
https://www.researchgate.net/profile/Deepak_Mishra19/publication/330912338/figure/fig1/AS:723363244810254@1549474645742/Basic-3D-CNN-architecture-the-3D-filter-is-convolved-with-the-video-in-three-dimensions.png

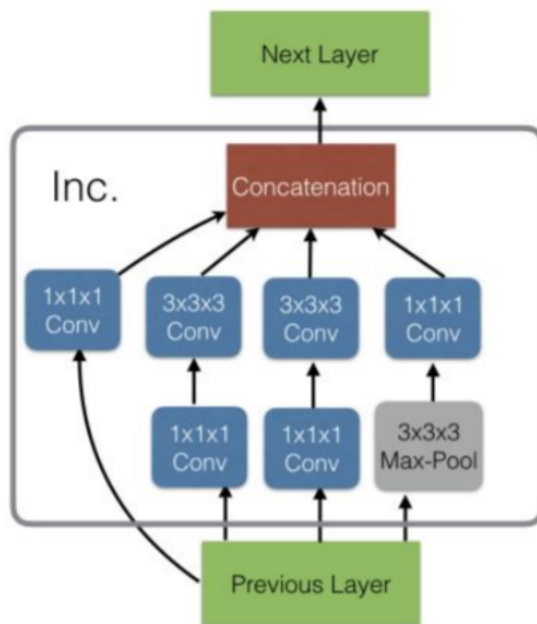
For video data, 3rd dimension is time

5. Advanced models for higher-dimensional (video) data



I3D: 3D convolutional network for video data

Inception Module (Inc.) w/
3D convolutions



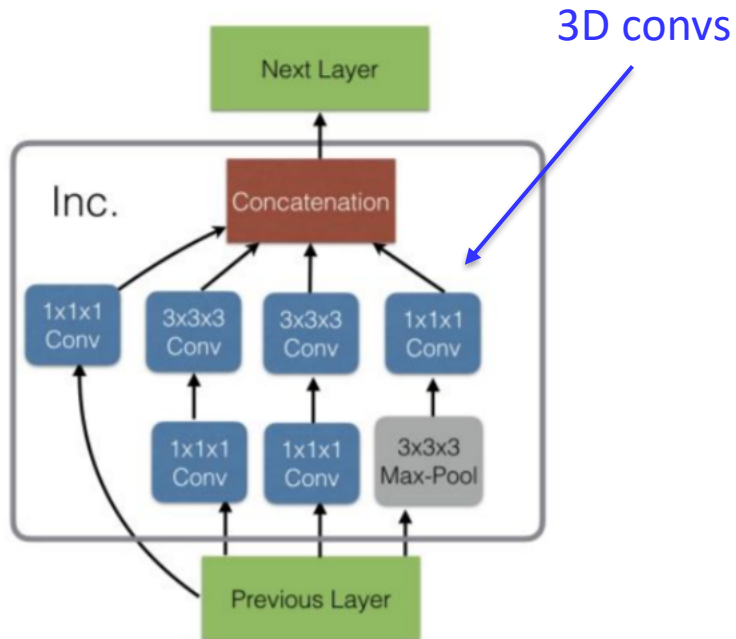
Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

5. Advanced models for higher-dimensional (video) data

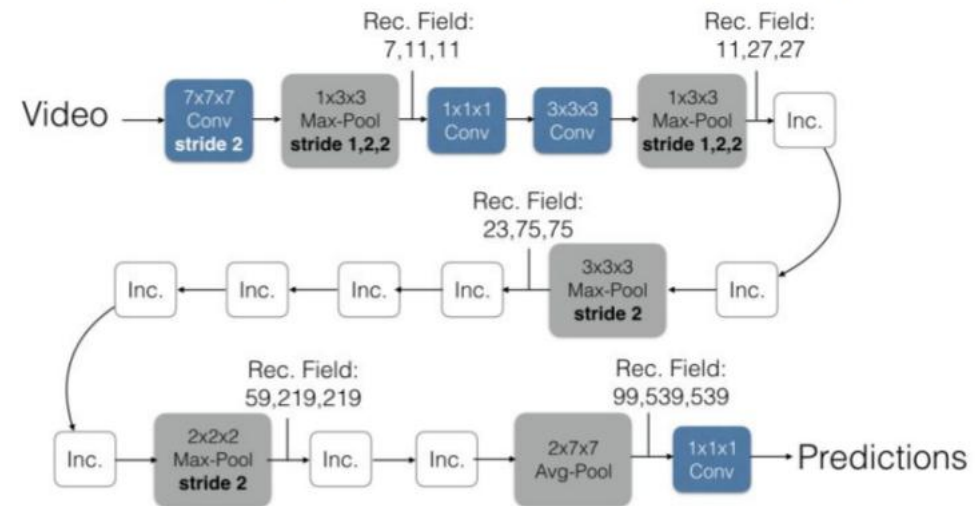


I3D: 3D convolutional network for video data

Inception Module (Inc.) w/
3D convolutions



3D Inception Module used in Inception
Network (also known as GoogLeNet)



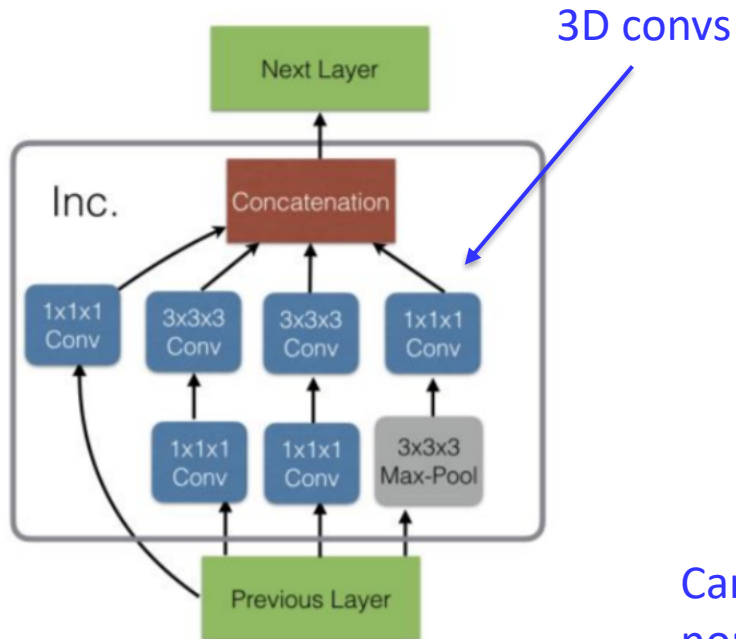
Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

5. Advanced models for higher-dimensional (video) data

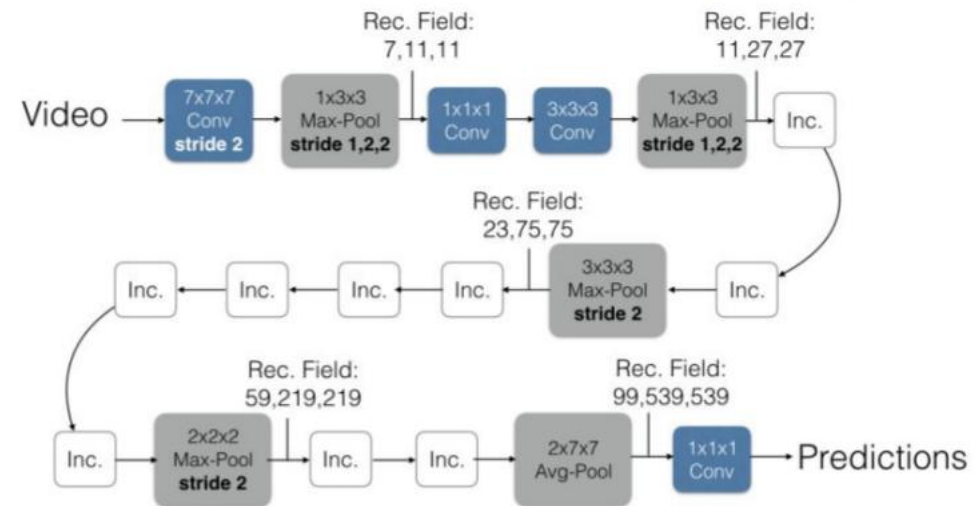


I3D: 3D convolutional network for video data

Inception Module (Inc.) w/
3D convolutions



3D Inception Module used in Inception
Network (also known as GoogLeNet)



Can pre-train from 2D datasets e.g. ImageNet by replicating and normalizing 2D weights over additional dimension!

Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

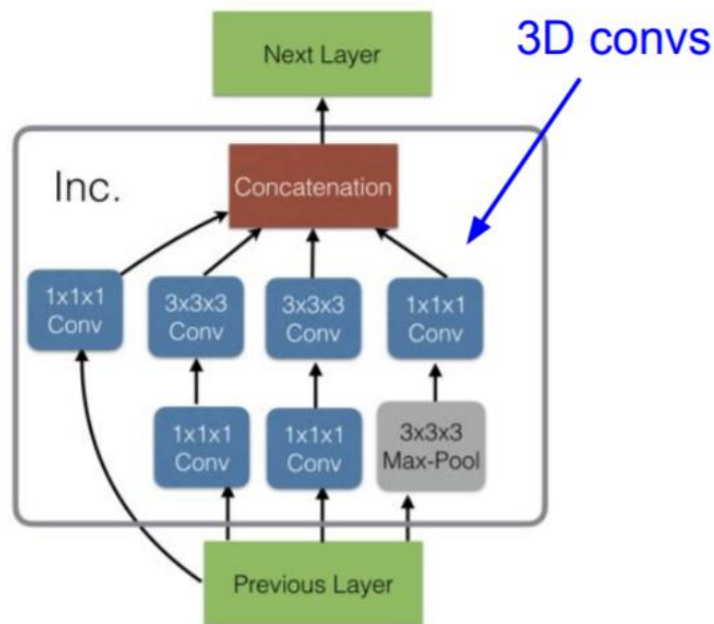
5. Advanced models for higher-dimensional (video) data



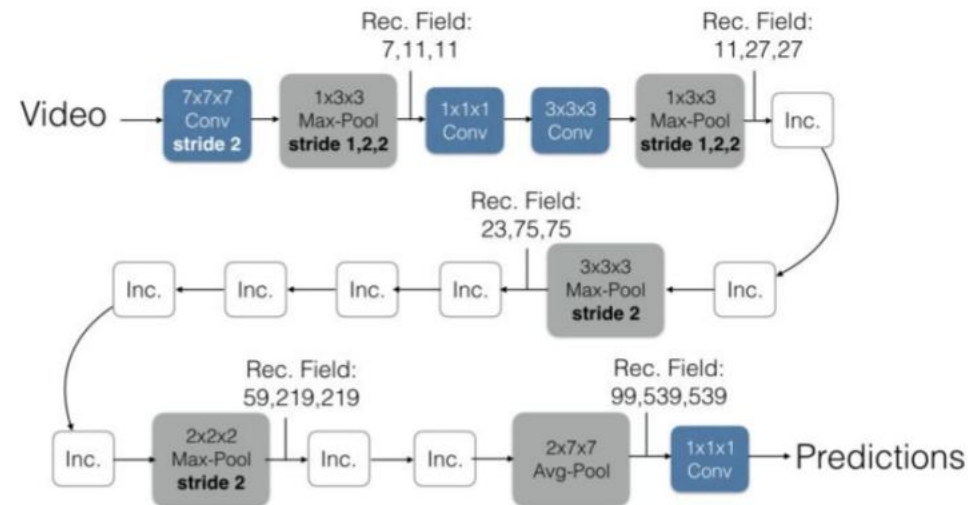
I3D: 3D convolutional network for video data

Note: in general,
can 3D-ify many
2D architectures!

Inception Module (Inc.) w/
3D convolutions



3D Inception Module used in Inception
Network (also known as GoogLeNet)



Can pre-train from 2D datasets e.g. ImageNet by replicating
and normalizing 2D weights over additional dimension!

Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow

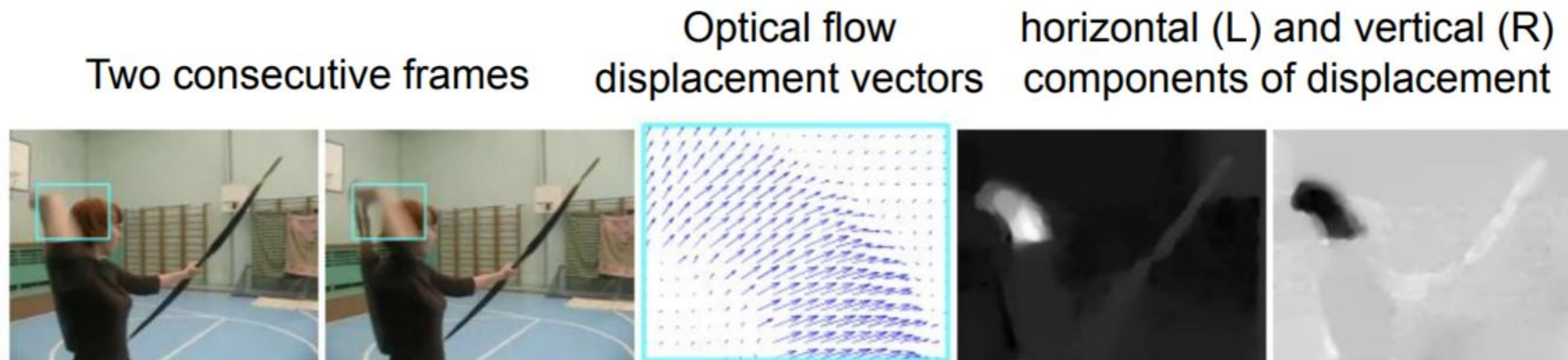
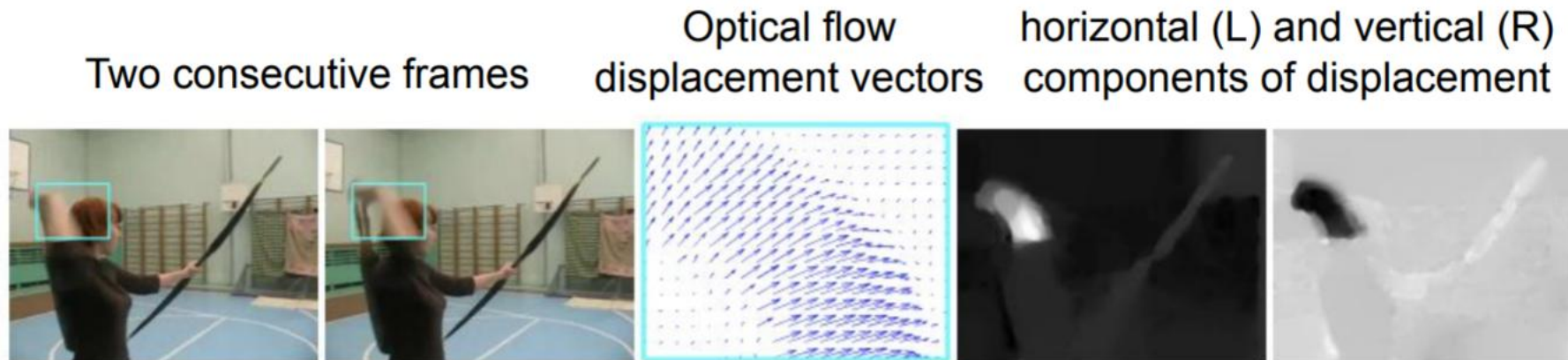


Figure credit: Simonyan and Zisserman. Two-Stream Convolutional Networks for Action Recognition in Videos. NeurIPS 2014.

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow



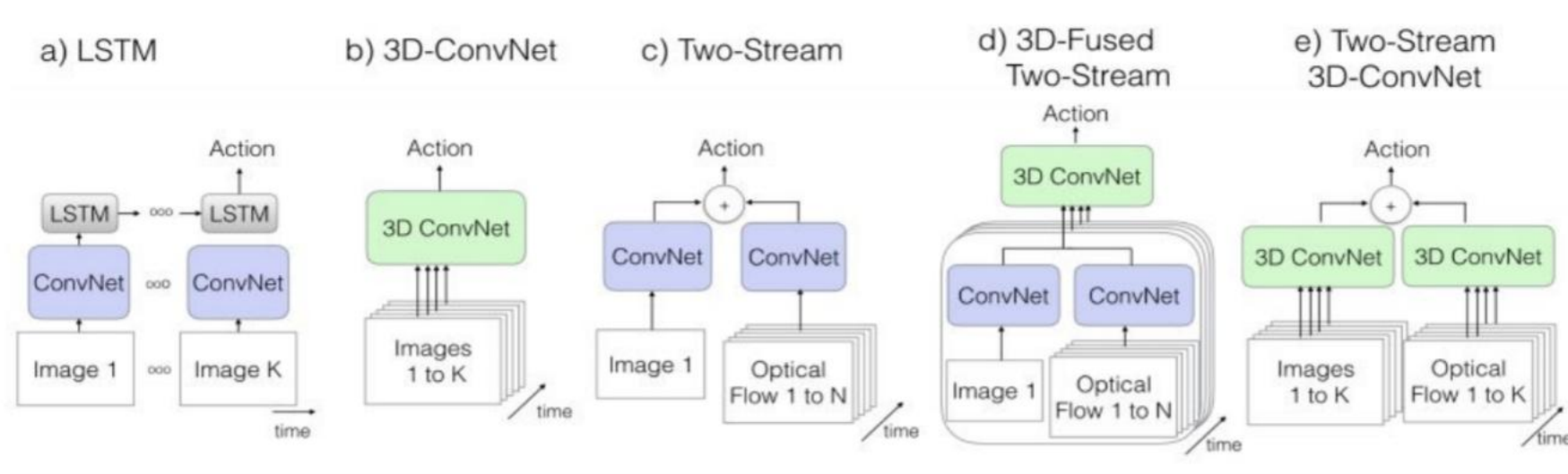
Directional components can be represented as images (or multiple channels of input volume!)

Figure credit: Simonyan and Zisserman. Two-Stream Convolutional Networks for Action Recognition in Videos. NeurIPS 2014.

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow

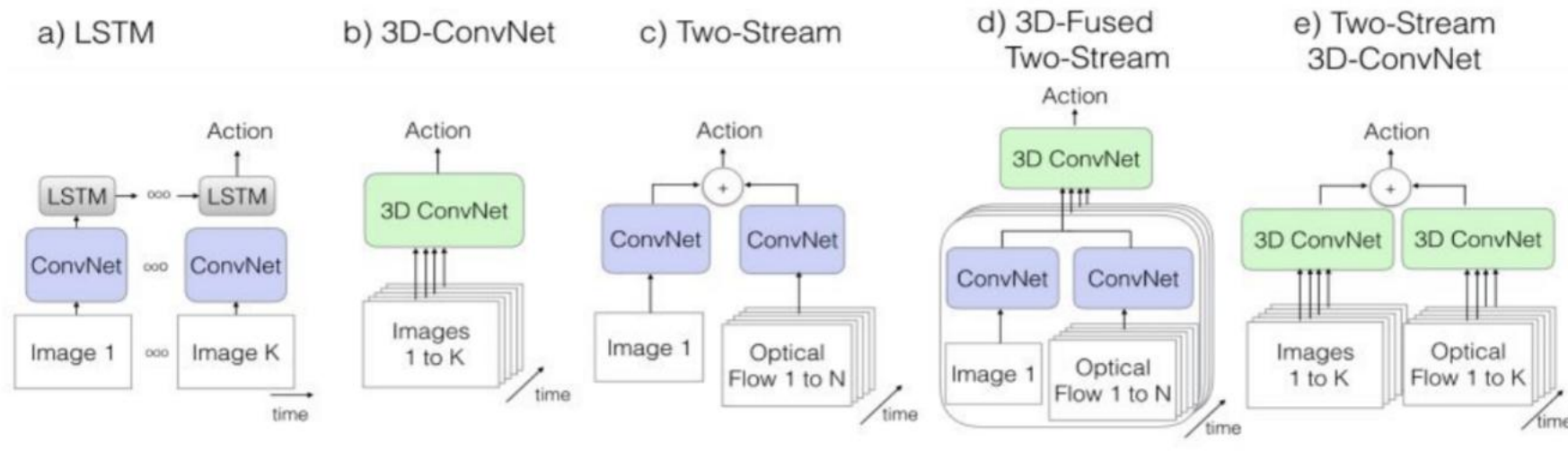


Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow



LSTM over RGB

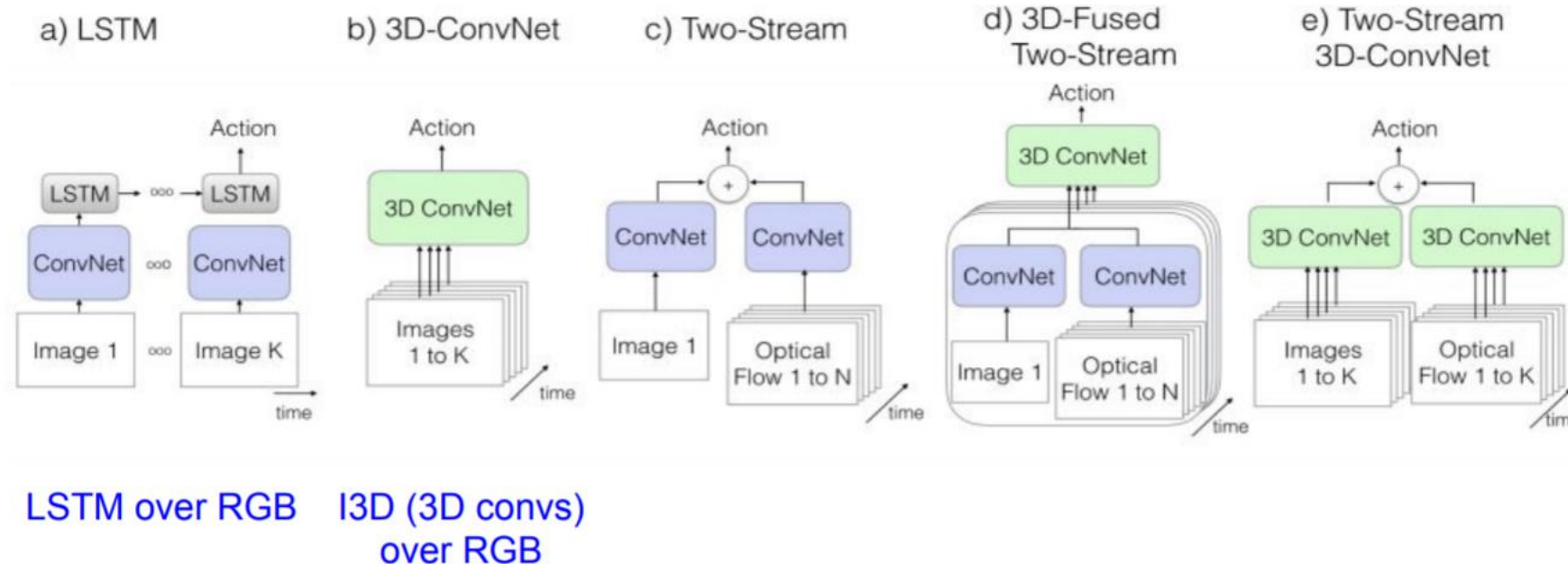
Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

(LSTM is a type of recurrent neural network. We will talk more about these soon!)

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow

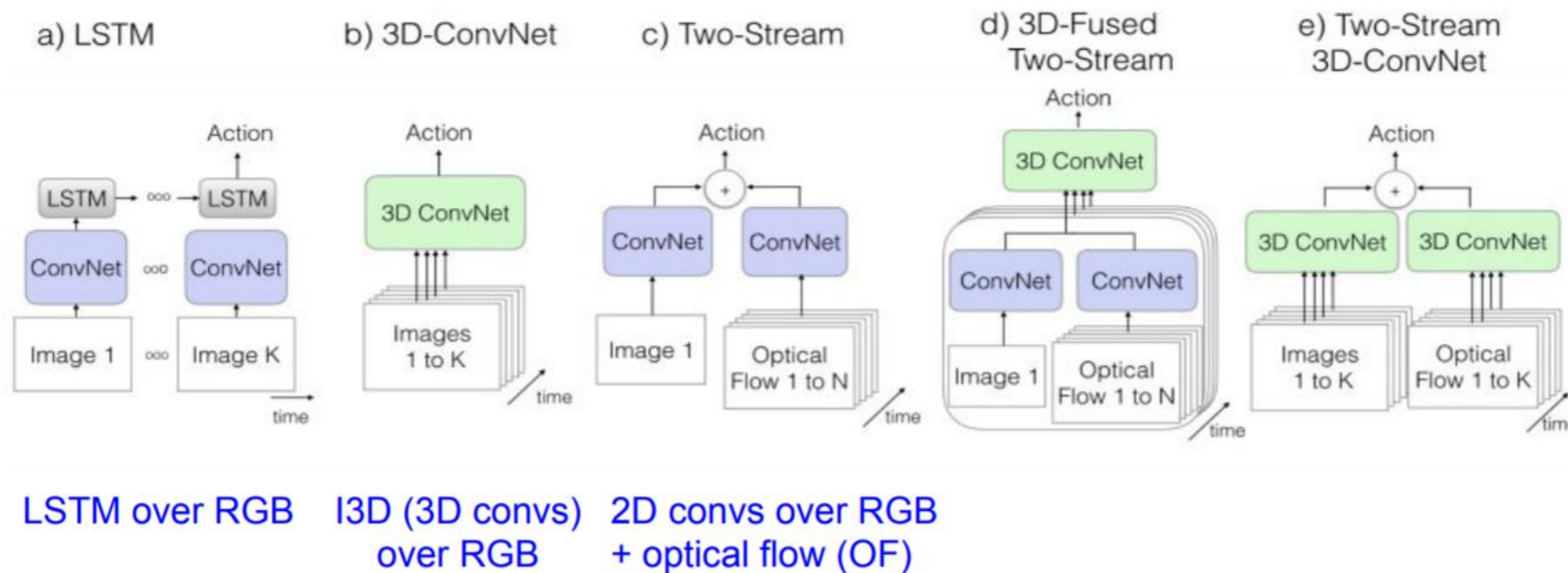


Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow

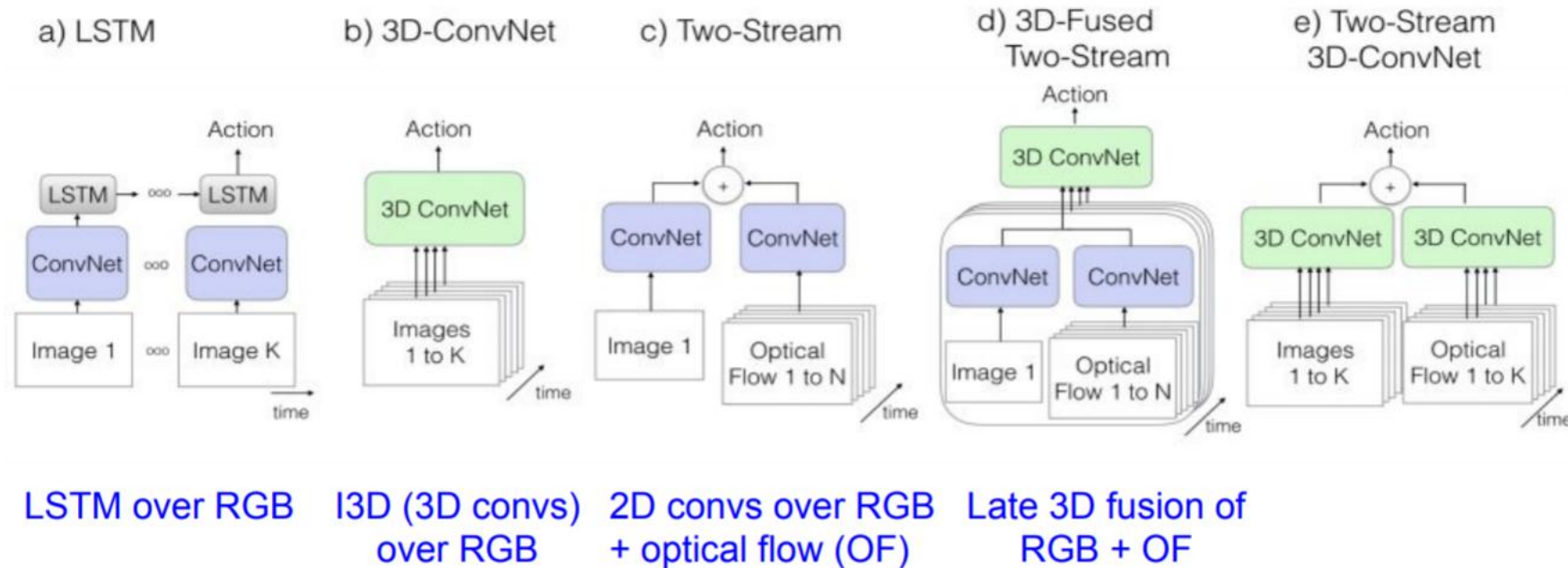


Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow

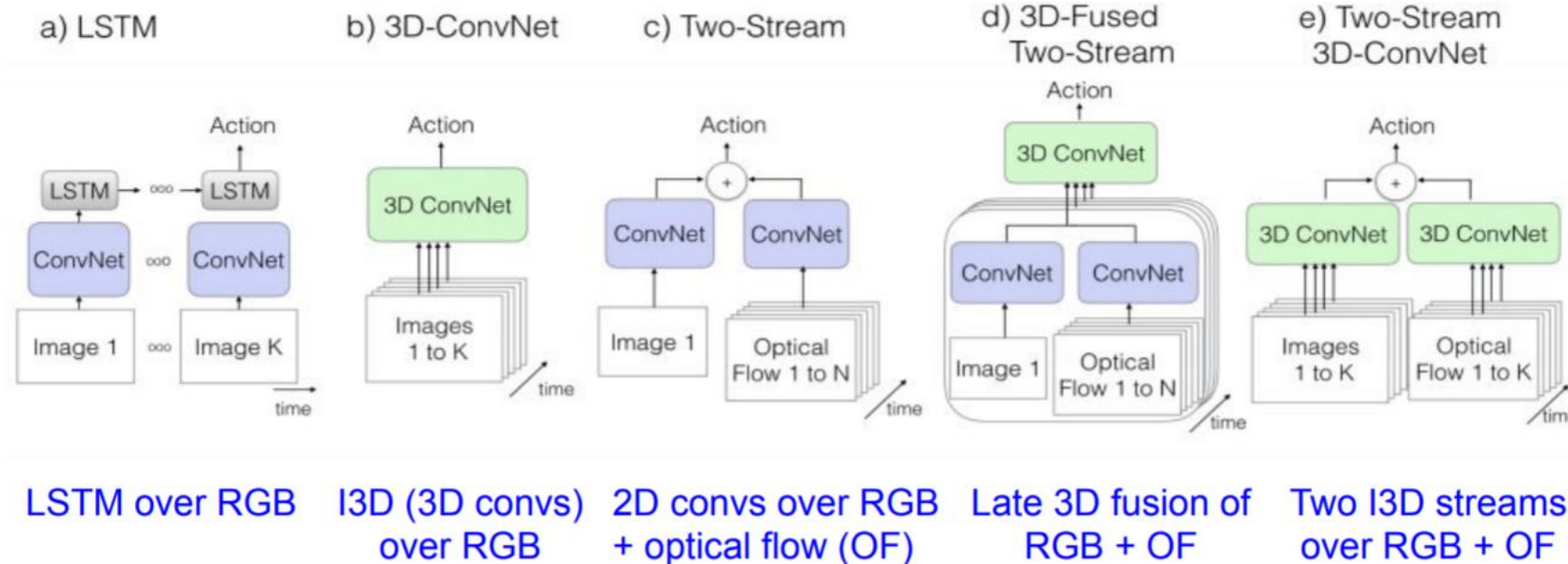


Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

5. Advanced models for higher-dimensional (video) data



Video classifiers (including I3D) often enhanced with optical flow

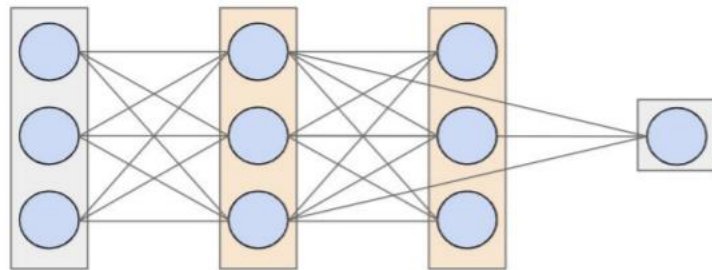


Carreira and Zisserman. Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. CVPR 2017.

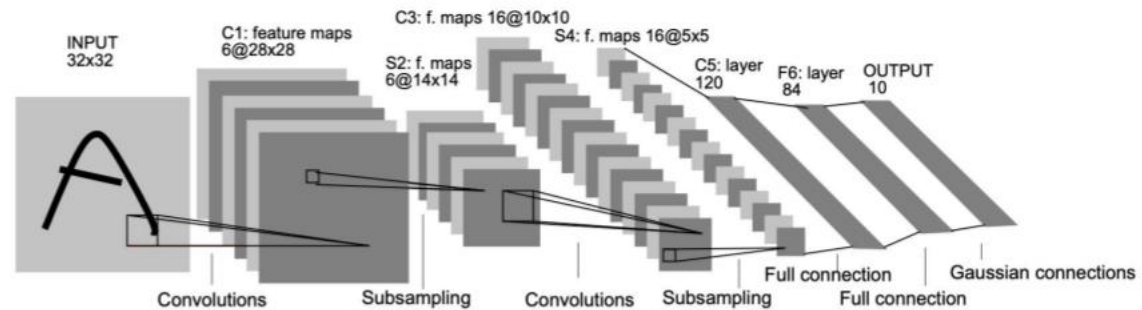
5. Advanced models for higher-dimensional (video) data



Preview: Recurrent neural networks

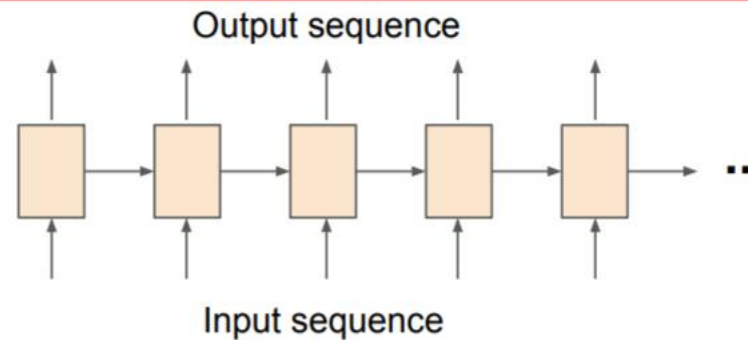


Fully connected neural networks
(linear layers, good for “feature vector” inputs)



Convolutional neural networks
(convolutional layers, good for image inputs)

Recurrent neural networks
(linear layers modeling recurrence relation across sequence, good for sequence inputs)



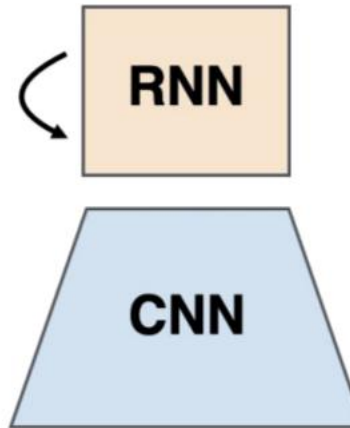
5. Advanced models for higher-dimensional (video) data



Videos are sequences: natural fit for recurrent networks

Abstracted overview: Use a CNN to extract features from each frame (e.g. final-layer features), then use RNN to perform temporal modeling over sequence of features

$$\mathbf{y} = \{y_0, y_1, \dots, y_T\}$$



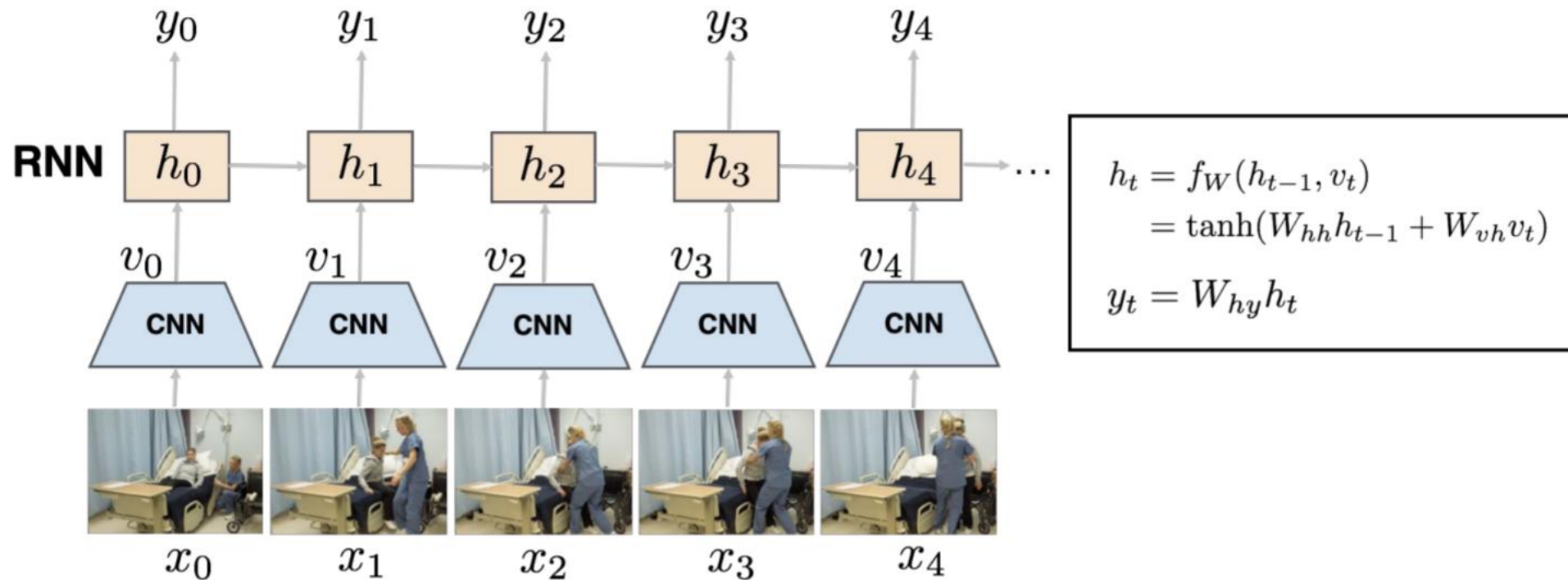
$$\mathbf{x} = \{x_0, x_1, \dots, x_T\}$$

5. Advanced models for higher-dimensional (video) data



Videos are sequences: natural fit for recurrent networks

Diagram of a CNN + RNN “rolled out” over time

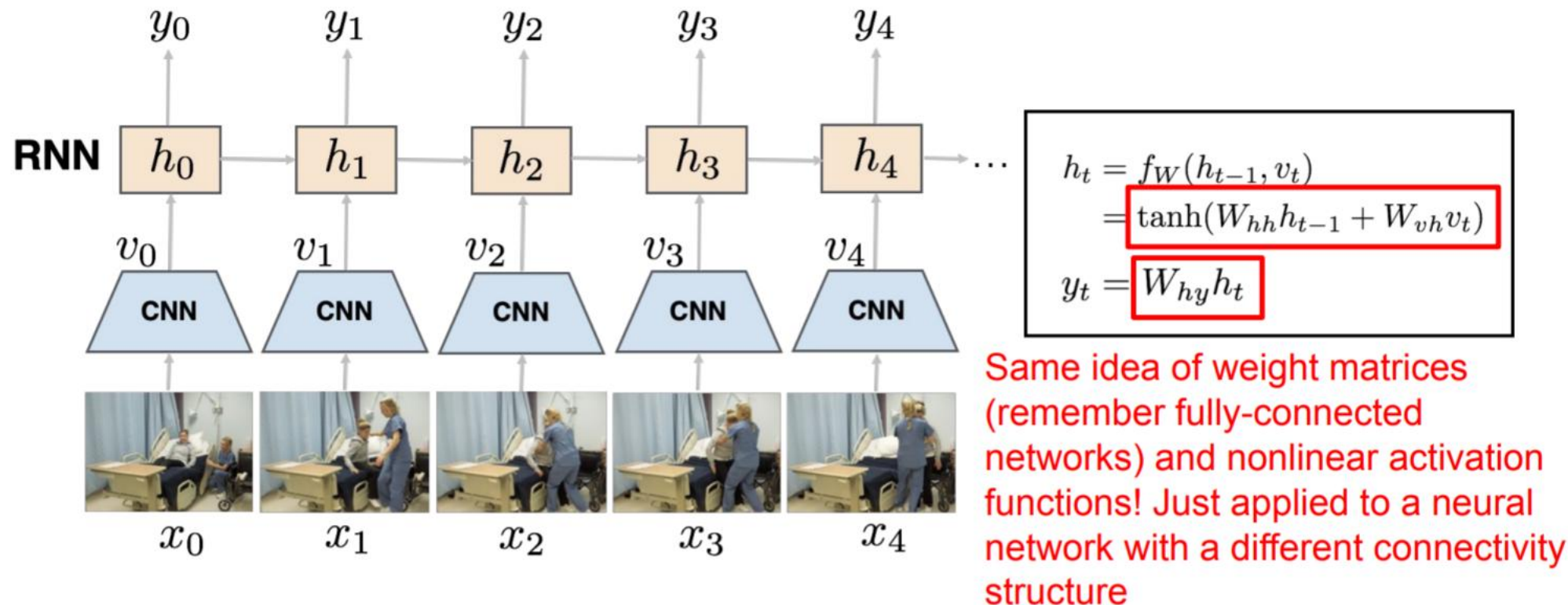


5. Advanced models for higher-dimensional (video) data



Videos are sequences: natural fit for recurrent networks

Diagram of a CNN + RNN “rolled out” over time



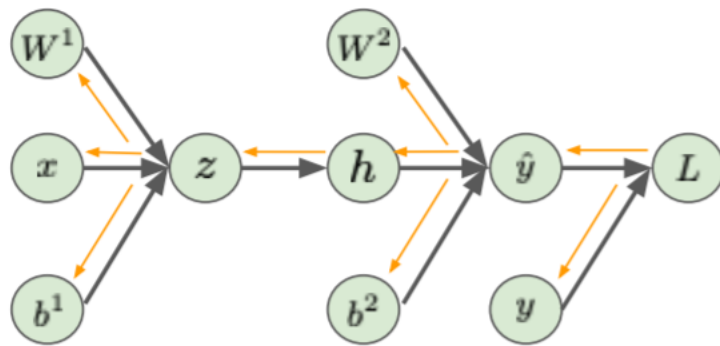
5. Advanced models for higher-dimensional (video) data



Aside: how do we compute gradient updates? Remember backpropagation.

Network output: $\hat{y} = W^2(\sigma(W^1x + b^1)) + b^2$

Think of computing loss function as staged computation of intermediate variables:



“Forward pass”:

$$z = W^1x + b^1$$
$$h = \sigma(z)$$
$$\hat{y} = W^2h + b^2$$
$$L = (\hat{y} - y)^2$$

Now, can use a repeated application of the chain rule, going backwards through the computational graph, to obtain the gradient of the loss with respect to each node of the computation graph.

“Backward pass”: $\frac{\partial L}{\partial \hat{y}} = 2(\hat{y} - y)$ (not all gradients shown)

Plug in from earlier computations via chain rule

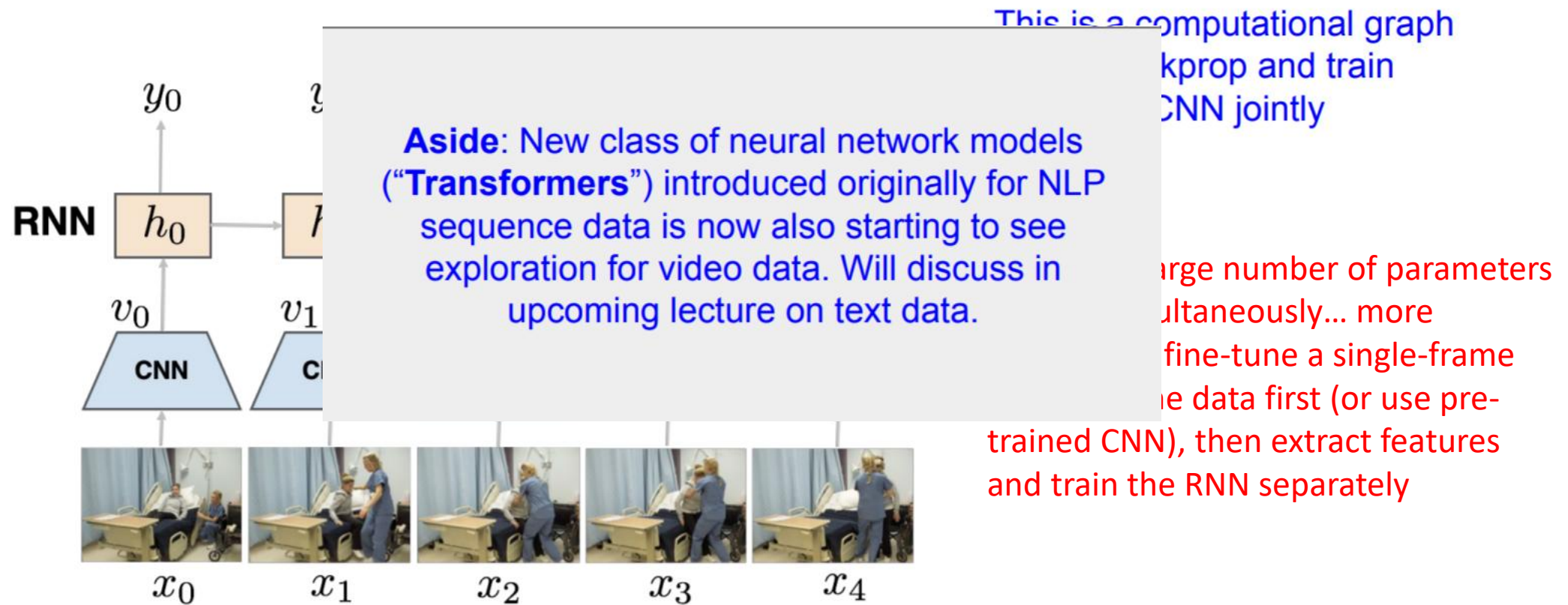
$$\frac{\partial L}{\partial W^2} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial W^2}$$
$$\frac{\partial L}{\partial H} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial H}$$
$$\frac{\partial L}{\partial Z} = \frac{\partial L}{\partial H} \frac{\partial H}{\partial Z}$$
$$\frac{\partial L}{\partial W^1} = \frac{\partial L}{\partial Z} \frac{\partial Z}{\partial W^1}$$

Local gradients to derive

5. Advanced models for higher-dimensional (video) data



Videos are sequences: natural fit for recurrent networks



5. Advanced models for higher-dimensional (video) data



Detecting patient mobilization activities in the ICU

Get patient
out of bed



Sit patient
in chair



Get patient
in bed



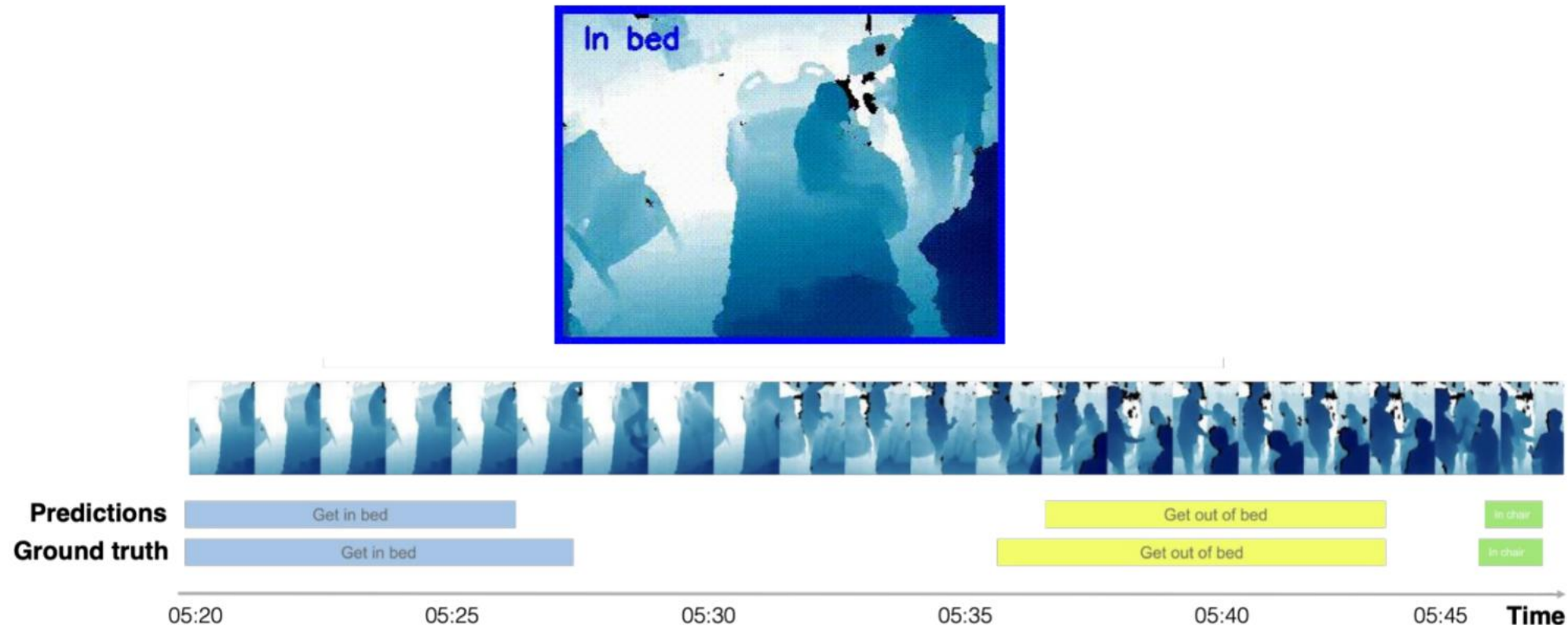
Get patient
out of chair



5. Advanced models for higher-dimensional (video) data



Detecting patient mobilization activities in the ICU

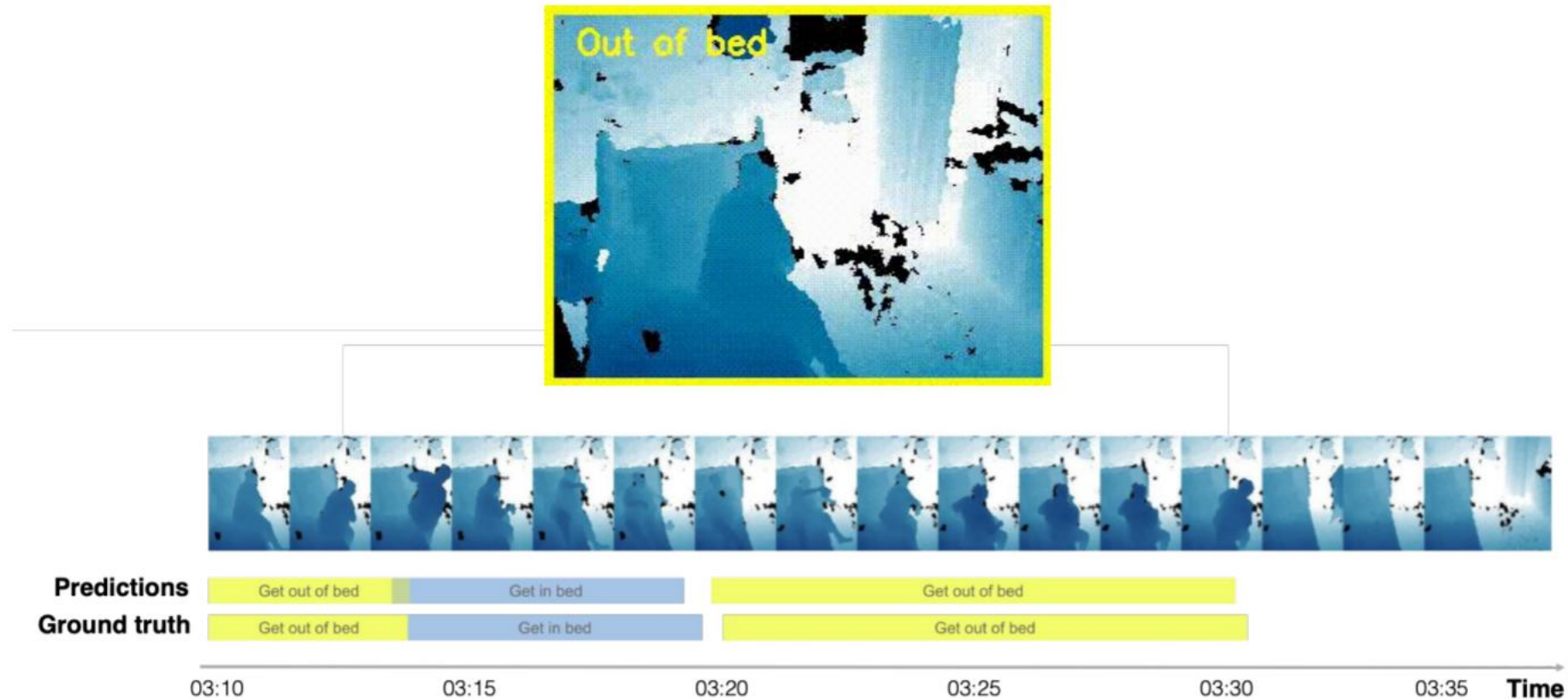


Yeung*, Salipur*, et al. A Computer Vision System for Deep Learning-Based Detection of Patient Mobilization Activities in the ICU. npj Digital Medicine, 2019.

5. Advanced models for higher-dimensional (video) data



Detecting patient mobilization activities in the ICU

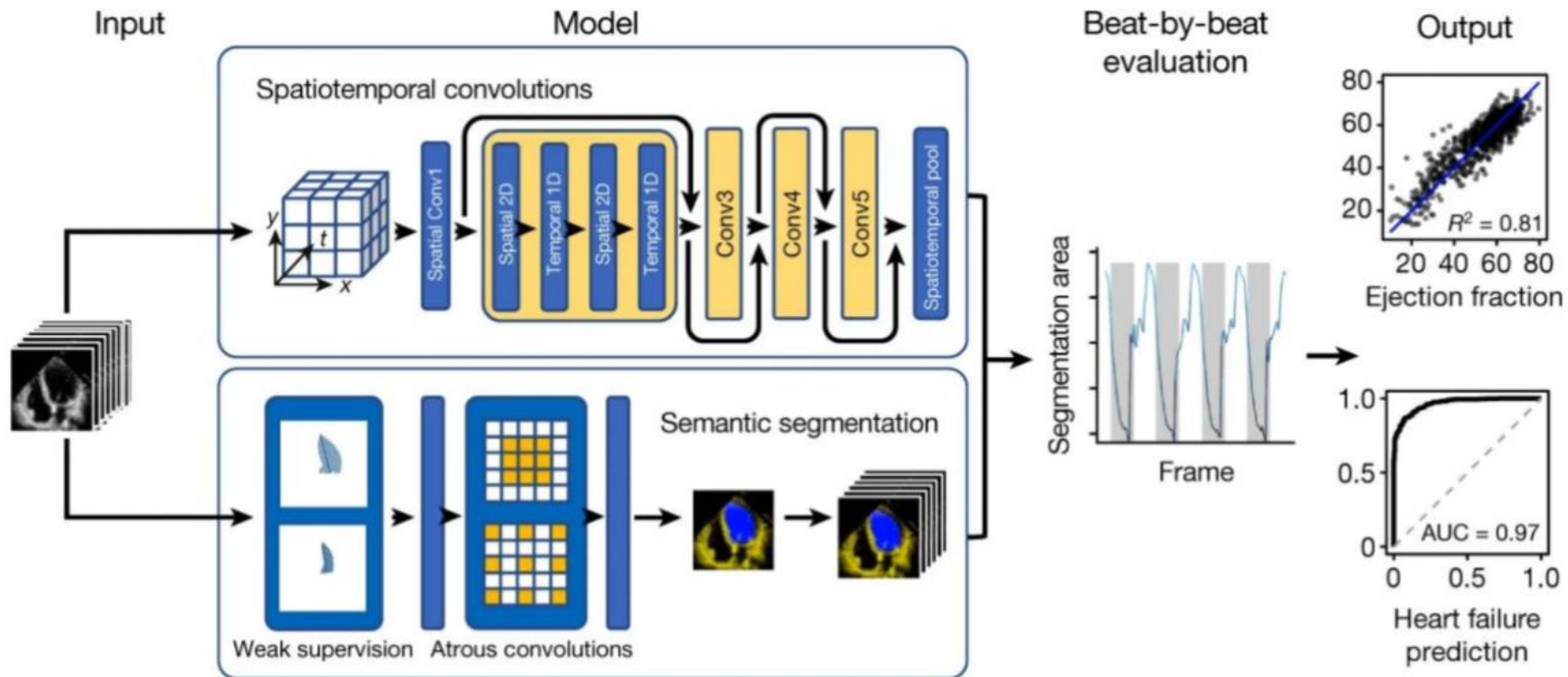


Yeung*, Salipur*, et al. A Computer Vision System for Deep Learning-Based Detection of Patient Mobilization Activities in the ICU. npj Digital Medicine, 2019.

5. Advanced models for higher-dimensional (video) data



Predicting ejection fraction in echocardiograms



Ouyang et al. Video-based AI for beat-to-beat assessment of cardiac function. Nature, 2020.

5. Advanced models for higher-dimensional (video) data

Surgical workflow recognition

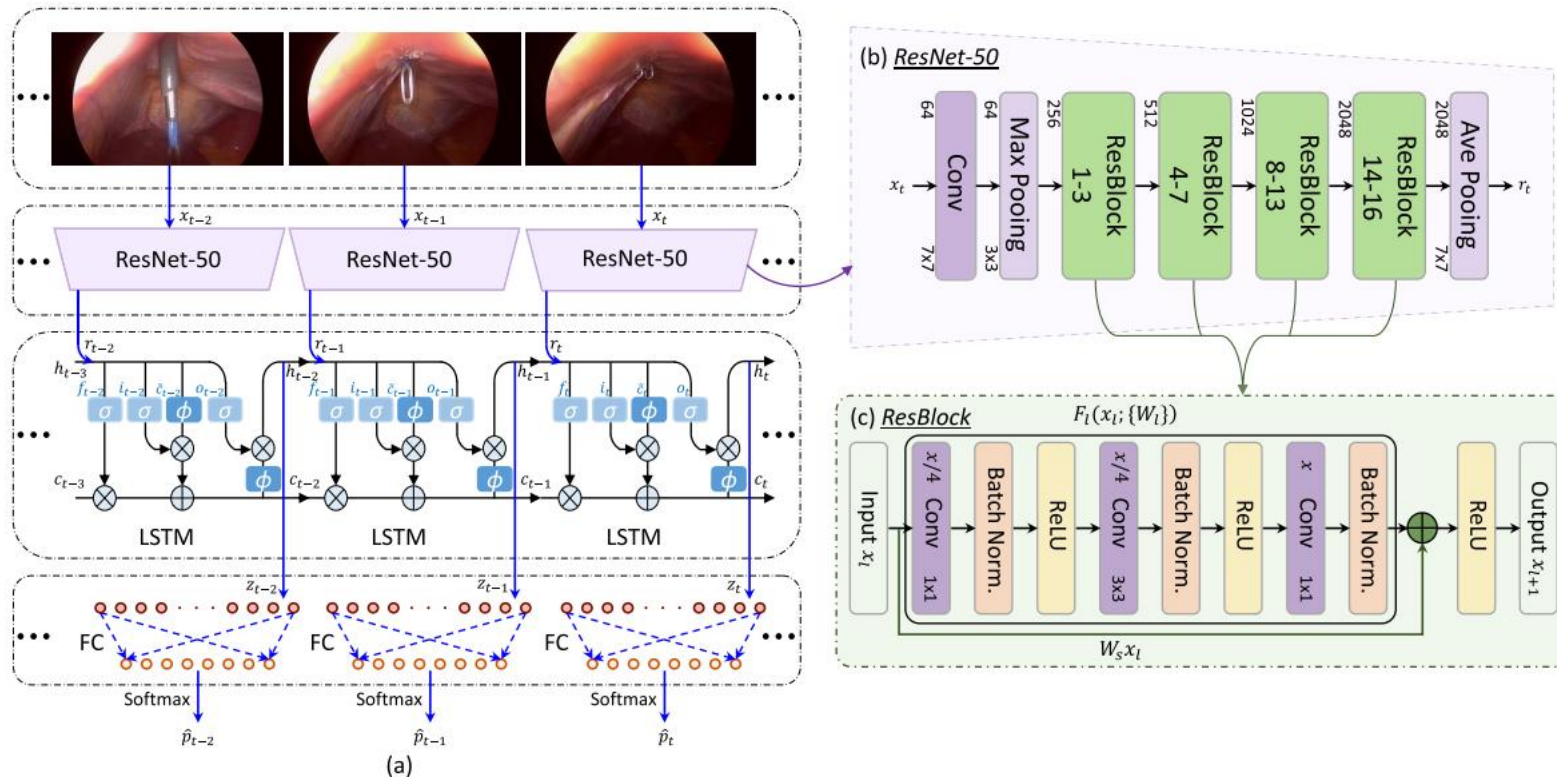
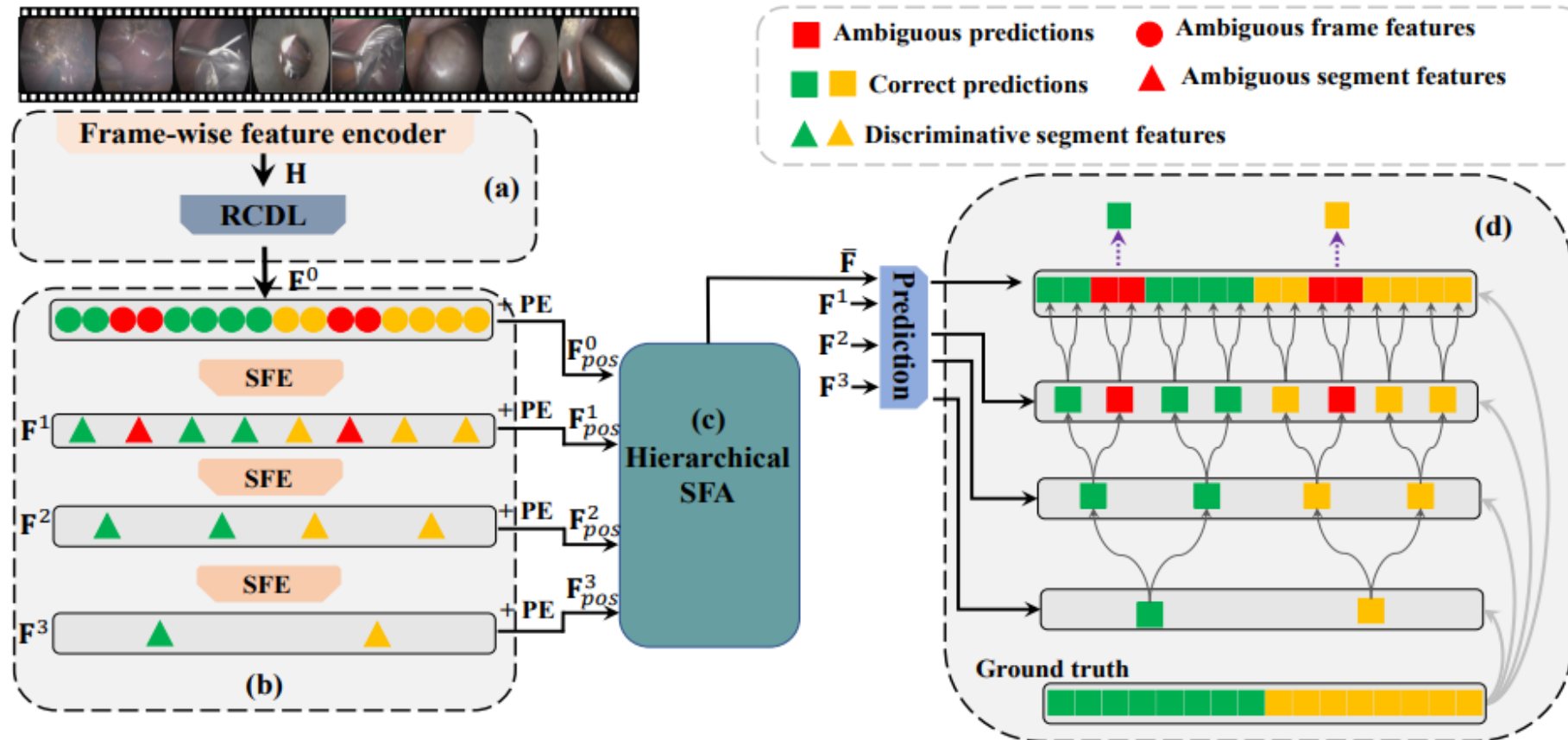


Fig. 2. (a) An overview of the proposed SV-RCNet for workflow recognition from surgical videos. The LSTM networks are instantiated by diagrams to indicate how temporal information is modeled. (b) The architecture of the ResNet to extract visual features from video frames. (c) Illustration of the structure within the residual block.

5. Advanced models for higher-dimensional (video) data



Surgical workflow recognition



5. Advanced models for higher-dimensional (video) data



Surgical instrument segmentation

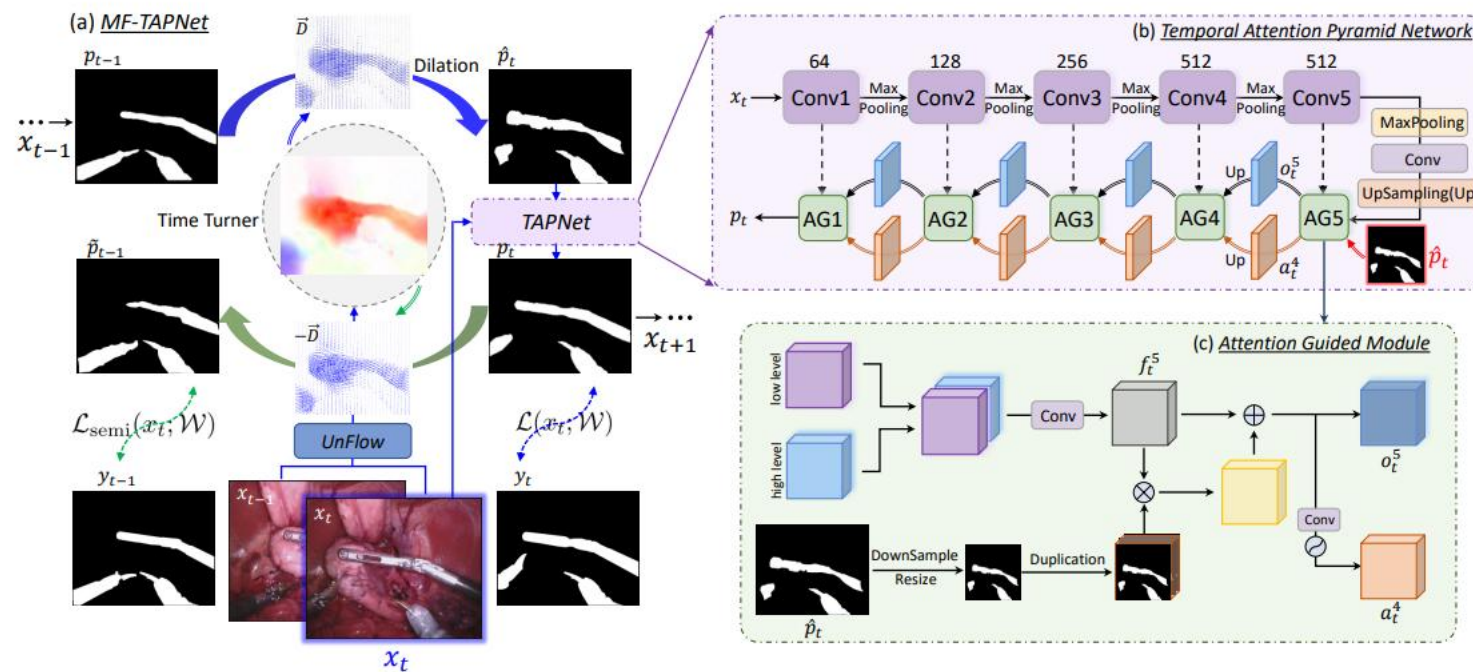


Fig. 1. Illustration of the proposed (a) MF-TAPNet for surgical instrument segmentation based on motion flow, with architecture of (b) temporal attention pyramid network and (c) attention guided module presented in detail.



Finished up advanced deep learning models for visual recognition tasks

- Case studies of CNNs for object detection
- Advanced models for higher-dimensional (3D) data
- Case studies of CNNs for medical image segmentation
- Advanced models for higher-dimensional (video) data

Questions are welcome!